arXiv:2605.06212v2 [cs.LG] 9 Oct 2026

Playing the Network Backward: A Game-Theoretic Attribution Framework

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Abstract

Attribution methods explain which input features drive a model's prediction, making them central to model debugging and mechanistic interpretability. Yet backward attribution methods, including gradients, Layer-wise Relevance Propagation (LRP), and transformer-specific rules, lack a shared framework in which to compare the underlying backward calculations. We introduce such a framework by recasting backward attribution as a two-player game on an extended network graph, building on Gaubert and Vlassopoulos' ReLU Net Game. Gradients and the full $\alpha\beta$-LRP family arise as integrals over game trajectories under specific equilibria, so attribution maps become projections of trajectory distributions rather than the primary object. Desired explanation properties, such as finer localization or robustness to input noise, become game-theoretic choices, such as policy regularization or risk aversion, and yield new adaptations of the known backward rules. In a vision case study, one such adaptation of $\alpha\beta$-LRP outperforms all baselines, including transformer-specific methods, in attribution localization on ViT-B/16. Code is available at `https://github.com/JayPiZimmermann/Playing_The_Network_Backward_Official_Repository`.

1 INTRODUCTION

Attribution has shifted from a tool for inspecting model decisions to a practical mechanism for improving and controlling models: recent work uses attributions to reveal and correct spurious behavior in vision models (Pahde et al., 2023), identify retrieval and parametric heads in language models (Kahardipraja et al., 2025), steer decoding toward instructions or evidence (Komorowski et al., 2026), and localize circuit components through attribution patching (Syed et al., 2024; Kramár et al., 2024; Hanna et al., 2024). This makes critiques of saliency more consequential: parameter insensitivity, input dependence, perturbation fragility, and misleading modified-gradient structure (Adebayo et al., 2018; Kindermans et al., 2018; Ghorbani et al., 2019; Nie et al., 2018) are no longer only failures of visual explanation, but potential failure modes of downstream procedures built on attribution. Understanding these methods therefore requires a common account of the backward calculation itself.

Backward attribution, however, is usually presented as a zoo of recipes—gradients as local sensitivities, Layer-wise Relevance Propagation (LRP) as redistribution, Deep Taylor as local decomposition, and transformer-specific rules as handcrafted extensions (Bach et al., 2015; Montavon et al., 2017; Binder et al., 2016; Kohlbrenner et al., 2020; Chefer et al., 2021b; Ali et al., 2022; Achtibat et al., 2024; Wróbel et al., 2026). Ancona et al. (2018) unify several gradient-based rules as modified gradients, a map-level unification that leaves out the $\alpha\beta$ rule; we instead seek a signed trajectory-space representation of gradients and the $\alpha\beta$ rule that exposes their routing laws. This need is especially acute on Vision Transformers (ViTs): LayerNorm's feature-wise coupling through its input-dependent mean and variance and the QK-softmax break LRP conservation (Ali et al., 2022).

Network calculation as a backward game. Gaubert and Vlassopoulos (2025) prove that a ReLU network's forward output is the game value of a two-

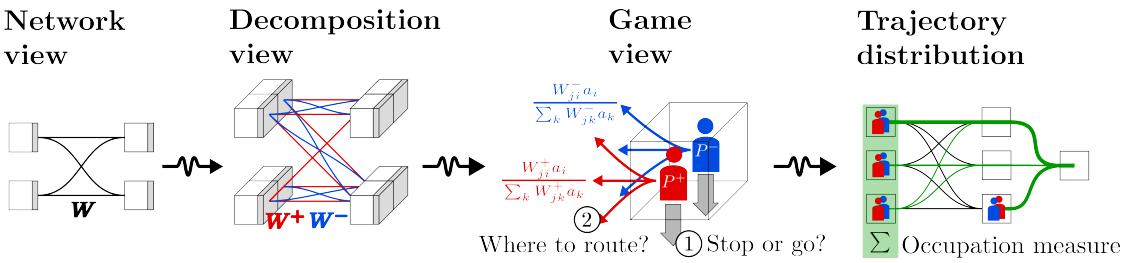

Figure 1: The backward pass as a two-player game. The Stopping Decomposition splits every weight by sign, $W = W^+ - W^-$; at each state of the extended graph the players decide (1) stop or continue and (2) where to route. Their equilibrium policies induce a distribution over backward trajectories, and the discounted occupation measure of that distribution at the input states is the attribution map.

player Stopping Game on its graph: layers are rounds, neurons are states, input values are terminal payoffs, and the scalar output $f(x)$ is the game value. We reinterpret backward attribution as a **distribution of weighted game trajectories** through the network (Figure 1), yielding a framework to compare different backward attribution methods. We adapt the game to a novel Stopping Decomposition, separating paths with positive and negative contributions to the prediction (Lemma 1)—information that the original game formulation averages out. In this framework, propagation rules become **equilibrium policies**, and attribution maps are recovered from the **discounted occupation measure** of the induced trajectory distribution—an integral over the path distribution assigning relevance to each game state. We introduce two games: the **Stopping Game (SG)**, whose occupation measures recover the Jacobian (Theorem 1), and the entropy-regularized **Routing Game (RG)** at temperature τ, whose occupation measures recover the full $\alpha\beta$-LRP family at its anchor $\tau{=}1$ (Theorem 2) on ReLU networks. Guiding the player's policies at attention blocks by reference policies of an oracle player recovering the QK-softmax extends the framework to attention modules; the same entropy regularization extends SG from Rectified Linear Unit (ReLU) to Softplus networks.

Consequences of the reformulation. *(i)* We interpret the trajectory distribution as the network's backward calculation. The Hellinger distance $\mathrm{H}(\Pi_A, \Pi_B)$ between two such distributions—computable by a linear-time backward pass—therefore gives a metric between calculations rather than between heatmaps. Under cascading randomization (Adebayo et al., 2018), this distance separates trained from randomized calculations even where their pixel projections stay similar: the two diagnostics detect different aspects of parameter sensitivity. *(ii)* The game framework yields **analytical modes**: ideas like risk aversion, action space augmentation or policy regularization map back to adaptations of classical backpropagation rules that change attribution maps in a controlled way. *(iii)* $\alpha\beta$-LRP parameters have game-space duals: (α, β) set occupation-measure path weights, while the ε-stabilizer becomes a stop option with payoff $\log(\varepsilon)$.

Contributions. *(i)* We present the Stopping Game (SG), adapting the ReLU Net Game of Gaubert and Vlassopoulos (2025) to the Stopping Decomposition, and the novel entropy-regularized Routing Game (RG). *(ii)* We prove that known methods arise as equilibria: SG occupation measures recover the Jacobian, and RG occupation measures at $\tau{=}1$ recover the full $\alpha\beta$-LRP family on ReLU networks. *(iii)* The Hellinger distance between trajectory distributions separates trained from randomized calculations under cascading weight randomization (Adebayo et al., 2018), even when their attribution maps look alike (§6). *(iv)* Finally, we show the utility of the framework beyond unification: the game-level changes such as risk aversion or policy regularization derive new attribution methods in a principled way, whose performance we demonstrate in a case study on ViT-B/16, VGG-16 and ResNet-50.

2 RELATED WORK

Game and trajectory views. The closest mathematical precursor is the ReLU Net Game of Gaubert and Vlassopoulos (2025), which identifies the forward evaluation of a ReLU network with the value of a backward Stopping Game but does not connect that game to the backward pass or to attribution. We reformulate the game to draw the connection to a novel Stopping Decomposition, whose inherent weight decomposition is related to the Difference-of-Convex (DC) decomposition of neural networks—the decomposition Zimmermann and Loho (2026) use directly for attribution. Our SG and RG are different games utilizing this weight decomposition, chosen to recover backward attribution rules rather than the pure DC decomposition.

Relevance propagation and stochastic explanations. LRP, Deep Taylor, and Excitation Backprop

constitute the core relevance propagation family (Bach et al., 2015; Montavon et al., 2017; Zhang et al., 2018), with later variants modifying redistribution for contrast, stability, or signed relevance (Binder et al., 2016; Kohlbrenner et al., 2020; Gu et al., 2019; Nam et al., 2020; Vukadin et al., 2024). We supply a common trajectory-space semantics for this family: SG recovers gradients, RG recovers the generalized $\alpha\beta$ rule, and its analytical modes map back to principled modifications of existing backward rules. Excitation Backprop (Zhang et al., 2018) is an absorbing Markov chain recovering the $\alpha\beta$-LRP rule for the special case $(\alpha, \beta) = (1, 0)$ known as the z^+-rule, which only considers the positive weights in the relevance propagation. Our game extends it with an opponent player (enabling negative-weight influence) and replaces input-dependent transition probabilities with soft Bellman equations. Deep Taylor (Montavon et al., 2017) justifies that same z^+-rule and explicitly leaves the mathematical modeling of the general $\alpha\beta$ rule as an open problem; RG closes this.

Shapley-value attribution methods. SHAP and KernelSHAP frame explanation as a cooperative game over feature coalitions (Lundberg and Lee, 2017) in a *black-box* setting; they are therefore a distinct, complementary game that explains marginal input-feature contributions relative to a baseline, whereas our backward game explains how relevance is routed through network internals in a *white-box* setting.

Robust explanations via Softplus. Dombrowski et al. (2022) propose Softplus replacement as a curvature-based defense against gradient-explanation manipulation. Appendix F shows that entropy-regularizing the activation-state stop/continue choice in SG recovers Softplus-network gradients, situating their construction within our framework.

Transformer and ViT explainability. Attention Rollout / Flow read attention as a routing graph (Abnar and Zuidema, 2020); transformer-LRP methods propagate through attention and skip connections (Chefer et al., 2021b; Ali et al., 2022; Achtibat et al., 2024); DAVE addresses ViT tokenization artifacts via conditioned operators and equivariant averaging (Wróbel et al., 2026). The conditioned-operator transformer constructions arise naturally inside our game semantics: the QK-softmax is the oracle player's reference policy, while attribution lives on the Value-routing trajectories. Hence, our method is compatible with Reynolds averaging, which in our RG+Eq mode removes the same grid artifact that DAVE targets—however, with more control via the different game modes.

3 BACKWARD ATTRIBUTION

Consider a ReLU network with layers $l = 1, \ldots, L$, input $a^{(0)} = x$, pre-activations $z_j^{(l)} = \sum_i a_i^{(l-1)} W_{ij}^{(l)} + b_j^{(l)}$ and activations $a_j^{(l)} = \mathrm{ReLU}(z_j^{(l)})$, where $W_{ij}^{(l)}$ connects neuron i of layer $l-1$ to neuron j of layer l. We explain a scalar output $f(x) = a_u^{(L)}$. We write $[u]^+ = \max(u, 0)$ and $[u]^- = \max(-u, 0)$, so $u = [u]^+ - [u]^-$, and $W^{(l,\pm)} = [W^{(l)}]^\pm$ entrywise.

Layer-wise relevance propagation. LRP (Bach et al., 2015) explains $f(x)$ by redistributing it backward, layer by layer: starting from $R_u^{(L)} = f(x)$, every neuron passes its relevance $R_j^{(l)}$ on to its inputs in proportion to their contributions $a_i^{(l-1)} W_{ij}^{(l)}$ to its pre-activation, so that relevance is conserved from layer to layer up to the share absorbed by biases and stabilizers. Rules differ in how the contributions are weighted. The $\alpha\beta$-rule is parameterized by $\alpha, \beta \geq 0$, with $\alpha - \beta = 1$ for conservation (attributions sum to $f(x)$), and a stabilization parameter $\varepsilon \geq 0$ (added to each sign denominator, as in common implementations (Anders et al., 2021)): the relevance of neuron j is split into a positive-contribution fraction scaled by α and a negative-contribution fraction scaled by β, each normalized by its own denominator,

$$R_i^{(l-1)} = \sum_j \Big(\alpha \, \frac{[a_i^{(l-1)} W_{ij}^{(l)}]^+}{\Sigma_j^{(l,+)} + \varepsilon} - \beta \, \frac{[a_i^{(l-1)} W_{ij}^{(l)}]^-}{\Sigma_j^{(l,-)} + \varepsilon} \Big) R_j^{(l)},$$
$$\Sigma_j^{(l,\pm)} = \sum_k [a_k^{(l-1)} W_{kj}^{(l)}]^\pm + [b_j^{(l)}]^\pm. \tag{1}$$

Relevance assigned to the bias and to ε is absorbed. $(\alpha, \beta) = (1, 0)$ gives the z^+-rule of Deep Taylor decomposition (Montavon et al., 2017), which turns out to be the same rule as Excitation Backprop (Zhang et al., 2018); other rules (LRP-ε, LRP-γ) change the redistribution fractions.

Transformers. In a Vision Transformer (ViT), LayerNorm couples the features of a token through its input-dependent mean and variance, and the attention matrix $A = \mathrm{softmax}(QK^\top/\sqrt{d})$ depends on the input, which breaks conservation (Ali et al., 2022). Transformer-specific rules therefore *condition* on the forward pass: conservative propagation (Ali et al., 2022) and DAVE (Wróbel et al., 2026) hold A and the LayerNorm statistics fixed and propagate relevance through the Value path only; AttnLRP (Achtibat et al., 2024) also propagates through the softmax and the QK product.

All these rules share one structure: a backward recursion that routes mass from each neuron to its inputs. The games below make the routing explicit.

4 ATTRIBUTION AS A GAME

Setup and common state-space structure. With the notation of Section 3, we build a finite-horizon two-player game (Osborne and Rubinstein, 1994) played *backward* from the output: a state records a position in the backward traversal (layer, neuron, plus game-specific bookkeeping for player turn and stream sign), and at each non-terminal state the active player either stops, moving to an absorbing cemetery state †, or continues. Skip connections become random transitions to summand states; max-pooling sends players greedily to maximal-payoff states. Appendix A collects the bookkeeping labels (Table 3). The SG value $V_p(s)$ is the γ-discounted expected payoff difference $r_p - r_{p'}$ of player p and opponent p' along trajectories from s under the policies, expressed for the player p in turn at s by the Bellman recursion

$$V_p(s) = \max_{a \in \mathcal{A}_x(s)} \Big\{ r_p(s,a) - r_{p'}(s,a) + \gamma(s) \textstyle\sum_{s'} \mathbb{P}_x(s' \mid s,a)\, V_p(s') \Big\}, \quad (2)$$

whereas the Routing Game replaces the hard maximization at routing states by an entropy-regularized maximization over mixed routing decisions, whose value is the soft game value (5). Any player policies induce a backward trajectory measure Π_x on trajectories $\zeta = (\zeta_0, \zeta_1, \dots)$ with a game-specific cumulative discount $d_t(\zeta)$ up to time t. The **discounted occupation measure** (Puterman, 1994)

$$\Gamma_x(v) \coloneqq \mathbb{E}_{\zeta \sim \Pi_x}\big[\textstyle\sum_t d_t(\zeta)\, \mathbf{1}\{\zeta_t = v\}\big] \quad (3)$$

quantifies how much weight the trajectories going through a state have.

4.1 The Stopping Game

With each neuron j in layer l, we associate two states $s^{(l)}_{j,+}$ and $s^{(l)}_{j,-}$, where the sign encodes which player is in turn. The biases become the immediate payoffs for continuing; under continuation, weight magnitudes drive the environment transitions with discount $\gamma^{(l)}_j = \sum_i |W^{(l)}_{ij}|$ (positive- and negative-weight predecessors follow $\mathbb{P} \propto W^{(l,\pm)}/\gamma^{(l)}_j$, and a negative weight passes the turn to the opponent). The players' expected payoffs realize the novel Stopping Decomposition, separating positive and negative paths to each neuron activation (Lemma 1). The gradient is then the difference of two *non-negative* occupation measures:

Theorem 1 (Gradient representation)**.** *The Stopping Game allows a subgame-perfect Nash equilibrium, unique up to tie-breaking at zero-activation neurons. Under the stop-at-tie-breaks equilibrium the discounted occupation measure* $\Gamma_x(s^{(l)}_{i,p}) \geq 0$ *equals the output derivative with respect to the corresponding* Stopping Decomposition *activation. The gradient of f with respect to the input is then the difference* $\Gamma_x(s^{(0)}_{i,+}) - \Gamma_x(s^{(0)}_{i,-})$.

Appendix B proves the theorem, works through an example, gives the full state space with skip connections and max-pooling, and compares the Stopping Decomposition with the DC decomposition.

Modeling $\alpha\beta$-LRP or attention modules via the SG would require activation-dependent transitions. The RG moves activation-dependence into the players' policies, opening a continuous family of deformations.

4.2 The Routing Game

In the RG we model every neuron by three game states: an activation state $s^{(l,\mathrm{act})}_{j,q}$ and two linear routing states $s^{(l,\mathrm{lin})}_{j,q,+}$ and $s^{(l,\mathrm{lin})}_{j,q,-}$, the + stream state and the − stream state, modeling the positive part $W^{(l,+)}$ and the negative part $W^{(l,-)}$ of the affine layer respectively. At an activation state the player in turn stops or continues to both streams with probability $\frac{1}{2}$ each; the − stream passes the turn to the opponent, and discounting the game value by 2 compensates the split. At a *linear routing state* $s^{(l,\mathrm{lin})}_{j,q,\kappa}$ for neuron j in layer l on sign stream $\kappa \in \{+,-\}$ with player q in turn, the active player mixes over predecessors i, a bias node and an outside option $\perp_{\mathrm{lin}}$ under a policy $\pi_q(s^{(l,\mathrm{lin})}_{j,q,\kappa})$; the bias node and $\perp_{\mathrm{lin}}$ absorb the trajectory, with log-payoffs $\log[b^{(l)}_j]^\kappa$ and $\log \varepsilon$. Writing the game in log space turns the active player r's advantage of moving to predecessor i into an additive action–value,

$$Q_r\big(s^{(l,\mathrm{lin})}_{j,q,\kappa}, i\big) = \begin{cases} \log W^{(l,\kappa)}_{ij} + V_r(s^{(l-1,\mathrm{act})}_{i,q}), & r{=}q,\ l{\geq}2, \\ \log\big[x_i W^{(1)}_{ij}\big]^\kappa, & r{=}q,\ l{=}1, \\ -\infty, & r \neq q, \end{cases} \quad (4)$$

where $V_r(\cdot)$ denotes the *game value* at the successor activation state from the perspective of player r. At each linear routing subgame we regularize the players' policies with Shannon entropy (Geist et al., 2019), which yields the following equilibrium value and Gibbs best response:

$$V_q\big(s^{(l,\mathrm{lin})}_{j,q,\kappa}\big) = \tau \log \textstyle\sum_i \exp\big(Q_q(s^{(l,\mathrm{lin})}_{j,q,\kappa}, i)/\tau\big), \quad (5)$$

$$\pi_q(s^{(l,\mathrm{lin})}_{j,q,\kappa})(i) \propto \exp\big(Q_q(s^{(l,\mathrm{lin})}_{j,q,\kappa}, i)/\tau\big), \quad (6)$$

where the sums include the bias node and $\perp_{\mathrm{lin}}$. At $\tau > 0$ the game admits an SPNE whose occupation measure at $\tau = 1$ is the $\alpha\beta$-LRP attribution:

Theorem 2 ($\alpha\beta$-LRP representation)**.** *For $\tau > 0$, the Gibbs policies of Eq.* (6) *and stopping decisions according to neuron activations yield SPNEs of the entropy-regularized RG, unique up to tie-breaking at zero-activation neurons. Under the stop-at-tie-breaks equilibrium at $\tau = 1$ the $\alpha\beta$-LRP attribution map in* (1) *equals $R_i^{(l)} = f(x)\left(\Gamma_x(s_{i,+}^{(l)}) - \Gamma_x(s_{i,-}^{(l)})\right)$.*

Appendix C gives the full game definition, the proofs, and a worked example of the local routing dynamics (Figure 4).

In practice we use τ as a *myopic* control: each linear subgame applies the τ-scaled Gibbs policy to its own action values, but inherits game values from the soft Bellman equations at τ=1. We do not re-solve at τ downstream: that would change the forward activations, while we want trajectories that explain the network's actual forward pass. The myopic setting at $\tau \neq 1$ is a controlled temperature deformation of the occupation measure. Low τ concentrates routing on the dominant pathway.

4.3 Attention via a detached-logit oracle

Two ViT subtleties break conservative transport: LayerNorm's input-dependent statistics and the data-dependent QK-softmax. LayerNorm statistics are detached so the layer becomes a pointwise linear map (Ali et al., 2022; Wróbel et al., 2026); the QK-softmax block is replaced by an *oracle game* emitting a detached reference policy, the softmax row $A_{t\cdot}$, with occupation-measure trajectories flowing only through the Value route. Like conservative propagation (Ali et al., 2022), this explains the network conditioned on its realized attention pattern.

Theorem 3 (Attention, informal)**.** *With reference policy $A_{t\cdot}$, the KL-regularized Value-routing subgame has an SPNE whose occupation measure at the block input equals $\alpha\beta$-LRP on the conditioned effective map $x \mapsto \sum_k A_{tk}\, x_k W_v + b_v$ (Theorem 9).*

Our implementation applies this rule to the attention product and the Value projection as one composite step; a full gradient fallback through the attention block degrades localization (AL 0.624 vs. 0.711 at the localization-selected configuration). Equivariant Reynolds averaging optionally suppresses tokenization-grid artifacts (Wróbel et al., 2026) (RG+Eq). Full KL formulation, oracle embedding, occupation-measure theorem and implementation are in Appendix D.

5 GAME-THEORETIC MODES

Game-theoretic concepts like risk aversion and entropy regularization map back to novel modes of the backward rules for probing and tuning explanation properties (Table 1). The exact representations of Section 4 are the defaults; every mode other than the LRP parameters $(\alpha, \beta, \varepsilon)$ is a deformation that is off at the anchor. We use five: the temperature τ (focus), risk aversion λ at activation states and its attention-logit variant λ_{sm}, the probit gate $\tilde{\Phi}$ (gate smoothing), and equivariant averaging over N input transformations (grid artifacts).

Focus. The myopic temperature of Section 4.2 rescales the Gibbs policy at every linear state while keeping the τ=1 game values. Low τ concentrates routing on dominant predecessors; high τ spreads mass more uniformly, exposing broader contextual features as well as architectural priors. *Risk aversion.* Assumed Density Filtering (ADF; Gast and Roth, 2018) propagates a Gaussian mean μ and variance σ^2 for every pre-activation z from an input variance σ^2 in one additional moment pass. The risk-averse player replaces the observed z at activation states by the lower-confidence bound $\mu + \lambda\sigma$ with $\lambda < 0$, the pessimistic counterpart of confidence-bound exploration (Auer et al., 2002; Fei et al., 2020), and so attenuates routes through high-variance neurons; linear-state routing is unchanged. On attention blocks, λ_{sm} applies the same shift to the oracle's attention logits before the softmax, downweighting high-variance key tokens. *Gate smoothing.* The risk-averse stop/continue rule is the hard gate $\mathbf{1}[\mu/\sigma + \lambda > 0]$. The probit gate $\tilde{\Phi}$ lets the player instead continue with probability $\Phi(\mu/\sigma + \lambda)$, the marginal of a player who perceives the continuation payoff as $z + \eta$, $\eta \sim \mathcal{N}(0, 1)$, and continues iff it is positive. At μ=z, σ=1, λ=0 this probit-regularized choice recovers the Gaussian-smoothed ReLU $\mathbb{E}[\text{ReLU}(z + \eta)]$, just as entropy regularization recovers Softplus (Appendix F); on ReLU networks it is a smoothing deformation. Because the gate uses the Assumed Density Filtering (ADF) σ, σ^2 is active whenever $\tilde{\Phi}$ is on. *Exact modes and cost.* (α, β) weight the two streams of the occupation measure, and the LRP stabilizer ε adds an outside option $\perp_{\text{lin}}$ with log-score $\log \varepsilon$ at every linear state, leaking weak-evidence neurons into the cemetery while leaving strong-evidence ones near-unaffected (Theorem 8). The exact modes and τ add no cost; λ, λ_{sm} and $\tilde{\Phi}$ add one ADF pass (0.10 → 0.17 s per ViT-B/16 image); averaging multiplies the cost by N (4.93 s at N=50, 8.66 s with λ_{sm}). Appendix E derives the modes and gives defaults, costs and an order of use.

6 TRAJECTORY SANITY CHECK

The cascading-randomization sanity check of Adebayo et al. (2018) asks whether an explanation meaningfully depends on the learned parameters, and answers by comparing attribution maps in pixel space. But the object of interest is the backward calculation itself—a

Table 1: Modes of the games. "Exact": the mode stays within a represented rule; "deform.": optional deformation, off (default value) at the anchor. σ^2: input variance of the ADF moment propagation; N: number of transformed inputs averaged by RG+Eq.

Mode	Effect on the game	Default	Property
(α, β)	branch discounts of the two streams (exact)	$(2, 1)$	support / oppose
ε	outside option $\perp_{\text{lin}}$, payoff $\log \varepsilon$ (exact)	0.5	damps low-inflow neurons
τ	entropy regularization of routing (deform.)	1	sharp / diffuse
λ, σ^2	values $\to$ lower-confidence bounds (deform.)	λ=0	noise / fragility
λ_{sm}	attention logits shifted by λ_{sm} times a per-key ADF std (deform.)	0	stable attention
$\tilde{\Phi}$	noise on player observations (deform.)	off	gate smoothing
N	average over input transformations (deform.)	off	grid artifacts

distribution Π over graph trajectories—and the attribution map is only its pixel projection. The **Hellinger distance** (Hellinger, 1909) $\mathrm{H}(\Pi_A, \Pi_B) = (1 - \mathrm{BC})^{1/2}$, with Bhattacharyya coefficient (Bhattacharyya, 1943) $\mathrm{BC} = \sum_\zeta \sqrt{\Pi_A(\zeta)\, \Pi_B(\zeta)}$, measures the routing logic directly rather than its visual projection. This section probes the representation claim: parameter randomization should move Π even when its pixel projection remains stable. We remark that $\mathrm{H} = 0$ iff $\Pi_A = \Pi_B$, and the metric takes its maximal value $\mathrm{H} = 1$ iff their supports are disjoint. Unlike KL-type divergences, H stays finite when randomization opens or closes routes, and since both laws are layered Markov chains on the same state graph, BC factorizes along layers: one backward pass computes H exactly, in time linear in the number of edges, and H^2 decomposes into per-input-state contributions (Appendix G).

6.1 Trajectories change, maps need not

Under cascading randomization (layers cumulatively re-initialized output→input), both SG and RG diverge from the first randomized layer and plateau near the fully-random-model reference of Table 7 (Figure 2). The reference itself sits below the ceiling $\mathrm{H} = 1$ because two fully randomized models still share some short trajectories: both may stop early at structurally dead neurons and route mass to the common cemetery state. Such short paths have a higher chance of coinciding across models than full trajectories reaching the input layer. Conditioning on survival removes this shared cemetery mass and gives the strict ceiling $\mathrm{H}_{\text{surv}} \approx 1$. Trajectory- and map-level readouts are per-image orthogonal (Spearman $|\rho| \leq 0.19$ between H and pixel-level Gradient or $\alpha\beta$-LRP similarity): images with similar attribution maps need not have similar calculations. Per-game values, the conditioning, the floor mechanism and the per-image analysis are in Appendix H.

What the map hides. Cascading randomization on the maps reproduces Adebayo et al. (2018); Sixt et al. (2020): some methods retain map structure under full randomization. On ViT-B/16, the RG anchor itself retains Spearman $|\rho| = 0.454$, the highest of all methods, and HOG $r = 0.469$; on VGG-16, LRP-ε retains 0.341 and 0.457 (Table 6). The localization-selected RG modes are slightly more model sensitive than the anchor on all three models. Read against Figure 2, the two readouts disagree: H sits at the fully random reference while the projection retains structure. The map-level failure stands; H locates the loss between calculation and projection.

7 CASE STUDY

Setup. We evaluate RG and SG against standard backward baselines (Appendix I.2) on VGG-16 (Simonyan and Zisserman, 2015), ResNet-50 (He et al., 2016), and ViT-B/16 (Dosovitskiy et al., 2021) with torchvision ImageNet-1K checkpoints (Paszke et al., 2019; Deng et al., 2009; Russakovsky et al., 2015), on a 50-class ImageNet-S subset (Gao et al., 2023) at 224×224 (50-image validation split, 565-image test split), and on Pascal VOC 2012 (Everingham et al., 2010). ImageNet-S provides the pixel masks required by the localization metrics; ImageNet boxes over-credit mass anywhere inside the box. All metrics are computed through Quantus (Hedström et al., 2023) (definitions in Appendix I.3): localization (Attribution Localization (AL), Pointing Game (PG), Top-k Intersection (TK); ↑), faithfulness (Pixel Flipping area under the curve over the first 5% and 20% of flips (PF_5, PF_{20}), Selectivity (Sel); ↓), robustness (Maximum Sensitivity (MaxS), Average Sensitivity (AvgS); ↓). Baselines run in their reference implementations (lxt, zennit, captum, pytorch-grad-cam, and the authors' code for TIBAV and GAE) at the authors' settings, plus declared grids (Achtibat et al., 2024; Anders et al., 2021; Kokhlikyan et al., 2020; Gildenblat and contributors, 2021; Chefer et al., 2021b,a). RG analytical modes are reported alongside each configuration: temperature τ, risk aversion λ (softmax-shift variant λ_{sm} on ViT attention logits), LRP stabilizer ε, equivariant-Reynolds

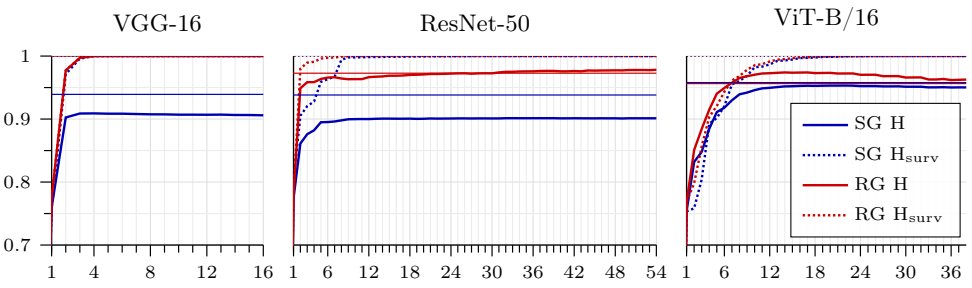

Figure 2: Hellinger trajectory distance under cascading parameter randomization (Adebayo et al., 2018). Hellinger distance H (vertical axis, from 0.7) against the number of cascadingly re-initialized layers, output to input (horizontal axis; H = 0 before the first layer is randomized). Solid: plain H; dotted: H_{surv}, conditioned on survival to an ordinary input-layer state. Thin horizontal reference lines mark the fully-random-model values from Table 7 (solid: plain H; dotted: H_{surv}), in the color of the corresponding game.

sample count N.

Protocol. Methods are evaluated under a two-stage protocol that treats baselines and our games alike. Stage 1 runs a grid search on the validation split (one image per class) for every method with tunable parameters, including DAVE (Appendix I.4, Table 9; 2,021 configurations in total). Stage 2 picks, for each method, the *loc* configuration with the best localization rank-sum over $\{\mathrm{AL}, \mathrm{PG}, \mathrm{TK}\}$ and, independently, the *faith* configuration with the best faithfulness rank-sum over $\{\mathrm{PF}_5, \mathrm{PF}_{20}, \mathrm{Sel}\}$, and evaluates both once on the test split, with 95% percentile-bootstrap confidence intervals (Efron and Tibshirani, 1993) and Holm-corrected (Holm, 1979) paired tests against the best row of each metric (Appendix I.6); for RG / RG+Eq we additionally show the $\alpha\beta$-LRP anchor $(\alpha, \beta, \varepsilon, \tau)=(2, 1, 0.5, 1)$ as a theorem-driven reference (Theorem 2). Localization is a natural target for modes designed to control routing focus, and object annotations provide external ground truth: Kohlbrenner et al. (2020) compare LRP rule composites by the share of relevance inside ground-truth bounding boxes, and we use pixel masks. The main table reports the loc row of every method; since mask agreement is not faithfulness, the faith rows of all methods are reported in Appendix J.

ViT-B/16 (Table 2). Under the localization-selected protocol, the RG family beats every baseline on attribution localization, including the prior transformer-specific backward methods AttnLRP (Achtibat et al., 2024), DAVE (Wróbel et al., 2026), Transformer Interpretability Beyond Attention Visualization (TIBAV) (Chefer et al., 2021b) and Generic Attention Explainability (GAE) (Chefer et al., 2021a): the selected mode reaches AL 0.711 against 0.644 for the best baseline (LRP-γ), and RG+Eq 0.698, both significantly ahead of every baseline row after Holm correction. RG+Eq also has the highest TK (0.836, not significantly above KernelSHAP at 0.823). The pointing game is led by KernelSHAP (0.961), whose maps are constant on superpixels: a hit needs only one pixel of the maximal region inside the mask. Scoring a single pixel drawn from that region instead, KernelSHAP falls below RG+Eq, which is then not significantly below the best baseline (LRP-γ; Appendix J.6). Faithfulness favors the LRP rules: AttnLRP is best on PF_5 and PF_{20} and LRP-γ on Sel (PF_{20} 6.43 versus 7.48 for RG), and the faithfulness-selected RG row narrows but does not close the gap (PF_{20} 6.78). The two anchors are the most robust rows (MaxS 0.166 and 0.170 versus 0.208 for AR), whereas the localization-selected mode reaches MaxS 0.729: sharper routing trades robustness and faithfulness for localization. ADF adds modest overhead (RG: 0.17 s vs. 0.10 s base), still below sampling baselines like SmoothGrad or IntGrad; Reynolds averaging is the only expensive variant (8.66 s), exceeding even the perturbation method KernelSHAP (6.71 s).

Higher τ spreads attribution into context. The localization-optimal modes are the sharpest ones: they place mass on the few patches that align with the segmentation mask. As τ increases, the same equilibrium spreads routing over more predecessors; the object remains in focus, but the heatmap also highlights environmental features the network used. These broader maps can be visually more informative, but on ViT-B/16 attribution localization worsens monotonically with τ around the localization-selected setting (AL 0.723 at τ=0.3 down to 0.461 at τ=1), while the faithfulness side is non-monotonic. High τ is the clearest example of the gap between metric-optimal and perception-optimal explanations within the same game.

Cross-dataset transfer (Pascal VOC). A natural worry with validation-tuned modes is that the

Table 2: ViT-B/16 on the ImageNet-S test split (n=565; MaxS/AvgS on a fixed random subset of 200 test images). **Bold**, dark shading: best in column; $^{\approx}$, light shading: not significantly different from the best (paired bootstrap / McNemar, Holm-corrected, $p \geq 0.05$); small numbers: 95% percentile-bootstrap confidence intervals over test images (Appendix I.6). One row per method, selected on the validation split by the localization rank-sum (faithfulness-selected rows and further baselines: Appendix J); the RG anchor is the exact $\alpha\beta$-LRP representation (Theorem 2). t/img: seconds per image on one GPU. Config: SmoothGrad averages N samples at noise level σ relative to the input range; LRP-γ has stabilizer ε and positive-weight boost γ; AttnLRP has layer-type parameters $\gamma_{\text{conv}}, \gamma_{\text{lin}}, \varepsilon_{\text{attn}}$; KernelSHAP draws N samples for M superpixels; RG modes as in Table 1.

Method	Config	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓	MaxS↓	AvgS↓	t/img
Gradient	–	0.411 [0.389,0.432]	0.361 [0.322,0.402]	0.386 [0.359,0.413]	8.45 [8.33,8.56]	8.08 [7.94,8.22]	6.033 [5.905,6.154]	1.290 [1.233,1.351]	0.922 [0.893,0.953]	**0.02**
SmGrad	N=50, σ/range=0.2	0.517 [0.498,0.536]	0.706 [0.667,0.743]	0.675 [0.654,0.694]	8.10 [7.96,8.24]	7.02 [6.83,7.20]	5.029 [4.894,5.163]	0.597 [0.569,0.630]	0.468 [0.451,0.485]	1.12
LRP-γ	ε=0.1, γ=2	0.644 [0.628,0.660]	0.873 [0.844,0.899]	0.816 [0.800,0.832]	7.93 [7.78,8.08]	6.64 [6.44,6.83]	**4.451** [4.324,4.572]	0.300 [0.287,0.314]	0.218 [0.210,0.226]	0.08
AttnLRP	γ_{conv}=0.25, γ_{lin}=0.05, $\varepsilon_{\text{attn}}$=0.25	0.644 [0.628,0.660]	0.855 [0.825,0.883]	0.809 [0.794,0.825]	**7.77** [7.61,7.93]	**6.43** [6.23,6.62]	4.582 [4.452,4.706]	0.456 [0.430,0.484]	0.342 [0.324,0.364]	0.10
TIBAV	–	0.615 [0.597,0.634]	0.878 [0.851,0.904]	0.782 [0.759,0.805]	8.13 [8.01,8.24]	7.60 [7.46,7.74]	4.906 [4.774,5.033]	1.062 [1.001,1.126]	0.512 [0.486,0.540]	0.15
GAE	–	0.533 [0.514,0.551]	0.743 [0.708,0.779]	0.649 [0.623,0.677]	8.13 [8.01,8.25]	7.62 [7.47,7.76]	5.098 [4.961,5.231]	0.525 [0.494,0.558]	0.368 [0.346,0.391]	0.03
KernelSHAP	N=2M+2048, 80 superpixels	0.565 [0.548,0.583]	**0.961** [0.945,0.977]	$0.823^{\approx}$ [0.799,0.848]	8.29 [8.18,8.40]	7.50 [7.35,7.65]	5.140 [5.012,5.261]	0.897 [0.868,0.928]	0.650 [0.632,0.667]	6.71
RG anchor	τ=1	0.509 [0.490,0.528]	0.724 [0.685,0.759]	0.699 [0.678,0.720]	8.19 [8.05,8.31]	6.97 [6.78,7.15]	4.854 [4.732,4.972]	**0.166** [0.161,0.172]	0.120 [0.117,0.124]	0.10
RG	τ=0.4, λ_{sm}=−10, σ^2=5, ε=0.1	**0.711** [0.692,0.730]	0.825 [0.793,0.855]	0.788 [0.768,0.808]	8.27 [8.14,8.39]	7.48 [7.31,7.64]	5.109 [4.987,5.226]	0.729 [0.685,0.777]	0.494 [0.471,0.518]	0.17
RG+Eq anchor	τ=1, N=50	0.520 [0.501,0.539]	0.761 [0.726,0.796]	0.752 [0.729,0.775]	8.41 [8.29,8.52]	7.72 [7.56,7.87]	4.864 [4.742,4.981]	$0.170^{\approx}$ [0.165,0.175]	**0.115** [0.112,0.118]	4.93
RG+Eq	τ=0.5, λ_{sm}=−10, σ^2=2, N=50	0.698 [0.680,0.717]	0.855 [0.827,0.883]	**0.836** [0.815,0.857]	8.39 [8.28,8.50]	7.80 [7.65,7.94]	5.010 [4.889,5.127]	0.441 [0.421,0.463]	0.301 [0.290,0.313]	8.66

chosen τ, λ_{sm} and σ^2 are overfitted to the ImageNet-S 50-class subset. We test transfer on a per-image disjoint split of Pascal VOC 2012 trainval (2,743 test (image, foreground-class) samples with pixel-accurate segmentation masks): RG and RG+Eq use the *same* mode parameters as Table 2 without any VOC-side re-tuning, while every baseline gets a fresh per-method validation grid on the new pool. This tests whether the selected modes capture model-specific rather than dataset-specific structure: the localization lead carries over (AL 0.459 for RG and 0.456 for RG+Eq versus 0.413 for TIBAV, both significant against every baseline row), while TK and PG are led by TIBAV and the perturbation methods, and faithfulness by the LRP rules (Appendix J.4).

VGG-16 and ResNet-50 (Appendix J). On the CNNs, where LRP-type rules are already strong, the localization-selected RG rows also lead on AL, clearly on VGG-16 (0.685 versus 0.613 for LRP-ε) and narrowly but significantly on ResNet-50 (0.719 versus 0.711 for LRP-γ). Pixel flipping favors the LRP rules (LRP-ε on VGG-16, LRP-γ on ResNet-50), while the Class Activation Mapping (CAM) methods and the perturbation methods lead TK and PG. On VGG-16 the risk-averse SG mode is the most robust method (MaxS 0.278 versus 0.314 for GCAM++).

8 DISCUSSION & CONCLUSION

The Stopping and Routing Games unify gradient and $\alpha\beta$-LRP attribution as equilibrium quantities on per-neuron states. Rule variants—temperature τ, risk aversion λ, smooth gate $\tilde{\Phi}$—arise as principled deformations of the game definition around these exact anchors, and attention modules drop in via an oracle routing policy. This also places robustness-motivated Softplus replacement (Dombrowski et al., 2022) inside the same picture: entropy regularization of the activation-state stop/continue choice recovers Softplus-network gradients (Appendix F). The games are representation theorems, not mechanistic inevitabilities: the architecture does not by itself determine one game, policy class, or discount, and other games may give other insights.

The modes improve the attribution metrics considered here, but the evidence is specific: exact results cover ReLU networks and attention conditioned on the realized attention pattern, LayerNorm statistics, and GELU gates. Human-mask agreement is only one evaluation axis, and sharper routing can remove network-relevant evidence outside the mask. Within this scope, one adapted ViT-B/16 $\alpha\beta$-LRP configuration outperforms all baselines, including prior transformer-specific backward methods, in attribution localization, while the LRP rules remain more faithful. Whether this transfers to other modalities, and whether metric gains translate to downstream tasks, remains future work.

The trajectory distribution Π is the object of study;

the attribution map is only its pixel projection. The Hellinger distance $\mathrm{H}(\Pi_A, \Pi_B)$ is computable in one backward pass and complements the parameter-insensitivity sanity check (Adebayo et al., 2018; Sixt et al., 2020) with a calculation-level readout: H detects randomization collapse even when projected maps remain similar. Its empirical utility will require calculation-level benchmarks, but first results are promising: H separates intra- from inter-class image pairs for at least one game on every architecture and increases monotonically with Gaussian input noise (Appendix G.4).

AI Use Statement

In this work, we used generative AI tools for implementing standard baseline methods and evaluation code, for producing plots and tables of results, for drafting parts of the text, and for editing (grammar, spelling, word choice, and reformatting). We have not used generative AI tools to generate research ideas, to formulate the games, or to produce the theorems and their proofs. All AI-assisted code was reviewed and tested by the authors, including unit tests that check the game-theoretic occupation measures against reference gradient and $\alpha\beta$-LRP implementations; all AI-assisted text was checked by the authors for correctness, and every reference was verified manually. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

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Playing the Network Backward: A Game-Theoretic Attribution Framework Supplementary Materials

A NOTATION AND CONVENTIONS

Throughout, $p, q, r \in \{+, -\}$ denote player or turn labels of the two players P^+ and P^-, and p' denotes the opponent of p ($+' = -$, $-' = +$); Table 3 lists all symbols, grouped as network, games, Routing Game and attention, modes and smooth activations, and Hellinger distance; each has a single meaning. We write $\delta_{p,q} = 1$ if $p = q$ and 0 otherwise, the sign operator $\xi_{p,q} = 2\delta_{p,q} - 1$ ($+1$ if $p = q$, -1 if $p \neq q$), and the XOR operator $p \oplus q = +$ if $p = q$ and $-$ if $p \neq q$; it is commutative and associative, with $p \oplus p = +$, $p \oplus + = p$, $p \oplus - = p'$, and $p' \oplus q = (p \oplus q)'$. **Boundary conventions.** We fix $\mathrm{ReLU}'(0) \coloneqq 0$; equivalently, the indicator $\mathbf{1}[z > 0]$ used throughout returns 0 at $z = 0$. For a finite tuple $(x_1, \dots, x_m)$, $\arg\max_k x_k \coloneqq \min\{k : x_k = \max_l x_l\}$, the smallest index attaining the maximum. We reserve the word *trajectory* for sequences of game states $\zeta = (\zeta_0, \zeta_1, \dots)$ and use *path* for sequences of neuron indices.

Table 3: Notation. Each symbol has one meaning throughout the paper and appendix, except the local attention symbols $Q_t, K_k, e_{tk}, s_{tk}, m, s, c, T_n$ and the token index t, defined in Appendix D.

Symbol	Meaning	Symbol	Meaning
l, L	layers; input 0 ($a^{(0)} = x$), output L	$W_{ij}^{(l)}$, $b_j^{(l)}$	weight from i (layer $l-1$) to j (layer l); bias
$z_j^{(l)}$, $a_j^{(l)}$	$\sum_i a_i^{(l-1)} W_{ij}^{(l)} + b_j^{(l)}$; $\mathrm{ReLU}(z_j^{(l)})$	$f(x) = a_u^{(L)}$	scalar output (target logit u)
$[t]^{\pm}$, $x_k^{\pm}$	$\max\{\pm t, 0\}$, so $t = [t]^+ - [t]^-$; $x_k^{\pm} = [x_k]^{\pm}$	$W^{(l,\pm)}$, $b^{(l,\pm)}$	$[W^{(l)}]^{\pm}$, $[b^{(l)}]^{\pm}$ (sign split)
$z_j^{(l,\pm)}$, $a_{\mathrm{stop},j}^{(l,\pm)}$	sign split of z; Stopping Decomposition	$\nabla^2 f$	Hessian
p, q, r; p'	players / turns in $\{+, -\}$; opponent	$\kappa \in \{+, -\}$	stream sign of a linear routing state
$s_{j,q}^{(l,\mathrm{act})}$, $s_{k,q}^{(0)}$	activation state (q in turn); input terminal	$s_{j,q,\kappa}^{(l,\mathrm{lin})}$	RG linear routing state
$\dagger$	cemetery (absorbing) state	$\perp_{\mathrm{lin}}$	outside action (log-payoff $\log \varepsilon$)
$r_p(s, a)$	payoff of player p	π, $\pi^\star$	policy; equilibrium (SPNE) policy
$\tilde{V}_p(s)$, $\tilde{Q}_p(s, a)$	SG player-specific value, action value	$V_p(s)$	game value (player p minus p')
$Q_r(s, i)$	RG log-gain of routing to predecessor i	$\Sigma_j^{(l,\kappa)}$	$\sum_i [a_i^{(l-1)} W_{ij}^{(l)}]^\kappa + [b_j^{(l)}]^\kappa$
$\gamma_j^{(l)}$	SG discount $\sum_i \lvert W_{ij}^{(l)} \rvert$	γ_{O}	Oracle discount (2 add./branch, 1 max-pool)
$\gamma(s)$	per-state discount: SG $\gamma_j^{(l)}$ or γ_{O}; RG $\in \{2\alpha, 2\beta, 1\}$	ζ	trajectory $(\zeta_0, \zeta_1, \dots)$
Π_x	equilibrium trajectory law at input x	$d_t(\zeta)$	cumulative trajectory discount
$\Gamma_x(s)$	discounted occupation measure of state s	$\mathcal{H}$, $\mathcal{H}_\theta$	Shannon entropy; binary form
τ	RG temperature (entropy weight)	α, β	$\alpha\beta$-LRP weights, $\alpha - \beta = 1$
ε	LRP stabilizer	$R_i^{(l)}$	LRP relevance
A	attention matrix (held fixed)	v, W_v	Value vectors, projection (sup. "val")
n_h	number of attention heads	N	equivariant-averaging samples
λ	activation-side risk aversion ($\lambda \leq 0$)	λ_{sm}	attention-logit risk aversion
μ, σ, σ^2	ADF mean, std.; input variance	λ_{ent}	fan-in entropy-bias mode
sp_θ, sp'_θ	Softplus $\theta \log(1+e^{z/\theta})$; its gate	Φ, ϕ	normal CDF, density
θ; $\tilde{\Phi}$	Softplus temperature; soft gate $\Phi(\mu/\sigma + \lambda)$	$\mathcal{R}_\Phi$	probit regularizer $\phi(\Phi^{-1}(\pi))$
H, $\mathrm{H}_{\mathrm{surv}}$	Hellinger distance; survival-conditioned	BC	Bhattacharyya coefficient
$T_M^{(l)}$	backward kernel of model M	$m_M^{(l)}(s)$	marginal occupation probability
$g^{(l)}(s)$	overlap mass (backward pass)	$c^{(l)}(s)$	continuation mass (forward pass)
$\mathrm{H}^2(s)$	per-terminal-state disagreement	$\psi_M^{(l)}(s)$	survival probability

B STOPPING GAME

This appendix realizes the activations of the Stopping Decomposition (10) as player-specific game values of a backward two-player Stopping Game on an extended network graph and demonstrates that gradients of the original network can be interpreted as integrals over game trajectories.

B.1 Stopping Decomposition

Throughout we consider a feedforward ReLU network of L layers with input layer 0, $a^{(0)} = x$, pre-activations $z_j^{(l)} = \sum_i a_i^{(l-1)} W_{ij}^{(l)} + b_j^{(l)}$, where $W_{ij}^{(l)}$ connects neuron i in layer $l-1$ to neuron j in layer l, activations $a_j^{(l)} = \mathrm{ReLU}(z_j^{(l)})$, and a *single scalar output* $f(x) = a_u^{(L)}$ (for multi-class networks, fix the target logit u).

Canonical sign split. We use the element-wise positive/negative decomposition of the original network's weights and biases,

$$W^{(l,+)} \coloneqq [W^{(l)}]^+, \qquad W^{(l,-)} \coloneqq [W^{(l)}]^-, \qquad b^{(l,+)} \coloneqq [b^{(l)}]^+, \qquad b^{(l,-)} \coloneqq [b^{(l)}]^-, \tag{7}$$

which maps non-negative activation pairs $(a_i^{(l-1,+)}, a_i^{(l-1,-)})$ to the pair of pre-activations at neuron j,

$$z_j^{(l,+)} \coloneqq b_j^{(l,+)} + \sum_i W_{ij}^{(l,+)} a_i^{(l-1,+)} + \sum_i W_{ij}^{(l,-)} a_i^{(l-1,-)}, \tag{8}$$

$$z_j^{(l,-)} \coloneqq b_j^{(l,-)} + \sum_i W_{ij}^{(l,+)} a_i^{(l-1,-)} + \sum_i W_{ij}^{(l,-)} a_i^{(l-1,+)}. \tag{9}$$

When $a_i^{(l-1,+)} - a_i^{(l-1,-)} = a_i^{(l-1)}$ for all i, this yields the identity $z_j^{(l,+)} - z_j^{(l,-)} = b_j^{(l)} + \sum_i (W_{ij}^{(l,+)} - W_{ij}^{(l,-)})(a_i^{(l-1,+)} - a_i^{(l-1,-)}) = z_j^{(l)}$, so the layer-$l$ pre-activation pair already encodes the original network's pre-activation through its difference.

The **Stopping decomposition function** $f_{\mathrm{stop}} \colon \mathbb{R}^2_{\geq 0} \to \mathbb{R}^2_{\geq 0}$ is

$$f_{\mathrm{stop}}(z^+, z^-) \;\coloneqq\; \big(\mathbf{1}\big[z^+ - z^- > 0\big]\, z^+,\;\; \mathbf{1}\big[z^+ - z^- > 0\big]\, z^-\big). \tag{10}$$

Applied at every neuron and propagated by (8)–(9), it defines the **Stopping Decomposition** activations $\big(a_{\mathrm{stop},j}^{(l,+)}, a_{\mathrm{stop},j}^{(l,-)}\big) = f_{\mathrm{stop}}\big(z_j^{(l,+)}, z_j^{(l,-)}\big)$, with the input-layer initialization

$$a_{\mathrm{stop},k}^{(0,\pm)} = x_k^{\pm}. \tag{11}$$

Theorem 4 (Layerwise Recovery of the Original Activations)**.** *For every layer l and neuron j,*

$$a_{\mathrm{stop},j}^{(l,+)} - a_{\mathrm{stop},j}^{(l,-)} = a_j^{(l)} = \mathit{ReLU}\big(z_j^{(l)}\big). \tag{12}$$

Proof. Induction on l. At $l = 0$, $x_k^+ - x_k^- = x_k = a_k^{(0)}$ by (11). At layer $l \geq 1$ the identity above gives $z_j^{(l,+)} - z_j^{(l,-)} = z_j^{(l)}$, hence $a_{\mathrm{stop},j}^{(l,+)} - a_{\mathrm{stop},j}^{(l,-)} = \mathbf{1}\{z_j^{(l)} > 0\}\big(z_j^{(l,+)} - z_j^{(l,-)}\big) = \mathrm{ReLU}(z_j^{(l)}) = a_j^{(l)}$. □

ReLU networks compute continuous piecewise linear functions (Arora et al., 2018), which admit a DC decomposition (Zimmermann and Loho, 2026); it uses the same sign split (8)–(9) with the pairwise activation $(\max\{z^+, z^-\}, z^-)$ in place of f_{stop} and yields components that are convex in x. The Stopping Decomposition instead gates both streams by the same indicator, so its components are not convex, but on each linear region they are non-negative affine, hence monotone, in (x^+, x^-).

Parity path decomposition of the split streams. We can decompose the Stopping Decomposition activations into small contributions coming from individual paths through the network: each non-negative decomposed activation at neuron j is the sum, over all backward paths from j to the input, of the absolute weight product along the path times either the positive part $x_{P_0}^+$ or the negative part $x_{P_0}^-$ of the input pixel where the path terminates, with the choice between $x_{P_0}^+$ and $x_{P_0}^-$ determined by the parity of the number of negative weights traversed.

Lemma 1 (Path view on the Stopping Decomposition)**.** *Fix a feedforward ReLU network with weights $W^{(l)}$ (biases ignored; see remark at end of proof). For neuron j in layer $l \geq 1$, call a* backward path *from j to the input layer a list of neuron indices*

$$P = (P_l, P_{l-1}, \ldots, P_0), \qquad P_l = j, \tag{13}$$

with P_t indexing a neuron in layer t. The edges of P use the weights $W^{(t)}_{P_{t-1},P_t}$ for $t = 1, \ldots, l$. Define

$$w(P) \coloneqq \prod_{t=1}^{l} W^{(t)}_{P_{t-1},P_t}, \qquad |w|(P) \coloneqq \prod_{t=1}^{l} \big|W^{(t)}_{P_{t-1},P_t}\big|, \tag{14}$$

the parity sign

$$\operatorname{sgn}(P) \in \{+,-\}, \qquad \operatorname{sgn}(P) = + \iff \#\big\{t : W^{(t)}_{P_{t-1},P_t} < 0\big\} \text{ is even}, \tag{15}$$

and the gate factor *(1 iff every gate on P at intermediate layers $1, \ldots, l-1$ is open)*

$$G_{<l}(P) \coloneqq \prod_{t=1}^{l-1} \mathbf{1}\big[z^{(t)}_{P_t} > 0\big]. \tag{16}$$

Then the decomposed pre-activations (8)–(9) *admit the path-sum representation*

$$z^{(l,+)}_j = \sum_{P:\, P_l = j} G_{<l}(P)\, |w|(P)\, x^{\operatorname{sgn}(P)}_{P_0}, \tag{17}$$

$$z^{(l,-)}_j = \sum_{P:\, P_l = j} G_{<l}(P)\, |w|(P)\, x^{\operatorname{sgn}(P)'}_{P_0}. \tag{18}$$

Proof. We prove (17)–(18) by induction on l.

Base case ($l = 1$): By the input convention (11), $a^{(0,+)}_{\text{stop},k} = x^+_k$ and $a^{(0,-)}_{\text{stop},k} = x^-_k$. Using (8) with biases omitted,

$$z^{(1,+)}_j = \sum_k \big(W^{(1,+)}_{kj} x^+_k + W^{(1,-)}_{kj} x^-_k\big). \tag{19}$$

Backward paths with $P_1 = j$ and $P_0 = k$ are in bijection with predecessors k; each has $|w|(P) = |W^{(1)}_{kj}|$, $\operatorname{sgn}(P) = +$ iff $W^{(1)}_{kj} > 0$, and $G_{<1}(P) = 1$ (empty product). Splitting by the sign of $W^{(1)}_{kj}$:

- $W^{(1)}_{kj} > 0$: $W^{(1,+)}_{kj} = |W^{(1)}_{kj}|$ contributes $|W^{(1)}_{kj}| x^+_k$ with $\operatorname{sgn}(P) = +$, matching $|w|(P)\, x^{\operatorname{sgn}(P)}_{P_0}$.
- $W^{(1)}_{kj} < 0$: $W^{(1,-)}_{kj} = |W^{(1)}_{kj}|$ contributes $|W^{(1)}_{kj}| x^-_k$ with $\operatorname{sgn}(P) = -$, again matching.

The argument for $z^{(1,-)}_j$ is identical with the parity swapped, establishing (18).

Inductive step ($l \geq 2$): The decomposition recursion (8) takes *Stopping Decomposition* activations from layer $l-1$ as inputs,

$$z^{(l,+)}_j = \sum_i \big(W^{(l,+)}_{ij}\, a^{(l-1,+)}_{\text{stop},i} + W^{(l,-)}_{ij}\, a^{(l-1,-)}_{\text{stop},i}\big), \tag{20}$$

and $a^{(l-1,\pm)}_{\text{stop},i} = \mathbf{1}[z^{(l-1)}_i > 0]\, z^{(l-1,\pm)}_i$. Hence, at each predecessor i, the gate indicator $\mathbf{1}[z^{(l-1)}_i > 0]$ multiplies the inductive representation of $z^{(l-1,\pm)}_i$, upgrading $G_{<l-1}(P')$ to $G_{\leq l-1}(P')$ for every sub-path P' ending at i.

Every backward path P with $P_l = j$ and $P_{l-1} = i$ is uniquely obtained by prepending j to a sub-path $P' = (P'_{l-1}, \ldots, P'_0)$ with $P'_{l-1} = i$; the prepended edge uses $W^{(l)}_{ij}$, contributing $|W^{(l)}_{ij}|$ to $|w|(P)$ and either preserving or flipping parity depending on $\operatorname{sgn}(W^{(l)}_{ij})$. Crucially, the new gate factor $G_{<l}(P)$ for P equals $G_{\leq l-1}(P')$: up to layer $l-1$ the gates are those along P', and there is no gate at layer l in the $G_{<l}$ product.

Case $W_{ij}^{(l)} > 0$. Then $W_{ij}^{(l,+)} = |W_{ij}^{(l)}|$ and $W_{ij}^{(l,-)} = 0$. Only $W_{ij}^{(l,+)}\, a_{\mathrm{stop},i}^{(l-1,+)}$ contributes, and the new edge preserves parity, $\mathrm{sgn}(P) = \mathrm{sgn}(P')$. Using the induction hypothesis under the updated gate factor,

$$\begin{aligned} W_{ij}^{(l,+)}\, a_{\mathrm{stop},i}^{(l-1,+)} &= |W_{ij}^{(l)}| \sum_{P' \colon P'_{l-1}=i} G_{\le l-1}(P')\, |w|(P')\, x_{P'_0}^{\mathrm{sgn}(P')} \\ &= \sum_{P \colon P_l=j,\, P_{l-1}=i} G_{<l}(P)\, |w|(P)\, x_{P_0}^{\mathrm{sgn}(P)}. \end{aligned} \tag{21}$$

Case $W_{ij}^{(l)} < 0$. Then $W_{ij}^{(l,+)} = 0$ and $W_{ij}^{(l,-)} = |W_{ij}^{(l)}|$. Only $W_{ij}^{(l,-)}\, a_{\mathrm{stop},i}^{(l-1,-)}$ contributes, and the new edge flips parity, $\mathrm{sgn}(P) = \mathrm{sgn}(P')'$, so $\mathrm{sgn}(P') = \mathrm{sgn}(P)'$, and the induction hypothesis for $z_i^{(l-1,-)}$ (with $x_{P'_0}^{\mathrm{sgn}(P')'} = x_{P_0}^{\mathrm{sgn}(P)}$) gives the same path sum as in the previous case. *Case* $W_{ij}^{(l)} = 0$. Both terms vanish and no path through i is prepended.

Summing over i yields (17). The same argument with positive/negative decomposed activations swapped gives (18).

Signed recovery. Using $x_k^+ - x_k^- = x_k$ and $\mathrm{sgn}(P) \in \{+, -\}$,

$$z_j^{(l,+)} - z_j^{(l,-)} = \sum_P G_{<l}(P)\, |w|(P) \left(x_{P_0}^{\mathrm{sgn}(P)} - x_{P_0}^{\mathrm{sgn}(P)'} \right) = \sum_P G_{<l}(P)\, w(P)\, x_{P_0}, \tag{22}$$

which is the standard active-path-sum representation of $z_j^{(l)}$ (paths through an inactive intermediate neuron contribute zero).

Biases. Adjoining a bias "stub path" at each intermediate layer l'—starting from (l, j), terminating at the bias node of neuron k' at layer l', with weight product taken over the $l - l'$ edges above and an appropriate gate factor $G_{<l}$ restricted to those edges—handles the bias terms by the same parity rule; the argument above goes through verbatim after absorbing these stub contributions into the path sums. □

B.2 Game Definition

Relation to the network games of Gaubert and Vlassopoulos (2025). The Stopping Game adapts the game of Gaubert and Vlassopoulos to the Stopping Decomposition. We keep their state-space structure—per-neuron ± states, the bias as the continuation and network input as final payoff, transitions driven by the positive and negative parts of the weights, a cemetery state, and a stop-versus-continue action. Two changes are load-bearing for attribution: (i) the original formulation tracks only one game value, which equals $\pm a_j^{(l)}$; we additionally maintain the non-negative payoff per player, and the original game value is recovered as their difference. (ii) We replace the signed terminal values $\pm x_k$ by the non-negative pair x_k^+, x_k^-, which keeps every player-specific payoff non-negative and exposes the parity decomposition of Lemma 1.

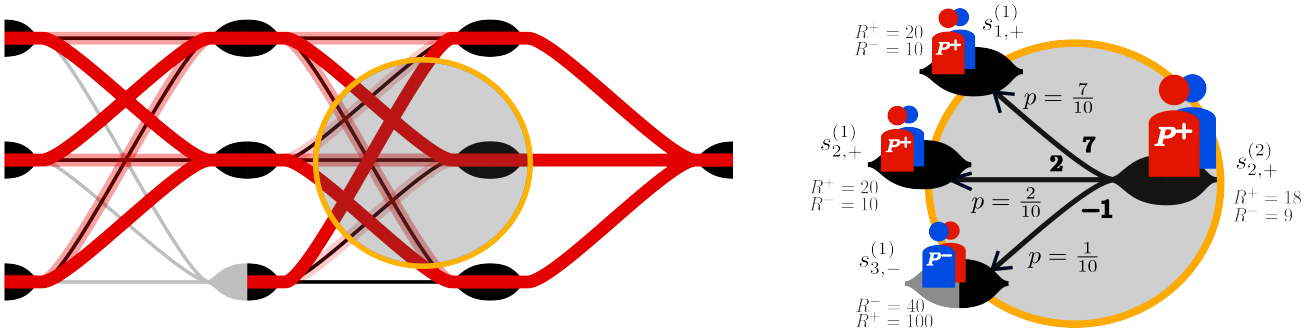

Figure 3: Stopping Game: trajectory distribution and local stopping decisions on a toy subnetwork. *Left:* trajectory distribution visualized on the network graph. *Right:* local game around $s^{(2)}_{2,+}$. Player P^+ decides whether to stop or continue. Under continuation, transitions are to $s^{(1)}_{1,+}$ with probability 7/10, to $s^{(1)}_{2,+}$ with probability 2/10 (both turn unchanged), and to $s^{(1)}_{3,-}$ with probability 1/10 (turn switches to P^-, reflecting the negative sign of the weight). The labels $R^{\pm}$ are the players' expected payoffs $\tilde{Q}_{\pm}(\cdot, \text{cont})$ (at $s^{(2)}_{2,+}$ without the discount $\gamma^{(2)}_2 = 10$). At $s^{(1)}_{3,-}$, continuing yields expected payoffs R^-=40 and R^+=100, giving continuation advantage $R^- - R^+ = -60 < 0$; therefore P^- stops (so $V_-(s^{(1)}_{3,-}) = 0$), shown by the grayed-out half-neuron. Correspondingly, in the left panel the trajectories terminate at that state (the cemetery state is omitted). Since P^- stops in $s^{(1)}_{3,-}$, traversals to this state contribute zero continuation payoff to both players. Hence, at $s^{(2,\text{act})}_{2,+}$ the positive player has larger expected payoff under continuation than his opponent and chooses to continue.

Definition 1 (Stopping Game State Space)**.** *The game is played on the following states:*

- ***Activation States:*** $s^{(l,\text{act})}_{j,q}$ *where layer* $l \in \{1, \dots, L\}$ *and player* $q \in \{+, -\}$ *is in turn at neuron* j*.*
- ***Terminal States:*** $s^{(0)}_{k,q}$ *where* k *is an input dimension and* $q \in \{+, -\}$ *is the final turn label.*
- ***Cemetery State:*** †*, an absorbing state reached upon stopping.*

We adopt the convention that, when a transition kernel or recursion at layer l *references the predecessor state* $s^{(l-1,\text{act})}_{i,r}$*, the symbol denotes the activation state at layer* $l - 1 \geq 1$ *as defined above and, in the boundary case* $l = 1$*, is interpreted as the terminal state* $s^{(0)}_{i,r}$*.*

Definition 2 (Payoffs, Transitions, and Value Recursion)**.** *Fix an input* x*.*

Payoffs. *The player-specific* payoff r_p *at activation states is paid out depending on the action chosen by the active player, and paid out at terminal and cemetery states without prerequisites:*

- *Cemetery:* $r_p(\dagger) \coloneqq 0$*.*
- *Active state:* $r_p(s^{(l,\text{act})}_{j,q}, \text{stop}) \coloneqq 0$ *(stopping pays nothing) and* $r_p(s^{(l,\text{act})}_{j,q}, \text{cont}) \coloneqq b^{(l,p\oplus q)}_j$ *(continuation pays a positive bias to the active player* q*—covered by the case* $p \oplus q = +$*—and the absolute value of a negative bias to its opponent* q'*—covered by the case* $p \oplus q = -$*).*
- *Terminal:* $r_p(s^{(0)}_{k,q}) \coloneqq x^{p\oplus q}_k$ *(at terminal states the payoff depends on state only—the game ends here. A positive input is paid to the active player* q*—covered by the case* $p \oplus q = +$*—whereas the absolute value of a negative input is paid to its opponent* q'*—covered by the case* $p \oplus q = -$*).*

Transition kernel. *Stopping at state* $s^{(l,act)}_{j,p}$ *deterministically sends the game to the cemetery; continuation transports backward by one layer with transition probabilities depending on the fraction of the respective absolute network weight compared to the sum of absolute incoming weights at neuron* j *of layer* l*. In case of a positive weight, the active player* p *stays in turn, in case of a negative weight, the turn switches to its opponent* p'*.*

$$\mathbb{P}\big(\dagger \,\big|\, s^{(l,\text{act})}_{j,q}, \text{stop}\big) \coloneqq 1, \tag{23}$$

$$\mathbb{P}\big(s_{i,r}^{(l-1,\mathrm{act})} \,\big|\, s_{j,q}^{(l,\mathrm{act})}, \mathrm{cont}\big) \coloneqq \frac{W_{ij}^{(l,q\oplus r)}}{\gamma_j^{(l)}}, \qquad \gamma_j^{(l)} \coloneqq \sum_i \big(W_{ij}^{(l,+)} + W_{ij}^{(l,-)}\big) = \sum_i \big|W_{ij}^{(l)}\big|, \tag{24}$$

where $\gamma_j^{(l)}$ is the local structural discount that restores unit row-sum (if $\gamma_j^{(l)} = 0$, continuation leads to †). For understanding the idea of the occupation measure, we remark that those $\gamma_j^{(l)}$ are going to weight the trajectories in the occupation measure.

Policy. *A pure, i.e., deterministic policy $\pi = (\pi_+, \pi_-)$ assigns each active state $s_{j,q}^{(l,\mathrm{act})}$ a binary stop/continue decision by the player in turn,*

$$\pi_q\big(s_{j,q}^{(l,\mathrm{act})}\big) \in \{0,1\}, \qquad 1 = \mathrm{cont},\ 0 = \mathrm{stop}. \tag{25}$$

Player-specific action and game values. *For each policy π, the player-specific **action value** $\tilde{Q}_p^\pi(s,a)$ is the expected discounted payoff for player p starting at s with action a and following π thereafter, defined by the standard Bellman equation*

$$\tilde{Q}_p^\pi(s,a) \coloneqq r_p(s,a) + \gamma(s) \textstyle\sum_{s'} \mathbb{P}\big(s' \,\big|\, s,a\big)\, \tilde{V}_p^\pi(s'), \tag{26}$$

where $\tilde{V}_p^\pi$ is the player-specific game value defined below. We use tildes to reserve V_p^π for the advantage further down, which already carries a subtracted $\tilde{V}_{p'}^\pi$. At active states (26) *specializes, using* (23)–(24)*, to*

$$\tilde{Q}_p^\pi\big(s_{j,q}^{(l,\mathrm{act})}, \mathrm{stop}\big) = 0 + 1 \cdot \tilde{V}_p^\pi(\dagger) = 0 \tag{27}$$

$$\tilde{Q}_p^\pi\big(s_{j,q}^{(l,\mathrm{act})}, \mathrm{cont}\big) = b_j^{(l,p\oplus q)} + \gamma_j^{(l)} \textstyle\sum_{s'} \mathbb{P}\big(s' \,\big|\, s_{j,q}^{(l,\mathrm{act})}, \mathrm{cont}\big)\, \tilde{V}_p^\pi(s') \tag{28}$$

$$= b_j^{(l,p\oplus q)} + \sum_{i,r} W_{ij}^{(l,q\oplus r)}\, \tilde{V}_p^\pi\big(s_{i,r}^{(l-1,\mathrm{act})}\big). \tag{29}$$

*The **player-specific game value** $\tilde{V}_p^\pi(s)$ is the action value averaged by the policy. At terminal and cemetery states it equals the local payoff there:*

$$\tilde{V}_p^\pi\big(s_{k,q}^{(0)}\big) \coloneqq x_k^{p\oplus q}, \qquad \tilde{V}_p^\pi(\dagger) \coloneqq 0. \tag{30}$$

At active states,

$$\tilde{V}_p^\pi\big(s_{j,q}^{(l,\mathrm{act})}\big) \coloneqq \pi_q\big(s_{j,q}^{(l,\mathrm{act})}\big)\, \tilde{Q}_p^\pi\big(s_{j,q}^{(l,\mathrm{act})}, \mathrm{cont}\big) + \big(1 - \pi_q\big(s_{j,q}^{(l,\mathrm{act})}\big)\big)\, \tilde{Q}_p^\pi\big(s_{j,q}^{(l,\mathrm{act})}, \mathrm{stop}\big). \tag{31}$$

Since π_q is binary, (31) *selects either $\tilde{Q}_p^\pi(\cdot, \mathrm{cont})$ or $\tilde{Q}_p^\pi(\cdot, \mathrm{stop}) = 0$.*

Game value. *The quantity the player in turn maximizes at each active state is the advantage of player p at s,*

$$V_p^\pi(s) \coloneqq \tilde{V}_p^\pi(s) - \tilde{V}_{p'}^\pi(s). \tag{32}$$

Figure 3 shows a worked example of the induced trajectory distribution and the local stop/continue logic on a small toy network.

Theorem 5 (Forward Equivalence for Stopping Game)**.** *The game admits a Subgame Perfect Nash Equilibrium (SPNE) where both players follow the same pure strategy:*

$$\pi_q^\star\big(s_{j,q}^{(l,\mathrm{act})}\big) = \mathbf{1}\big[z_j^{(l)} > 0\big]. \tag{33}$$

Under this equilibrium, the player-specific game values and the game values satisfy

$$\tilde{V}_p^{\pi^\star}\big(s_{j,q}^{(l,\mathrm{act})}\big) = a_{\mathrm{stop},j}^{(l,p\oplus q)}, \qquad V_p^{\pi^\star}\big(s_{j,q}^{(l,\mathrm{act})}\big) = \xi_{p,q}\, a_j^{(l)}. \tag{34}$$

Remark 1 (Uniqueness of the SPNE up to tie-breaking)**.** *It is easy to see in the following proof that the SPNE* (33) *is unique up to tie-breaking at states where continuing yields zero advantage $\tilde{Q}_q^{\pi^\star}(\cdot, \mathrm{cont}) - \tilde{Q}_{q'}^{\pi^\star}(\cdot, \mathrm{cont}) = z_j^{(l)} = 0$ (the implementation continues at exact ties, which changes nothing unless $z_j^{(l)} = 0$ exactly)—representing neurons of preactivation zero. We refer to the policy* (33) *by **stop-at-tie-breaks** equilibrium.*

Proof. We proceed by backward induction on the layer index l.

Base Case ($l = 0$): By (30), $\tilde{V}_p^{\pi^\star}(s_{k,q}^{(0)}) = x_k^{p\oplus q} = a_{\text{stop},k}^{(0,p\oplus q)}$ (input convention (11)). The game value is

$$V_p^{\pi^\star}\big(s_{k,q}^{(0)}\big) = x_k^{p\oplus q} - x_k^{p'\oplus q} = \xi_{p,q}\, x_k, \tag{35}$$

using $x_k^+ - x_k^- = x_k$ and $x_k^- - x_k^+ = -x_k$.

Inductive Step: Assume the subgames starting at states up to layer $l-1$ admit the claimed SPNE and (34) holds at layer $l-1$. Consider an active state $s_{j,q}^{(l,\text{act})}$. The active player q chooses $a \in \{\text{stop}, \text{cont}\}$ to maximize the advantage $\tilde{Q}_q^{\pi^\star}(s_{j,q}^{(l,\text{act})}, a) - \tilde{Q}_{q'}^{\pi^\star}(s_{j,q}^{(l,\text{act})}, a)$. For $a = \text{stop}$ this advantage is 0 by (27); for $a = \text{cont}$, using (28),

$$\tilde{Q}_q^{\pi^\star}\big(s_{j,q}^{(l,\text{act})}, \text{cont}\big) - \tilde{Q}_{q'}^{\pi^\star}\big(s_{j,q}^{(l,\text{act})}, \text{cont}\big) = \big(b_j^{(l,q\oplus q)} - b_j^{(l,q'\oplus q)}\big) + \sum_{i,r} W_{ij}^{(l,q\oplus r)}\, V_q^{\pi^\star}\big(s_{i,r}^{(l-1,\text{act})}\big) \tag{36}$$

$$= \big(b_j^{(l,+)} - b_j^{(l,-)}\big) + \sum_{i,r} W_{ij}^{(l,q\oplus r)}\, V_q^{\pi^\star}\big(s_{i,r}^{(l-1,\text{act})}\big). \tag{37}$$

By the inductive hypothesis $V_q^{\pi^\star}(s_{i,r}^{(l-1,\text{act})}) = \xi_{q,r}\, a_i^{(l-1)}$, with $\xi_{q,q} = +1$ and $\xi_{q,q'} = -1$,

$$\tilde{Q}_q^{\pi^\star}\big(s_{j,q}^{(l,\text{act})}, \text{cont}\big) - \tilde{Q}_{q'}^{\pi^\star}\big(s_{j,q}^{(l,\text{act})}, \text{cont}\big) = b_j^{(l)} + \sum_i \big(W_{ij}^{(l,q\oplus q)}\, \xi_{q,q} + W_{ij}^{(l,q\oplus q')}\, \xi_{q,q'}\big) a_i^{(l-1)} \tag{38}$$

$$= b_j^{(l)} + \sum_i W_{ij}^{(l)}\, a_i^{(l-1)} = z_j^{(l)}. \tag{39}$$

Thus, player q prefers *continue* in case $z_j^{(l)} > 0$ and prefers *stop* in case $z_j^{(l)} < 0$, and is indifferent at $z_j^{(l)} = 0$, where we tie-break in favor of *stop*. The rule $\pi_q^\star(s_{j,q}^{(l,\text{act})}) = \mathbf{1}[z_j^{(l)} > 0]$ together with the player policies of earlier layers thus constitutes an SPNE of the subgame starting at state $s_{j,q}^{(l,\text{act})}$.

To verify (34), substitute $\pi_q^\star$ into (31) and use the inductive hypothesis $\tilde{V}_p^{\pi^\star}(s_{i,r}^{(l-1,\text{act})}) = a_{\text{stop},i}^{(l-1,p\oplus r)}$ in (28):

$$\tilde{V}_p^{\pi^\star}\big(s_{j,q}^{(l,\text{act})}\big) = \mathbf{1}\big[z_j^{(l)} > 0\big]\, \tilde{Q}_p^{\pi^\star}\big(s_{j,q}^{(l,\text{act})}, \text{cont}\big) \tag{40}$$

$$= \mathbf{1}\big[z_j^{(l)} > 0\big] \left[b_j^{(l,p\oplus q)} + \sum_i \big(W_{ij}^{(l,+)}\, a_{\text{stop},i}^{(l-1,p\oplus q)} + W_{ij}^{(l,-)}\, a_{\text{stop},i}^{(l-1,p\oplus q')}\big) \right]. \tag{41}$$

If $p = q$, the bracket is $b_j^{(l,+)} + \sum_i \big(W_{ij}^{(l,+)}\, a_{\text{stop},i}^{(l-1,+)} + W_{ij}^{(l,-)}\, a_{\text{stop},i}^{(l-1,-)}\big) = z_j^{(l,+)}$. If $p = q'$, the bracket is $b_j^{(l,-)} + \sum_i \big(W_{ij}^{(l,+)}\, a_{\text{stop},i}^{(l-1,-)} + W_{ij}^{(l,-)}\, a_{\text{stop},i}^{(l-1,+)}\big) = z_j^{(l,-)}$. In both cases, $\tilde{V}_p^{\pi^\star}(s_{j,q}^{(l,\text{act})}) = \mathbf{1}[z_j^{(l)} > 0]\, z_j^{(l,p\oplus q)} = a_{\text{stop},j}^{(l,p\oplus q)}$. Finally,

$$V_p^{\pi^\star}\big(s_{j,q}^{(l,\text{act})}\big) = a_{\text{stop},j}^{(l,p\oplus q)} - a_{\text{stop},j}^{(l,(p\oplus q)')} = \xi_{p,q}\, a_j^{(l)}. \quad \square \tag{42}$$

□

B.3 Skip Connections and Max Pooling

Architectures such as ResNets and CNNs with pooling contain structural operators beyond single-parent affine maps. To handle these natively in the Stopping Game, we add an **Addition Oracle** for skip connections and a deterministic **Max-Pooling Decision State** for pooling, mirroring the analogous constructions in the Routing Game (Appendix C.1.1).

Definition 3 (Addition Oracle)**.** *For a residual node $z = x + y$, the* addition state $s_{z,q}^{\text{add}}$ *has active player $q \in \{+,-\}$ and operand states $s_{x,q}, s_{y,q}$ at the two summands. The transition kernel is the uniform Oracle*

$$\mathbb{P}\big(s_{x,q} \,\big|\, s_{z,q}^{\text{add}}\big) \coloneqq \tfrac{1}{2}, \qquad \mathbb{P}\big(s_{y,q} \,\big|\, s_{z,q}^{\text{add}}\big) \coloneqq \tfrac{1}{2}, \tag{43}$$

with player q retained at the chosen operand. The immediate payoff is $r_p(s_{z,q}^{\text{add}}) \coloneqq 0$ for both $p \in \{+,-\}$, and the structural discount is $\gamma_{\text{O}} \coloneqq 2$.

Proposition 1 (Forward Equivalence for Addition)**.** *The player-specific state value at the addition state equals the sum of the operand state values,*

$$\tilde{V}_p^{\pi}\big(s_{z,q}^{\mathrm{add}}\big) \;=\; \tilde{V}_p^{\pi}(s_{x,q}) + \tilde{V}_p^{\pi}(s_{y,q}). \tag{44}$$

Consequently, if the operand states recover the Stopping-Decomposition activations—$\tilde{V}_p^{\pi^\star}(s_{x,q}) = x^{(p\oplus q)}$ and $\tilde{V}_p^{\pi^\star}(s_{y,q}) = y^{(p\oplus q)}$—then $\tilde{V}_p^{\pi^\star}(s_{z,q}^{\mathrm{add}}) = z^{(p\oplus q)}$, preserving the forward equivalence of Theorem 5 across the residual sum.

Proof. Apply the Bellman recursion (26)–(31) at $s_{z,q}^{\mathrm{add}}$ with $r_p = 0$, the uniform Oracle kernel, and $\gamma_{\mathrm{O}} = 2$ from Definition 3:

$$\tilde{V}_p^{\pi}\big(s_{z,q}^{\mathrm{add}}\big) \;=\; 0 + \gamma_{\mathrm{O}} \cdot \Big(\tfrac{1}{2}\,\tilde{V}_p^{\pi}(s_{x,q}) + \tfrac{1}{2}\,\tilde{V}_p^{\pi}(s_{y,q})\Big) \;=\; \tilde{V}_p^{\pi}(s_{x,q}) + \tilde{V}_p^{\pi}(s_{y,q}), \tag{45}$$

which is the first identity. Substituting the operand assumption and using $z^{(p\oplus q)} = x^{(p\oplus q)} + y^{(p\oplus q)}$ from the additive convention (8)–(9) of Appendix B.1 yields the second. □

On the uniform $1/2$ split. The $(\frac{1}{2}, \frac{1}{2})$ Oracle with $\gamma_{\mathrm{O}} = 2$ is not forced by the architecture; it is a convention, the simplest choice inside a one-parameter family of conservation-preserving Oracles $(\rho, 1-\rho)$ with matching discounts $(1/\rho, 1/(1-\rho))$ attached to the branch taken (with the single state discount γ_{O} of (26) and d_t, $\rho\gamma_{\mathrm{O}} = (1-\rho)\gamma_{\mathrm{O}} = 1$ forces $\rho = \frac{1}{2}$). The uniform split is the maximum-entropy choice—the one that imposes *no prior* on which summand carries more responsibility. In the Stopping Game, Proposition 1 and Theorems 5 and 6 hold for any such split with branch discounts, since the proofs only use that routing probability times Oracle discount equals 1 on each branch; only the trajectory law Π_x (and with it the Hellinger distance of Appendix G) depends on the split. This invariance does not extend to the Routing Game (Appendix C.1.1), where the split changes the relevance assigned to each summand.

Max pooling is simpler, because the forward operator already selects a single predecessor.

Definition 4 (Max-Pooling Decision State)**.** *For a pooled scalar output $z = \max\{x_1, \dots, x_m\}$, the* pooling state *$s_{z,q}^{\max}$ has active player $q \in \{+,-\}$ and admissible action set $\mathcal{A}(s_{z,q}^{\max}) = \{1, \dots, m\}$ indexing the operand states $s_{x_1,q}, \dots, s_{x_m,q}$. Under action k, the transition is deterministic,*

$$\mathbb{P}\big(s_{x_k,q} \,\big|\, s_{z,q}^{\max}, k\big) \coloneqq 1, \tag{46}$$

with player q retained at the chosen operand. The immediate payoff is $r_p(s_{z,q}^{\max}, k) \coloneqq 0$ for both $p \in \{+,-\}$, and the structural discount is $\gamma_{\mathrm{O}} \coloneqq 1$.

Tie-breaking at the pooling state. Whenever multiple operands x_k attain the maximum, the active player is indifferent over the corresponding actions, and any pure best response over the tied indices yields the same forward value but a different equilibrium policy and hence a different occupation measure on the operand states. The *stop-at-tie-breaks equilibrium*, fixed by the boundary conventions of Appendix A, resolves this by the convention $k^\star \coloneqq \min\{k : x_k = \max_l x_l\}$, i.e., the active player picks the smallest index among the operands attaining the maximal payoff. All gradient and occupation-measure statements below are stated for this equilibrium.

Proposition 2 (Forward Equivalence for Max Pooling)**.** *Assume the operand states recover the Stopping-Decomposition activations, $\tilde{V}_p^{\pi^\star}(s_{x_k,q}) = x_k^{(p\oplus q)}$ for all k, with non-negative operand activations $x_k \geq 0$. Then at the SPNE of the subgame rooted at $s_{z,q}^{\max}$,*

$$\tilde{V}_p^{\pi^\star}\big(s_{z,q}^{\max}\big) \;=\; z^{(p\oplus q)}, \tag{47}$$

so the pooling state inherits the same Stopping-Decomposition identity as its winning operand and the forward equivalence of Theorem 5 extends across the pooling node.

Proof. The active player q chooses the operand that maximizes its game value $V_q^{\pi^\star} = \tilde{V}_q^{\pi^\star} - \tilde{V}_{q'}^{\pi^\star}$. By the operand assumption, $V_q^{\pi^\star}(s_{x_k,q}) = \xi_{q,q}\, x_k = x_k$ for every k, so the maximizer is $k^\star \in \arg\max_k x_k$ (ties resolved by the smallest index). The Bellman recursion (26)–(31) with $r_p = 0$, the deterministic transition, and $\gamma_{\mathrm{O}} = 1$ from Definition 4 gives

$$\tilde{V}_p^{\pi^\star}\big(s_{z,q}^{\max}\big) \;=\; 0 + 1 \cdot \tilde{V}_p^{\pi^\star}(s_{x_{k^\star},q}) \;=\; x_{k^\star}^{(p\oplus q)}. \tag{48}$$

Since $z = x_{k^\star}$, the right-hand side equals $z^{(p\oplus q)}$. □

B.4 Discounted Occupation Measures and Gradient Representations

To connect game trajectories with neural network gradients, we define the **discounted occupation measure**. This measure quantifies the expected influence of a neuron on the game's outcome, weighted by the discount factors accumulated along all possible game trajectories (including those created by skip connections and max-pooling nodes).

Definition 5 (Game trajectory and trajectory law)**.** *Fix an input x and a root state s. A* game trajectory *rooted at s is a finite sequence of game states*

$$\zeta \;=\; (\zeta_0, \zeta_1, \ldots, \zeta_T), \qquad \zeta_0 = s, \tag{49}$$

that follows the admissible transitions of the game (Definition 2 together with the addition/max-pooling Oracle splits of Definitions 3–4) until it terminates either at a layer-0 terminal state $\zeta_T = s^{(0)}_{k,r}$ (reaching an input dimension k) or at the cemetery state $\zeta_T = \dagger$ (reached the first time a player chooses stop*). The set of all such trajectories is the trajectory* ground set *$\mathcal{T}_s$. Composing the transition kernel (23)–(24), the Oracle splits, and a (possibly stochastic) policy π at every activation state defines a Markov kernel on game states, which together with the deterministic start at s induces a probability distribution $\Pi^{\pi}_x(\cdot \mid s)$ on $\mathcal{T}_s$. Trajectories that pass through closed gates—neurons j with $\mathbf{1}[z^{(l)}_j > 0] = 0$—are admissible elements of $\mathcal{T}_s$, but under the equilibrium policy $\pi^\star$ the player at the activation state stops there, so any trajectory continuing past a closed gate carries probability zero. We write $\Pi_x(\cdot \mid s) \coloneqq \Pi^{\pi^\star}_x(\cdot \mid s)$ for the equilibrium* trajectory law *and $\Pi_x \coloneqq \Pi_x(\cdot \mid s_0)$ for the law rooted at the output state $s_0 \coloneqq s^{(L,\mathrm{act})}_{u,+}$.*

Definition 6 (Discounted Occupation Measure)**.** *Let $\Pi_x(\cdot \mid s)$ be the equilibrium trajectory law of Definition 5. Define the cumulative discount*

$$d_t(\zeta) \coloneqq \prod_{m=0}^{t-1} \gamma(\zeta_m), \qquad d_0(\zeta) = 1, \tag{50}$$

where $\gamma(\zeta_m) = \gamma_{\mathrm{O}} = 2$ for addition Oracle states, $\gamma(\zeta_m) = \gamma_{\mathrm{O}} = 1$ for max-pooling states, and $\gamma(\zeta_m) = \gamma^{(l)}_j$ for continuation at activation states $\zeta_m = s^{(l,\mathrm{act})}_{j,q}$ (per-neuron weight-magnitude discount). The discounted occupation measure *of state v for trajectories rooted at s is*

$$\Gamma_x(v \mid s) \coloneqq \mathbb{E}_{\zeta \sim \Pi_x(\cdot \mid s)} \left[\sum_t d_t(\zeta)\, \mathbf{1}\{\zeta_t = v\} \right], \qquad \Gamma_x(v) \coloneqq \Gamma_x(v \mid s_0). \tag{51}$$

Theorem 6 (Activation–Occupation Correspondence)**.** *Fix a neuron state $s = s^{(m,\mathrm{act})}_{j,q}$ and a layer $l \le m$ (for $l = 0$, $s^{(0,\mathrm{act})}_{i,\cdot}$ denotes the terminal state $s^{(0)}_{i,\cdot}$). Under the stop-at-tie-breaks equilibrium fixed by the boundary conventions of Appendix A (uniquely fixing the SPNE at ReLU gates with $z = 0$ and at max-pooling states with multiple maximizers), the discounted occupation measures exactly represent, in a multi-trajectory ReLU network (e.g., a ResNet or a CNN with max pooling), the local Stopping-Decomposition and scalar activation gradients:*

1. ***Stopping-Decomposition Gradient:***

$$\frac{\partial a^{(m,p\oplus q)}_{\mathrm{stop},j}}{\partial a^{(l,k)}_{\mathrm{stop},i}} \;=\; \Gamma_x\big(s^{(l,\mathrm{act})}_{i,p\oplus k} \;\big|\; s\big). \tag{52}$$

2. ***Network Activation Gradient.*** *The ordinary-network gradient of $a^{(m)}_j$ with respect to the scalar activation $a^{(l)}_i$ is*

$$\frac{\partial a^{(m)}_j}{\partial a^{(l)}_i} \;=\; \xi_{q,+}\big(\Gamma_x(s^{(l,\mathrm{act})}_{i,+} \mid s) - \Gamma_x(s^{(l,\mathrm{act})}_{i,-} \mid s)\big). \tag{53}$$

Proof. We establish Part 1 by backward induction on the layer gap $m - l$ and derive Part 2 from it. Since x is fixed, we write $\Gamma(v \mid s) \coloneqq \Gamma_x(v \mid s)$.

Part 2. By Theorem 5, $a_i^{(l)} = a_{\mathrm{stop},i}^{(l,+)} - a_{\mathrm{stop},i}^{(l,-)}$ and $a_j^{(m)} = a_{\mathrm{stop},j}^{(m,+)} - a_{\mathrm{stop},j}^{(m,-)}$. Differentiating $a_j^{(m)}$ along the $+$ direction (perturb $a_{\mathrm{stop},i}^{(l,+)}$, keep $a_{\mathrm{stop},i}^{(l,-)}$ fixed) and applying Part 1 at $k = +$,

$$\frac{\partial a_j^{(m)}}{\partial a_i^{(l)}} = \frac{\partial a_{\mathrm{stop},j}^{(m,+)}}{\partial a_{\mathrm{stop},i}^{(l,+)}} - \frac{\partial a_{\mathrm{stop},j}^{(m,-)}}{\partial a_{\mathrm{stop},i}^{(l,+)}} = \Gamma\big(s_{i,q}^{(l,\mathrm{act})} \mid s\big) - \Gamma\big(s_{i,q'}^{(l,\mathrm{act})} \mid s\big) = \xi_{q,+}\big(\Gamma(s_{i,+}^{(l,\mathrm{act})} \mid s) - \Gamma(s_{i,-}^{(l,\mathrm{act})} \mid s)\big), \tag{54}$$

using $q \oplus + = q$ and $q' \oplus + = q'$ in the second equality.

Part 1. The decomposition recursion of Theorem 5 writes the upper-layer Stopping-Decomposition activation as

$$a_{\mathrm{stop},j}^{(m,p\oplus q)} = \mathbf{1}\big[z_j^{(m)} > 0\big]\left[b_j^{(m,p\oplus q)} + \sum_h \Big(W_{hj}^{(m,+)}\, a_{\mathrm{stop},h}^{(m-1,p\oplus q)} + W_{hj}^{(m,-)}\, a_{\mathrm{stop},h}^{(m-1,p\oplus q')}\Big)\right]. \tag{55}$$

Base case ($m = l$). If $s = s_{i,r}^{(l,\mathrm{act})}$ already lies at the target neuron and decomposed activation, then

$$\frac{\partial a_{\mathrm{stop},i}^{(l,p\oplus r)}}{\partial a_{\mathrm{stop},i}^{(l,k)}} = \delta_{p\oplus r,\,k} = \mathbf{1}\{r = p \oplus k\} = \Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{i,r}^{(l,\mathrm{act})}\big). \tag{56}$$

Inductive step (linear routing, $m > l$). Differentiating (55) with respect to $a_{\mathrm{stop},i}^{(l,k)}$ (the bias $b_j^{(m,p\oplus q)}$ does not depend on lower-layer activations) and substituting the inductive hypothesis at layer $m-1$,

$$\frac{\partial a_{\mathrm{stop},j}^{(m,p\oplus q)}}{\partial a_{\mathrm{stop},i}^{(l,k)}} = \mathbf{1}\big[z_j^{(m)} > 0\big] \sum_h \Big(W_{hj}^{(m,+)}\, \Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{h,q}^{(m-1,\mathrm{act})}\big) + W_{hj}^{(m,-)}\, \Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{h,q'}^{(m-1,\mathrm{act})}\big)\Big). \tag{57}$$

On the game side, the one-step occupation recursion at $s = s_{j,q}^{(m,\mathrm{act})}$ uses transition probabilities $W_{hj}^{(m,q\oplus r)}/\gamma_j^{(m)}$ together with the matching cumulative-discount factor $\gamma_j^{(m)}$ and the equilibrium gate $\mathbf{1}[z_j^{(m)} > 0]$:

$$\Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s\big) = \mathbf{1}\big[z_j^{(m)} > 0\big] \sum_{h,r} W_{hj}^{(m,q\oplus r)}\, \Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{h,r}^{(m-1,\mathrm{act})}\big). \tag{58}$$

The two right-hand sides agree, completing the inductive step at linear routing layers.

Addition-node step. For an addition node $z = x + y$, the additivity of the Stopping Decomposition from Appendix B.1 gives $a_{\mathrm{stop},z}^{(p\oplus q)} = a_{\mathrm{stop},x}^{(p\oplus q)} + a_{\mathrm{stop},y}^{(p\oplus q)}$, so by the inductive hypothesis,

$$\frac{\partial a_{\mathrm{stop},z}^{(p\oplus q)}}{\partial a_{\mathrm{stop},i}^{(l,k)}} = \Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{x,q}\big) + \Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{y,q}\big). \tag{59}$$

The Oracle split of Definition 3 sends mass $\frac{1}{2}$ to each branch and applies the discount $\gamma_{\mathrm{O}} = 2$, so the occupation measure satisfies the matching identity

$$\Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{z,q}^{\mathrm{add}}\big) = 2 \cdot \tfrac{1}{2}\, \Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{x,q}\big) + 2 \cdot \tfrac{1}{2}\, \Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{y,q}\big). \tag{60}$$

Max-pooling step. For a max-pooling node $z = \max\{x_1, \ldots, x_m\}$ with winner $k^\star$, Proposition 2 gives $a_{\mathrm{stop},z}^{(p\oplus q)} = a_{\mathrm{stop},x_{k^\star}}^{(p\oplus q)}$, and the inductive hypothesis yields

$$\frac{\partial a_{\mathrm{stop},z}^{(p\oplus q)}}{\partial a_{\mathrm{stop},i}^{(l,k)}} = \Gamma\big(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{x_{k^\star},q}\big). \tag{61}$$

The deterministic transition with unit discount in Definition 4 gives the matching identity $\Gamma(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{z,q}^{\max}) = \Gamma(s_{i,p\oplus k}^{(l,\mathrm{act})} \mid s_{x_{k^\star},q})$. Hence, the correspondence also holds across skip connections and max-pooling nodes, completing the induction. □

C ROUTING GAME: LOG-SPACE FORMULATION

This appendix makes the two-player payoff structure of the Routing Game explicit. The recursion used in Section 4.2 is the logarithmic representation of an entropy-regularized two-player game, and admits a Gibbs subgame-perfect equilibrium for every $\tau > 0$.

C.1 Exact Routing Kernel Behind $\alpha\beta$-LRP

The linear routing subgames are entropy-regularized one-step maximization problems whose closed form is given by the Legendre–Fenchel identity for Shannon entropy $\mathcal{H}(\pi) \coloneqq -\sum_i \pi(i) \log \pi(i)$ (Boyd and Vandenberghe, 2004): for every $\tau > 0$, $n \in \mathbb{N}$, and payoff vector $r \in \mathbb{R}^n$,

$$\max_{\pi \in \Delta_n} \Big\{ \textstyle\sum_i \pi(i)\, r_i + \tau\, \mathcal{H}(\pi) \Big\} = \tau \log \textstyle\sum_i \exp(r_i/\tau), \qquad \pi^\star(i) \propto \exp(r_i/\tau). \tag{62}$$

We write

$$\ell(z) \coloneqq \begin{cases} \log z, & z > 0, \\ -\infty, & z \le 0, \end{cases} \qquad \exp(-\infty) \coloneqq 0, \tag{63}$$

so that $\exp(\ell(x)) = \mathrm{ReLU}(x)$ for any $x \in \mathbb{R}$; we use this fact for notational convenience. Zero or negative routed mass thus receives the stopping value $-\infty$; where a player compares stopping with continuing (activation states), the comparison is made on the argument of ℓ, the signed routed mass, so that negative mass is strictly worse than stopping (mass 0). The ReLU-gating itself is fully modeled by the player policies at states modeling the ReLU gates—just like in the Stopping Game.

State Space and Game Mechanics. To correctly model the interaction between player turns and stream signs, we define a structured state space:

- **Activation States:** $\mathcal{S}_{act}^{(l)} = \{s_{j,p}^{(l,act)}\}$, where $p \in \{+,-\}$ is the player in turn.
- **Linear Routing States:** $\mathcal{S}_{lin}^{(l)} = \{s_{j,q,\kappa}^{(l,lin)}\}$, where q is the player turn and $\kappa \in \{+,-\}$ is the stream sign.

At $s_{j,p}^{(l,act)}$, the active player p chooses between *stop* and *continue*. Under the *continue* action, the environment's transition kernel is uniform over the two sign branches,

$$\mathbb{P}\big(s_{j,p,+}^{(l,lin)} \,\big|\, s_{j,p}^{(l,act)}, \text{continue}\big) = \tfrac{1}{2}, \qquad \mathbb{P}\big(s_{j,p',-}^{(l,lin)} \,\big|\, s_{j,p}^{(l,act)}, \text{continue}\big) = \tfrac{1}{2}, \tag{64}$$

where the first branch keeps player p in control evaluating $\kappa = +$ weights and the second switches the turn to opponent p' evaluating $\kappa = -$ weights. To compensate for the $1/2$ branching, the environment applies an Oracle structural discount of $\gamma_{\mathrm{O}} = 2$ (distinct from the per-neuron weight-magnitude discount $\gamma_j^{(l)}$ used in the Stopping Game).

Definition 7 (Entropy-Regularized Two-Player Routing Game)**.** *Fix an input x and a stabilizer $\varepsilon \ge 0$. At each activation state $s_{j,p}^{(l,act)}$, the player in turn chooses* stop *or* continue *with similar mechanics as in the Stopping Game (Appendix B) reflecting the open-closed state of a ReLU gate: stopping sends the game to the cemetery $\dagger$ giving immediate payoff $\ell(0) = -\infty$—and continuation transitions backward with structural discount $\gamma = 2$ and the uniform $1/2$ sign-branch kernel given above. Here happens the critical step: gains of the opponent on the negative stream state $s_{j,p',-}^{(l,lin)}$ shall be counted as losses of the active player p. At each linear routing state $s_{j,q,\kappa}^{(l,lin)}$, the player in turn instead chooses a mixed action $\nu \in \Delta(\mathrm{Pred}_j^{(l)} \cup \{\mathrm{bias}, \perp_{\mathrm{lin}}\})$ over the predecessors of neuron j augmented by a terminal* bias node *with immediate payoff $\ell([b_j^{(l)}]^\kappa)$ and a fictitious* outside option $\perp_{\mathrm{lin}}$ *which routes the game's mass to the cemetery with immediate payoff $\ell(\varepsilon)$; trajectories entering either are absorbed. Let $\mathcal{H}(\nu) \coloneqq -\sum_a \nu(a) \log \nu(a)$ be Shannon entropy over this augmented action set. At $\varepsilon = 0$ the outside option carries $\ell(0) = -\infty$ and is dominated by every positive-mass predecessor, so the augmented game reduces to the Routing Game over* $\mathrm{Pred}_j^{(l)} \cup \{\mathrm{bias}\}$.

Throughout, $\mathbf{V}(\cdot) = (V_p(\cdot), V_{p'}(\cdot))^\top$ *denotes the game's backward-induction player-specific log-value function, defined jointly at linear routing states by* (69) *and at activation states by* (72) *below. The recursion is well-posed because the layered state graph is a finite DAG: the* $l = 1$ *case of* (65) *needs no lower-layer* $\mathbf{V}$*, and every subsequent layer uses only* $\mathbf{V}$ *at strictly lower layers.*

For a ***linear routing state*** *at* $l \geq 2$ *the active player continuing to the* i*th neuron's activation state receives an immediate payoff of the logarithm of the respective weight summed with the respective log-value function of the next state, whereas in the first layer the final payoff is paid out to the active player. The inactive player always obtains an immediate payoff of* $\ell(0) = -\infty$*. Let us denote by* $Q_r(s, a)$ *player* r*'s gain for the game value in case action* a *actually happens, whereas* $\bar{Q}_r$ *denotes the mixture of the differences between the payoffs of the opponents including the entropy regularization term given the mixed strategy chosen by the active player.*

$$Q_r\big(s^{(l,lin)}_{j,q,\kappa}, i\big) \coloneqq \begin{cases} \ell\big(W^{(l,\kappa)}_{ij}\big) + V_r\big(s^{(l-1,act)}_{i,q}\big), & r = q,\ l \geq 2, \\ \ell\big([x_i W^{(1)}_{ij}]^\kappa\big), & r = q,\ l = 1, \\ -\infty, & r = q'. \end{cases} \tag{65}$$

The outside option $\perp_{\mathrm{lin}}$ *and the bias node have log-scores* $Q_r(s^{(l,\mathrm{lin})}_{j,q,\kappa}, \perp_{\mathrm{lin}}) \coloneqq \log \varepsilon$ *and* $Q_r(s^{(l,\mathrm{lin})}_{j,q,\kappa}, \mathrm{bias}) \coloneqq \ell([b^{(l)}_j]^\kappa)$ *for* $r = q$ *and* $-\infty$ *for* $r = q'$*; they route the* ε*-fraction and the bias share of the mass to absorbing terminals instead of to any predecessor. If player* q *uses mixed action* ν *at this state, the induced* individual routed masses *are*

$$\begin{aligned} \bar{Q}_r\big(s^{(l,lin)}_{j,q,\kappa}; \nu\big) \coloneqq \sum_a \nu(a)\, \ell\left(\exp\left(Q_r\big(s^{(l,lin)}_{j,q,\kappa}, a\big)\right) - \exp\left(Q_{r'}\big(s^{(l,lin)}_{j,q,\kappa}, a\big)\right)\right) \\ + \tau \mathcal{H}(\nu), \qquad r \in \{q, q'\}, \end{aligned} \tag{66}$$

which, as $\exp(-\infty) = 0$*, evaluates to*

$$\bar{Q}_q\big(s^{(l,lin)}_{j,q,\kappa}; \nu\big) = \sum_a \nu(a)\, Q_q\big(s^{(l,lin)}_{j,q,\kappa}, a\big) + \tau \mathcal{H}(\nu) \tag{67}$$

$$\bar{Q}_{q'}\big(s^{(l,lin)}_{j,q,\kappa}; \nu\big) = -\infty. \tag{68}$$

If $W^{(l,\kappa)}_{ij} = 0$ *for every predecessor* i *and* $[b^{(l)}_j]^\kappa = 0$ *(i.e. the stream-*κ *part of the fan-in to neuron* j *is identically zero), then* $Q_q(s^{(l,\mathrm{lin})}_{j,q,\kappa}, i) = -\infty$ *for all* i *as well and the value* (69) *equals* $\ell(\varepsilon)$*. If* $\varepsilon = 0$ *and every action scores* $-\infty$ *(e.g.* $\Sigma^{(l,\kappa)}_j = 0$ *at* $\tau = 1$*), every* ν *is a best response; we fix* $\nu = \delta_{\perp_{\mathrm{lin}}}$ *(absorption), matching the convention* $0/0 \coloneqq 0$ *in* (90)*. We define the corresponding equilibrium linear-game values by*

$$\begin{aligned} V^\star_q\big(s^{(l,lin)}_{j,q,\kappa}\big) &\coloneqq \max_{\nu \in \Delta(\mathrm{Pred}^{(l)}_j \cup \{\mathrm{bias}, \perp_{\mathrm{lin}}\})} \bar{Q}_q\big(s^{(l,lin)}_{j,q,\kappa}; \nu\big) \\ &= \max_{\nu \in \Delta(\mathrm{Pred}^{(l)}_j \cup \{\mathrm{bias}, \perp_{\mathrm{lin}}\})} \left\{ \sum_a \nu(a)\, Q_q\big(s^{(l,lin)}_{j,q,\kappa}, a\big) + \tau \mathcal{H}(\nu) \right\} \\ &= \tau \cdot \ell\left(\varepsilon^{\frac{1}{\tau}} + \big([b^{(l)}_j]^\kappa\big)^{\frac{1}{\tau}} + \sum_i \exp\left(Q_q(s^{(l,lin)}_{j,q,\kappa}, i)\right)^{\frac{1}{\tau}} \right), \end{aligned} \tag{69}$$

$$V^\star_{q'}\big(s^{(l,lin)}_{j,q,\kappa}\big) \coloneqq -\infty, \tag{70}$$

where we used (62)*. At* $\tau = 1$ *the associated Gibbs policy assigns the outside option the probability* $\varepsilon/(\varepsilon + \sum_{a \neq \perp_{\mathrm{lin}}} \exp Q_q(\cdots, a))$ *and distributes the remaining mass over the predecessors and the bias node in proportion to* $\exp Q_q$*.*

For an ***activation state****, continuing exposes both sign branches. Let* $s_+ \coloneqq s^{(l,lin)}_{j,p,+}$ *and* $s_- \coloneqq s^{(l,lin)}_{j,p',-}$*. If player* p *chooses* continue *at* $s^{(l,act)}_{j,p}$*, the game transitions to* s_+ *or to* s_- *with equal probability, while* stop *sends the game to the cemetery with game value* $V_r(\dagger) = \ell(0) = -\infty$*. Taking the expectation over the uniform sign-branch kernel*

($\frac{1}{2}$ *each) and respecting the discount* $\gamma_O = 2$ *interpreted in logarithmic space, the continuation value becomes*

$$Q_r(s_{j,p}^{(l,act)}, \text{continue}) := \ell\left(\sum_{s'\in\{s_+,s_-\}} \mathbb{P}(s' \mid s_{j,p}^{(l,act)}) \cdot \gamma_O \cdot \Big(\exp V_r(s') - \exp V_{r'}(s')\Big)\right)$$
$$= \ell\left(\xi_{r,p}\Big(\exp\left(V_p\left(s_+\right)\right) - \exp\left(V_{p'}\left(s_-\right)\right)\Big)\right). \tag{71}$$

Observe that at least one of the two equals $-\infty$*. On the other hand, stopping yields* $-\infty$ *for both. The equilibrium activation-game values are*

$$V_p(s_{j,p}^{(l,act)}) := \max\{-\infty, Q_p(s_{j,p}^{(l,act)}, \textit{continue})\}, \tag{72}$$

and, since player p *continues only if* $Q_p(s_{j,p}^{(l,act)}, \textit{continue}) > -\infty$*, in which case* $Q_{p'}(s_{j,p}^{(l,act)}, \textit{continue}) = -\infty$*, we have*

$$V_{p'}(s_{j,p}^{(l,act)}) = -\infty. \tag{73}$$

Figure 4 shows a worked example of the local routing subgame.

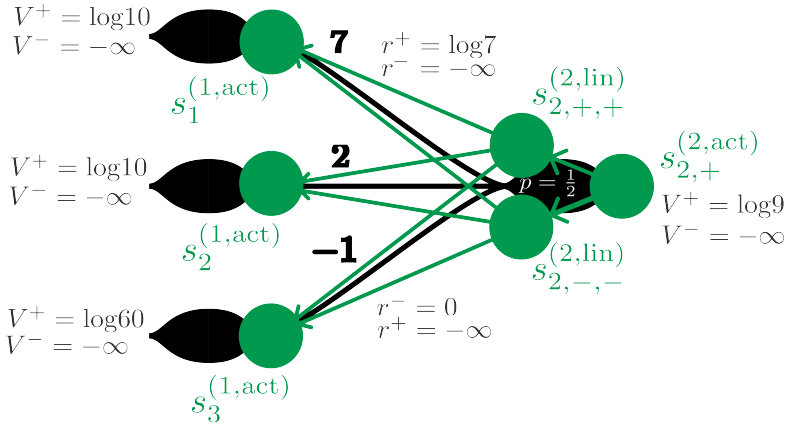

Figure 4: Routing Game: local routing subgame around $s_{2,+}^{(2,\text{act})}$. This uses the same toy subnetwork and numerical values as Figure 3. At this activation state, P^+ is in turn and continues because the routed mass is positive ($V^+ = \log 9 > -\infty$ in the depicted example). The game then branches with probability $1/2$ to $s_{2,+,+}^{(2,\text{lin})}$ (turn stays with P^+) and with probability $1/2$ to $s_{2,-,-}^{(2,\text{lin})}$ (turn switches to P^-). In each linear state, the active player samples a predecessor according to the Gibbs policy from the immediate log-payoff and game value. Example: along $s_{2,+,+}^{(2,\text{lin})} \to s_{1,+}^{(1,\text{act})}$, the player on the positive stream state gets immediate payoff $\log W_{12}^{(2,+)} = \log 7$ and P^- gets $-\infty$. Edges with zero routed mass have payoff $\log W_{12}^{(2,-)} = \log 0 = -\infty$. For readability, the $\pm$ activation copies in layer 1 are collapsed, and the intermediate payoff $\log 2$ on the transition from position 2 to 2 is omitted.

In the following let us denote, with $a^{(0)} = x$,

$$\Sigma_j^{(l,\kappa)} := \textstyle\sum_i [a_i^{(l-1)} W_{ij}^{(l)}]^\kappa + [b_j^{(l)}]^\kappa, \tag{74}$$

so that $z_j^{(l)} = \Sigma_j^{(l,+)} - \Sigma_j^{(l,-)}$; for $l \geq 2$, $a^{(l-1)} \geq 0$ gives $[a_i^{(l-1)} W_{ij}^{(l)}]^\kappa = a_i^{(l-1)} W_{ij}^{(l,\kappa)}$.

Theorem 7 (SPNE and Forward Equivalence of the Routing Game)**.**

(i) *SPNE existence for every* $\tau > 0$ *and* $\varepsilon \geq 0$*. For every* $\tau > 0$ *and every* $\varepsilon \geq 0$*, the entropy-regularized two-player Routing Game of Definition 7 (with linear-state entropy weight* $\tau\,\mathcal{H}$ *and outside-option stabilizer* ε*) admits a subgame-perfect Nash equilibrium (see* (79)–(80) *in the proof for the explicit form of the equilibrium policy). We write* $\cdot^\star$ *for equilibrium quantities throughout.*

(ii) *Forward equivalence at the anchor $\tau = 1$, $\varepsilon \geq 0$. At $\tau = 1$ and $\varepsilon \geq 0$, the equilibrium quantities recover the network forward pass: for every layer l, neuron j, player labels p, q, r, and stream sign $\kappa \in \{+, -\}$,*

$$V_r\big(s^{(l,lin)}_{j,q,\kappa}\big) = \ell\big(\delta_{r,q}\,(\Sigma^{(l,\kappa)}_j + \varepsilon)\big), \tag{75}$$

$$Q_r\left(s^{(l,act)}_{j,p}, \mathit{continue}\right) = \ell\left(\xi_{r,p} \cdot z^{(l)}_j\right), \tag{76}$$

$$V_r\big(s^{(l,act)}_{j,p}\big) = \ell\big(\delta_{r,p}\ \cdot a^{(l)}_j\big), \tag{77}$$

$$\pi^\star_p\big(s^{(l,act)}_{j,p}\big) = \mathbf{1}[z^{(l)}_j > 0]. \tag{78}$$

Remark 2 (Uniqueness)**.** *The linear-state Gibbs best response* (79) *is the unique maximizer of the strictly concave entropy-regularized objective whenever some action has a finite score (otherwise Definition 7 fixes $\perp_{\mathrm{lin}}$). At activation states the active player is indifferent only when the signed continuation mass* $\exp V_p(s_+) - \exp V_{p'}(s_-)$ *is zero ($z^{(l)}_j = 0$ at $\tau = 1$): both stopping and continuing are best responses, so multiple SPNEs can coexist under different tie-breaking. The theorem specifies the tie-breaker $\pi^\star_p = \mathbf{1}[Q_p(\cdot, \mathrm{continue}) > -\infty]$, which at $\tau = 1$ reduces to $\mathbf{1}[z^{(l)}_j > 0]$. Just like in the SG (Appendix B), we refer to this SPNE as the* ***stop-at-tie-breaks*** *equilibrium.*

Proof. **SPNE existence for every $\tau > 0$.** The layered state graph is a finite DAG: linear routing states at layer l transition to activation states at layer $l-1$ (or to terminal input positions at $l = 1$), and activation states at layer l transition to linear routing states at the same layer l or to the cemetery †. We construct the equilibrium by backward induction on the layer index l.

At layer $l = 1$ the linear routing subgames have no unresolved game values, so the action values Q_q in (65) are fully specified by the input x. At each linear state the active player solves $\max_{\nu \in \Delta(\mathrm{Pred}^{(1)}_j)} \{\sum_i \nu(i)\, Q_q(s^{(1,lin)}_{j,q,\kappa}, i) + \tau\, \mathcal{H}(\nu)\}$, which is strictly concave on the simplex. Identity (62) yields the unique Gibbs best response, and simplifying $\exp \circ \ell$ in (65) gives the explicit form

$$\pi^\star_q\big(i \,\big|\, s^{(l,lin)}_{j,q,\kappa}\big) \propto \begin{cases} \big(W^{(l,\kappa)}_{ij}\big)^{1/\tau}\, \exp\big(V_q(s^{(l-1,act)}_{i,q})/\tau\big), & l \geq 2, \\ \big([x_i W^{(1)}_{ij}]^\kappa\big)^{1/\tau}, & l = 1, \end{cases} \tag{79}$$

with zero mass on routes of zero or wrong sign and closed-form value $V_q(s^{(1,lin)}_{j,q,\kappa}) = \tau \log \sum_i \exp(Q_q(s^{(1,lin)}_{j,q,\kappa}, i)/\tau)$. With these values in hand, the activation-state subgame at layer $l = 1$ is a binary choice between stop (log-payoff $\ell(0) = -\infty$) and continue (71); comparing signed masses and stopping at ties, the active player's best response is the pure stop/continue strategy

$$\pi^\star_p\big(s^{(l,act)}_{j,p}\big) = \mathbf{1}\big[Q_p(s^{(l,act)}_{j,p}, \mathrm{continue}) > -\infty\big]. \tag{80}$$

Assume the equilibrium has been constructed at all layers $\leq l-1$, yielding definite values $V_r(s^{(l-1,act)}_{i,q})$. At layer l the linear-state action values Q_q in (65) are then specified, and the same strict-concavity argument gives the Gibbs best response (79) with the analogous log-sum-exp value. The layer-l activation-state best response is again the binary hard comparison (80). Because each subgame's best response is taken to be optimal given the game values from lower layers, the composed strategy profile is subgame-perfect. This proves existence of an SPNE for every $\tau > 0$. The argument is unchanged under the bias-node and outside-option augmentation of Definition 7: the linear-state action set gains the extra elements bias and $\perp_{\mathrm{lin}}$ with constant log-scores $\ell([b^{(l)}_j]^\kappa)$ and $\log \varepsilon$ (finite or $-\infty$), the strict-concavity / Legendre–Fenchel step goes through verbatim on the augmented simplex, and the closed-form value picks up the extra terms $([b^{(l)}_j]^\kappa)^{1/\tau}$ and $\varepsilon^{1/\tau}$ inside the log-sum-exp. Hence, an SPNE exists for every $(\tau, \varepsilon) \in \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0}$.

Forward equivalence at $\tau = 1$. We now specialize to $\tau = 1$ and show by induction on l that the equilibrium quantities recover the network forward pass.

Base case ($l = 1$): Identical to the inductive step below, with the fused-input action values $Q_q(s^{(1,lin)}_{j,q,\kappa}, i) = \ell([x_i W^{(1)}_{ij}]^\kappa)$ of (65) in place of $\ell(a^{(l-1)}_i W^{(l,\kappa)}_{ij})$.

Inductive step ($l \geq 2$): Assume $V_q(s_{i,q}^{(l-1,act)}) = \ell(a_i^{(l-1)})$ for all predecessors i and both player labels q. Since standard activations satisfy $a_i^{(l-1)} \geq 0$, the linear routing-state action values become

$$Q_q\big(s_{j,q,\kappa}^{(l,lin)}, i\big) = \ell\big(W_{ij}^{(l,\kappa)}\big) + \ell\big(a_i^{(l-1)}\big) = \ell\big(a_i^{(l-1)} W_{ij}^{(l,\kappa)}\big). \tag{81}$$

Applying (62) to the linear routing subgame gives the Gibbs best response (79) and

$$V_q\big(s_{j,q,\kappa}^{(l,lin)}\big) = \ell\left(\sum_a \exp\big(Q_q(s_{j,q,\kappa}^{(l,lin)}, a)\big)\right) = \ell\left(\sum_i a_i^{(l-1)} W_{ij}^{(l,\kappa)} + [b_j^{(l)}]^\kappa + \varepsilon\right) = \ell(\Sigma_j^{(l,\kappa)} + \varepsilon), \tag{82}$$

At the activation state the active player's continuation advantage is again $z_j^{(l)}$

$$Q_p\big(s_{j,p}^{(l,act)}, \text{continue}\big) = \ell\left(\Sigma_j^{(l,+)} - \Sigma_j^{(l,-)}\right) = \ell(z_j^{(l)}), \tag{83}$$

whereas for the opponent

$$Q_{p'}\left(s_{j,p}^{(l,\text{act})}, \text{continue}\right) = \ell\left(\Sigma_j^{(l,-)} - \Sigma_j^{(l,+)}\right) = \ell(-z_j^{(l)}), \tag{84}$$

again confirming (76). Thus, the best response is (80). Finally,

$$V_r\big(s_{j,p}^{(l,act)}\big) = \ell\big(\delta_{r,p}\, \mathbf{1}[z_j^{(l)} > 0]\, z_j^{(l)}\big) = \ell\big(\delta_{r,p}\, a_j^{(l)}\big), \tag{85}$$

proving (77). Since each subgame has an optimal action chosen by the player in turn, the resulting strategy profile is subgame-perfect. This completes the induction. □

C.1.1 Log-Space Addition Subgame, Skip Connections, and Max Pooling

Residual additions $z = x + y$ and max-pooling nodes $z = \max_k x_k$ enter the RG through the addition state $s_{z,p}^{\text{add}}$ (uniform Oracle to the operand states, player p retained, $\gamma_{\text{O}} = 2$) and the pooling state $s_{z,p}^{\max}$ (deterministic choice of an operand, player p retained, $\gamma_{\text{O}} = 1$, ties broken by the smallest index) of Appendix B.3, with the values written in log space:

$$V_r(s_{z,p}^{\text{add}}) = \ell\big(\exp V_r(s_{x,p}) + \exp V_r(s_{y,p})\big), \qquad V_p(s_{z,p}^{\max}) = \max_k V_p(s_{x_k,p}). \tag{86}$$

If the operand states recover their (non-negative) activations, $V_p(s_{x,p}) = \ell(a_x)$ etc., then $V_p(s_{z,p}^{\text{add}}) = \ell(a_x + a_y) = \ell(a_z)$ and $V_p(s_{z,p}^{\max}) = \ell(a_{x_{k^\star}}) = \ell(a_z)$, so forward equivalence extends across both nodes. Two things differ from the SG. First, $\gamma_{\text{O}} = 2$ enters only the values: the trajectory discount (88) of the addition transition is 1, so each operand receives half of the addition-state occupation (Theorem 8, item 2). Second, the relevance split therefore equals the Oracle probabilities: a $(\rho, 1-\rho)$ Oracle with value discounts $(1/\rho, 1/(1-\rho))$ leaves all forward values unchanged but splits the relevance as $(\rho, 1-\rho)$, so the uniform split is a convention of the RG rule. The implementation passes the full relevance to both operands (trajectory discount 2); since every backward trajectory of a ResNet or ViT that reaches the input crosses each of its B addition nodes exactly once, this rescales the input occupation by the constant 2^B and leaves normalized maps unchanged.

C.2 Backward Path Integral and $\alpha\beta$-LRP Recovery

We now show that the projected discounted occupation measure induced by the SPNE on the layered routing graph realizes the $\alpha\beta$-LRP-ε relevance vector of (1). To keep the game side and the attribution side strictly separate, we use distinct symbols throughout this subsection:

- the *game-side* object is the discounted occupation measure $\Gamma_x(\,\cdot\,)$ on the states of the routing graph (Definition 6), now driven by the SPNE trajectory law Π_x rooted at $s_{u,+}^{(L,\text{act})}$ with unit initial mass: $\Gamma_x(s_{u,+}^{(L,\text{act})}) = 1$ and $\Gamma_x(s_{j,p}^{(L,\text{act})}) = 0$ for every other output activation state. With non-negative branch discounts (cf. (88)) Γ_x is itself non-negative; we then form the per-neuron difference of the two player copies,

$$\Gamma_j^{(l)} \;\coloneqq\; \Gamma_x\big(s_{j,+}^{(l,\text{act})}\big) \;-\; \Gamma_x\big(s_{j,-}^{(l,\text{act})}\big). \tag{87}$$

- the *attribution-side* object is the $\alpha\beta$-LRP-ε relevance vector $R^{(l)}$ (the $\alpha\beta$ rule of Bach et al. (2015) with the stabilization convention adopted here, see below), a deterministic backward DP on the network's linear layers seeded by a chosen output mass $R_u^{(L)}$ and recursing via (90) below.

The recovery theorem (Theorem 8) shows that, under the SPNE policy and the following trajectory discount (88), the per-neuron difference $\Gamma_j^{(l)}$ satisfies the same recursion as $R_j^{(l)}$, hence equals $R_j^{(l)}/R_u^{(L)}$.

Trajectory discount. We track sign-dependent branch discounts governed by the stream-balance mode (α, β) $(\alpha - \beta = 1)$. Let $\zeta = (\zeta_0, \zeta_1, \dots)$ be a backward trajectory on linear routing, activation, addition or max-pooling states. Define the per-state discount γ as a non-negative function of the entered state,

$$\gamma(s) \coloneqq \begin{cases} 2\alpha, & s = s_{j,q,+}^{(l,lin)} \text{ for some } j,q,l, \\ 2\beta, & s = s_{j,q,-}^{(l,lin)} \text{ for some } j,q,l, \\ 1, & \text{otherwise (activation, addition Oracle, max-pool).} \end{cases} \tag{88}$$

The cumulative discount along the trajectory up to time t is the product

$$d_t(\zeta) \coloneqq \prod_{m=1}^{t} \gamma(\zeta_m), \qquad d_0(\zeta) = 1 \text{ (empty product).} \tag{89}$$

The signed minus of the $\alpha\beta$-LRP recursion is produced by the difference (87) between the two player copies.

Remark 3 (Choice of starting player)**.** *Rooting Π_x at $s_{u,-}^{(L,\mathrm{act})}$ instead of $s_{u,+}^{(L,\mathrm{act})}$ swaps the two output-layer occupations and therefore negates $\Gamma_j^{(l)}$ at every layer. We fix the canonical convention that the game starts at the $+$-state.*

Remark 4 (Equivalent signed-discount sum formulation)**.** *The same recovery theorem can be phrased with the signed trajectory discount $\gamma(s_{j,q,+}^{(l,\mathrm{lin})}) = 2\alpha$, $\gamma(s_{j,q,-}^{(l,\mathrm{lin})}) = -2\beta$ and the per-neuron* sum $\Gamma_j^{(l)} = \Gamma_x(s_{j,+}^{(l,\mathrm{act})}) + \Gamma_x(s_{j,-}^{(l,\mathrm{act})})$ *in place of* (87)*. We have chosen the non-negative-discount version because it is easier to comprehend—Γ_x remains a genuine non-negative occupation measure, and the LRP sign is generated by an explicit algebraic difference rather than absorbed into the trajectory weights.*

Attribution-side: $\alpha\beta$-LRP-ε relevance. Independently of the game, the $\alpha\beta$-LRP-ε relevance vector $R^{(l)}$ is defined recursively from $R_j^{(L)} = R_u^{(L)}\,\delta_{j,u}$ at the output by

$$R_i^{(l-1)} \;=\; \sum_j \left(\alpha \frac{[a_i^{(l-1)} W_{ij}^{(l)}]^+}{\varepsilon + \Sigma_j^{(l,+)}} - \beta \frac{[a_i^{(l-1)} W_{ij}^{(l)}]^-}{\varepsilon + \Sigma_j^{(l,-)}} \right) R_j^{(l)}, \tag{90}$$

where $\Sigma_j^{(l,\pm)}$ is (74): the bias enters the denominators as a terminal bias node, and the relevance routed to it (and to the outside option) is absorbed, as in the implementation. Recursion (90) is the $\alpha\beta$ rule of Bach et al. (2015) with the stabilization convention adopted throughout this paper: the stabilizer ε of their LRP-ε rule is added separately to each sign denominator. Bach et al. (2015) present the two rules as alternative constructions and do not state this combined form; it is the convention of common LRP implementations (Anders et al., 2021), and Theorem 8 is exact for this form. At $\varepsilon = 0$ it is the original $\alpha\beta$ rule.

Theorem 8 (Recovery of Multi-Path $\alpha\beta$-LRP)**.** *For every $\varepsilon \geq 0$ and every input with $f(x) = a_u^{(L)} > 0$ (if $f(x) = 0$ the root player stops and $R^{(l)} = f(x)\Gamma^{(l)} = 0$ trivially), under the stop-at-tie-breaks equilibrium (Remark 2; ties at max-pooling states broken by the smallest index) at the anchor temperature $\tau = 1$ and stabilizer ε, with the cumulative trajectory discount* (89) *and the occupation measure $\Gamma_j^{(l)}$ from* (87)*:*

1. ***Linear Map.*** *The per-neuron difference $\Gamma_j^{(l)}$ of* (87) *satisfies*

$$\Gamma_i^{(l-1)} \;=\; \sum_j \left(\alpha \frac{[a_i^{(l-1)} W_{ij}^{(l)}]^+}{\varepsilon + \Sigma_j^{(l,+)}} - \beta \frac{[a_i^{(l-1)} W_{ij}^{(l)}]^-}{\varepsilon + \Sigma_j^{(l,-)}} \right) \Gamma_j^{(l)}. \tag{91}$$

Because (91) *and* (90) *share their computational step,* $\Gamma_j^{(l)} \cdot R_u^{(L)} = R_j^{(l)}$ *at every layer; in particular* $R_j^{(l)} = f(x)\,\Gamma_j^{(l)}$ *for the seed* $R_u^{(L)} = f(x)$*. The same formula covers the fused input layer with* $a^{(0)} = x$*. At* $\varepsilon = 0$ *this reduces to the unstabilized* $\alpha\beta$*-LRP redistribution (not conservative: bias nodes absorb relevance); at* $\varepsilon > 0$ *it is the* $\alpha\beta$ *rule with the stabilization convention adopted here, i.e.* (1) *with* ε *added to each sign denominator separately.*

2. ***Residual Addition.*** *Let* $z = x_{\mathrm{op}} + y_{\mathrm{op}}$ *be an addition node with addition state* $s_{z,p}^{\mathrm{add}}$ *and operand states* $s_{x_{\mathrm{op}},p}, s_{y_{\mathrm{op}},p}$ *as in Appendix C.1.1. For every player label* $p \in \{+,-\}$*, with the edge occupation* $\Gamma_x(s \to s') \coloneqq \mathbb{E}_{\zeta\sim\Pi_x}\big[\sum_t d_{t+1}(\zeta)\mathbf{1}\{\zeta_t = s, \zeta_{t+1} = s'\}\big]$ *(an operand's total occupation adds the edges from its other parents),*

$$\Gamma_x\big(s_{z,p}^{\mathrm{add}} \to s_{x_{\mathrm{op}},p}\big) \;=\; \tfrac{1}{2}\,\Gamma_x\big(s_{z,p}^{\mathrm{add}}\big), \qquad \Gamma_x\big(s_{z,p}^{\mathrm{add}} \to s_{y_{\mathrm{op}},p}\big) \;=\; \tfrac{1}{2}\,\Gamma_x\big(s_{z,p}^{\mathrm{add}}\big), \tag{92}$$

so the operand pair carries the full addition-state mass with no duplication.

3. ***Max Pooling.*** *For a pooled output* $z = \max\{x_1, \dots, x_m\}$ *with winner* $k^\star$ *and pooling state* $s_{z,p}^{\max}$ *(Appendix C.1.1), the deterministic value-maximizing transition concentrates all mass on the winner: for every player label* $p \in \{+,-\}$*,*

$$\Gamma_x\big(s_{z,p}^{\max} \to s_{x_{k^\star},p}\big) \;=\; \Gamma_x\big(s_{z,p}^{\max}\big), \qquad \Gamma_x\big(s_{z,p}^{\max} \to s_{x_k,p}\big) \;=\; 0 \text{ for } k \neq k^\star. \tag{93}$$

Proof. We prove $R_u^{(L)} \cdot \Gamma_j^{(l)} = R_j^{(l)}$ at every layer l by backward induction on l.

Base case ($l = L$*):* The trajectory law Π_x has unit mass at $s_{u,+}^{(L,\mathrm{act})}$ by definition, so $\Gamma_x(s_{u,+}^{(L,\mathrm{act})}) = 1$ and $\Gamma_x(s_{j,p}^{(L,\mathrm{act})}) = 0$ for every $(j,p) \neq (u,+)$. By (87),

$$\Gamma_j^{(L)} \;=\; \Gamma_x\big(s_{j,+}^{(L,\mathrm{act})}\big) - \Gamma_x\big(s_{j,-}^{(L,\mathrm{act})}\big) \;=\; \delta_{j,u}, \tag{94}$$

i.e. the difference is 1 at the chosen output neuron u and 0 elsewhere. Multiplying by the output mass gives $R_u^{(L)} \cdot \Gamma_j^{(L)} = R_u^{(L)}\,\delta_{j,u} = R_j^{(L)}$, matching the LRP root specification of (90).

Inductive step ($l \Rightarrow l-1$*):* Assume $R_u^{(L)} \cdot \Gamma_j^{(l)} = R_j^{(l)}$ for every neuron j in layer l. From the SPNE construction in Theorem 7 together with the activation-game value reduction $V_p(s_{j,p}^{(l,\mathrm{act})}) = \ell(a_j^{(l)})$ (which holds for every $\varepsilon \geq 0$ because the additive ε-shifts on the two sign branches cancel in the continuation advantage), the linear-state Gibbs best response on the alive support is

$$\pi_{j,\kappa}^\star(\perp_{\mathrm{lin}}) = \frac{\varepsilon}{\varepsilon + \Sigma_j^{(l,\kappa)}}, \qquad \pi_{j,\kappa}^\star(\mathrm{bias}) = \frac{[b_j^{(l)}]^\kappa}{\varepsilon + \Sigma_j^{(l,\kappa)}}, \qquad \pi_{j,\kappa}^\star(i) = \frac{a_i^{(l-1)} W_{ij}^{(l,\kappa)}}{\varepsilon + \Sigma_j^{(l,\kappa)}}. \tag{95}$$

A trajectory reaches the layer-$(l-1)$ activation state $s_{i,r}^{(l-1,\mathrm{act})}$ from a layer-l activation state through one of two routes. By the activation-state continuation kernel of Definition 7, departing from $s_{j,p}^{(l,\mathrm{act})}$ (the player continues: at $l = L$ since $f(x) > 0$; at $l < L$, $\Gamma_x(s_{j,\cdot}^{(l,\mathrm{act})}) = 0$ unless $a_j^{(l)} > 0$ by (95)) the trajectory takes the $+$ branch to $s_{j,p,+}^{(l,\mathrm{lin})}$ with probability $\frac{1}{2}$ (player p retains the turn), and the $-$ branch to $s_{j,p',-}^{(l,\mathrm{lin})}$ with probability $\frac{1}{2}$ (turn switches to opponent p'); from each linear state the policy (95) routes to predecessor i with the player label of that linear state. Combining the two branch discount factors $2\alpha, 2\beta$ from (88) with these $\frac{1}{2}$-probabilities, the trajectory contributions to $\Gamma_x(s_{i,r}^{(l-1,\mathrm{act})})$ are

$$\begin{aligned}\Gamma_x\big(s_{i,r}^{(l-1,\mathrm{act})}\big) &= \sum_j \Big(\tfrac{1}{2}(2\alpha)\,\pi_{j,+}^\star(i)\,\Gamma_x\big(s_{j,r}^{(l,\mathrm{act})}\big) \;+\; \tfrac{1}{2}(2\beta)\,\pi_{j,-}^\star(i)\,\Gamma_x\big(s_{j,r'}^{(l,\mathrm{act})}\big)\Big) \\ &= \sum_j \Big(\alpha\,\pi_{j,+}^\star(i)\,\Gamma_x\big(s_{j,r}^{(l,\mathrm{act})}\big) \;+\; \beta\,\pi_{j,-}^\star(i)\,\Gamma_x\big(s_{j,r'}^{(l,\mathrm{act})}\big)\Big),\end{aligned} \tag{96}$$

where $r' \in \{+,-\}$ denotes the opposite player label and the absorbing actions bias and $\perp_{\mathrm{lin}}$ contribute no mass to any predecessor.

Differencing (96) over $r \in \{+,-\}$ (i.e. $r{=}+$ minus $r{=}-$) flips the role of r and r' between the two terms inside the bracket. Using $\Gamma_x(s^{(l,\mathrm{act})}_{j,r}) - \Gamma_x(s^{(l,\mathrm{act})}_{j,r'}) = +\Gamma^{(l)}_j$ when $r{=}+$, $r'{=}-$, and the same expression with the opposite sign when $r{=}-$, $r'{=}+$, the per-neuron difference at layer $l-1$ satisfies

$$\begin{aligned}
\Gamma^{(l-1)}_i &= \Gamma_x\big(s^{(l-1,\mathrm{act})}_{i,+}\big) - \Gamma_x\big(s^{(l-1,\mathrm{act})}_{i,-}\big) \\
&= \sum_j \Big(\alpha\, \pi^\star_{j,+}(i)\, \big[\Gamma_x(s^{(l,\mathrm{act})}_{j,+}) - \Gamma_x(s^{(l,\mathrm{act})}_{j,-})\big] \;+\; \beta\, \pi^\star_{j,-}(i)\, \big[\Gamma_x(s^{(l,\mathrm{act})}_{j,-}) - \Gamma_x(s^{(l,\mathrm{act})}_{j,+})\big]\Big) \\
&= \sum_j \big(\alpha\, \pi^\star_{j,+}(i) - \beta\, \pi^\star_{j,-}(i)\big)\, \Gamma^{(l)}_j \\
&= \sum_j \left(\alpha \frac{[a^{(l-1)}_i W^{(l)}_{ij}]^+}{\varepsilon + \Sigma^{(l,+)}_j} - \beta \frac{[a^{(l-1)}_i W^{(l)}_{ij}]^-}{\varepsilon + \Sigma^{(l,-)}_j}\right) \Gamma^{(l)}_j,
\end{aligned} \tag{97}$$

where the third equality regroups the two square brackets as $+\Gamma^{(l)}_j$ and $-\Gamma^{(l)}_j$ respectively, and the fourth substitutes (95) together with $a^{(l-1)}_i W^{(l,\kappa)}_{ij} = [a^{(l-1)}_i W^{(l)}_{ij}]^\kappa$ for non-negative $a^{(l-1)}_i$. Multiplying both sides by $R^{(L)}_u$ and using the inductive hypothesis $R^{(L)}_u \cdot \Gamma^{(l)}_j = R^{(l)}_j$,

$$R^{(L)}_u \cdot \Gamma^{(l-1)}_i \;=\; \sum_j \left(\alpha \frac{[a^{(l-1)}_i W^{(l)}_{ij}]^+}{\varepsilon + \Sigma^{(l,+)}_j} - \beta \frac{[a^{(l-1)}_i W^{(l)}_{ij}]^-}{\varepsilon + \Sigma^{(l,-)}_j}\right) R^{(l)}_j \;=\; R^{(l-1)}_i. \tag{98}$$

This closes the induction and proves $R^{(L)}_u \cdot \Gamma^{(l)}_j = R^{(l)}_j$ for every layer l and neuron j.

Input layer ($l = 1$): The same calculation applies with the fused first-layer masses $[x_i W^{(1)}_{ij}]^\kappa$ in place of $a^{(l-1)}_i W^{(l,\kappa)}_{ij}$, yielding the identical formula with $a^{(0)} = x$.

Addition node: Let $z = x_{\mathrm{op}} + y_{\mathrm{op}}$ be an addition node. By Appendix C.1.1, from $s^{\mathrm{add}}_{z,p}$ the Oracle transitions to $s_{x_{\mathrm{op}},p}$ and $s_{y_{\mathrm{op}},p}$ with probability $\frac{1}{2}$ each, retaining player label p. Under the trajectory-discount convention (88)–(89), addition Oracle transitions carry trajectory-discount factor 1. Hence

$$\Gamma_x\big(s^{\mathrm{add}}_{z,p} \to s_{x_{\mathrm{op}},p}\big) \;=\; \tfrac{1}{2}\,\Gamma_x\big(s^{\mathrm{add}}_{z,p}\big), \qquad \Gamma_x\big(s^{\mathrm{add}}_{z,p} \to s_{y_{\mathrm{op}},p}\big) \;=\; \tfrac{1}{2}\,\Gamma_x\big(s^{\mathrm{add}}_{z,p}\big), \tag{99}$$

proving Item 2 of Theorem 8 for both player labels.

Max-pooling node: Let $z = \max\{x_1, \ldots, x_m\}$ with winner $k^\star$ and pooling state $s^{\max}_{z,p}$. By Appendix C.1.1, the active player has the unique value-maximizing action $x_{k^\star}$ at $s^{\max}_{z,p}$, so the transition to $s_{x_{k^\star},p}$ is deterministic, retains player label p, and (88) assigns unit trajectory factor. Therefore $\Gamma_x(s_{x_{k^\star},p}) = \Gamma_x(s^{\max}_{z,p})$ and $\Gamma_x(s_{x_k,p}) = 0$ for $k \neq k^\star$, proving Item 3. □

C.3 Temperature τ as a Controlled Policy Deformation

Theorem 7(i) establishes an SPNE for every $\tau > 0$ when the entropy weight τ is used throughout the backward recursion, but exact $\alpha\beta$-LRP recovery (Theorem 8) holds only at $\tau = 1$. In practice, we deform τ only *locally* at each linear subgame's in-round Gibbs readout,

$$\pi^{(\tau)\star}_{j,\kappa}(i) \propto \exp\big(Q_q(s^{(l,lin)}_{j,q,\kappa}, i)/\tau\big), \tag{100}$$

while keeping the game values $V(s^{(l-1,act)}_{i,q})$ at their $\tau{=}1$ equilibrium—i.e. at the network activations recovered by Theorem 7(ii). The players are therefore *unaware* of the τ-deformation (a *myopic* policy): their decisions rest on the Bellman equations at temperature 1 (the network forward pass), and only the local routing readout is perturbed. Under non-negative predecessor activations this gives $\pi^{(\tau)\star}_{j,\kappa}(i) \propto (a^{(l-1)}_i W^{(l,\kappa)}_{ij})^{1/\tau}$, so $\tau < 1$ sharpens and $\tau > 1$ flattens the route distribution. This controlled deformation is used in the experiments; it carries no exactness claim at $\tau \neq 1$. The implementation raises activations, weights and bias-node masses to the power $1/\tau$ but keeps the outside-option weight at ε rather than $\varepsilon^{1/\tau}$. On ResNet-50 it passes relevance through the global average pool uniformly (not in proportion to the activations) and merges batch normalization into the preceding convolution with an ε-stabilized rule, so the ResNet-50 anchor is $\alpha\beta$-LRP up to these two operations; VGG-16 (whose average pool is the identity at 224×224) and ViT-B/16 are not affected.

D EXTENSION: VISION TRANSFORMER ATTENTION

This appendix develops the full state-space formulation for attention modules in the RG. The development proceeds in two stages. §D.1 extends the backward walk of Theorem 8 through the attention block by treating the softmax row $A_{t,\cdot}$ as a fixed reference policy on the keys, derives the resulting Value-routing equilibrium from the Legendre–Fenchel transform of the KL divergence, and proves $\alpha\beta$-LRP recovery for the conditioned effective map $X \mapsto O$ in the occupation-measure notation Γ of Theorem 8. §D.2 then identifies the fixed reference $A_{t,\cdot}$ with the policy of an auxiliary entropy-regularized QK oracle player, closing the loop with the rest of the game-theoretic framework. The risk-averse attention variant is in §D.3; forward-pass detachments and gradient-hook conventions used by all ViT experiments are collected in §D.4; the equivariant spatial averaging is in §D.5.

D.1 Attention as fixed-reference KL Value routing

D.1.1 Target rule: $\alpha\beta$-LRP with detached logits

We first state the rule that the Routing-Game state-space construction below is designed to recover. We focus on a single attention head; the multi-head extension carries an additional head index h and the recovery is per-head. Note that ViT-B/16 implements the Value projection as a single $D \times D$ linear $v = XW_v$ followed by a reshape into n_h heads (the per-head matrix $W_v^{(h)} \in \mathbb{R}^{D \times d_h}$ is the head-h column block of the full W_v), so one input embedding $X_{k,e}$ feeds Value entries of every head and the per-head R^X contributions sum at $X_{k,e}$ to give the total. The per-head description below applies unchanged. We omit the ε stabilizer inside the attention block — the surrounding ReLU recursion of §C.2 carries its own ε on every linear-routing state and that mechanism transfers to the $Av{+}W_v$ composite without modification, but tracking it through the tensor indexing here only obscures the structure.

Forward pass. The Value projection is applied per token: for each $k \in \{1, \dots, S\}$,

$$v_{k,d} \;=\; \sum_e W_{v,e,d}\, X_{k,e} + b_{v,d}, \qquad d \in \{1, \dots, d_h\}, \tag{101}$$

where the same matrix $W_v \in \mathbb{R}^{D \times d_h}$ and bias $b_v \in \mathbb{R}^{d_h}$ are applied independently to every token. The detached attention matrix $A \in \mathbb{R}_{\geq 0}^{S \times S}$ (with $A_{t,k} \geq 0$ and $\sum_k A_{t,k} = 1$ from softmax) then mixes Values across tokens *independently per feature dimension* d: for each output token $t \in \{1, \dots, S\}$ and feature d,

$$O_{t,d} \;=\; \sum_k A_{t,k}\, v_{k,d} \;=\; \sum_{k,e} A_{t,k}\, W_{v,e,d}\, X_{k,e} + b_{v,d}. \tag{102}$$

Composite linear map and κ-stream split. Holding A, the LayerNorm statistics and the GeLU gates at their forward values (§D.4) is a modeling assumption about the explanation question—given the realized attention policy, through which Value routes does relevance flow?—and not a theorem about relevance through Q and K. Our attention extension is therefore a Value-only attribution conditioned on the realized attention matrix, resembling the conservative propagation of Ali et al. (2022) (CP-LRP), which holds the attention matrix constant, and DAVE's conditioned operators (Wróbel et al., 2026). It differs from AttnLRP (Achtibat et al., 2024), which also propagates relevance through the attention-generating operations (the softmax and the bilinear QK product) and so attributes the Query and Key paths. With A so conditioned, the attention block reduces to the affine *conditioned effective map* $X \mapsto O$ with composite weights $W^{\text{eff}}_{(k,e),\,(t,d)} = A_{t,k}\, W_{v,e,d}$ and bias $b_{v,d}$. Since $A_{t,k} \geq 0$, the sign of the composite weight is the sign of $W_{v,e,d}$:

$$\big[\, W^{\text{eff}}_{(k,e),\,(t,d)} \,\big]^{\kappa} \;=\; A_{t,k}\, W^{\kappa}_{v,e,d}, \qquad \kappa \in \{+, -\}. \tag{103}$$

For signed inputs X (LayerNorm outputs), the per-token κ-stream split of the Value tensor (as in the fused input layer of the RG, with the bias as a bias node) and the per-output-feature stream sum are

$$\tilde{v}^{\kappa}_{k,d} \;\coloneqq\; \sum_e [X_{k,e} W_{v,e,d}]^{\kappa} + [b_{v,d}]^{\kappa}, \qquad Z^{\kappa}_{t,d} \;\coloneqq\; \sum_k A_{t,k}\, \tilde{v}^{\kappa}_{k,d} \;=\; \sum_{k,e} A_{t,k}\, [X_{k,e} W_{v,e,d}]^{\kappa} + [b_{v,d}]^{\kappa}. \tag{104}$$

Note that $\tilde{v}^{\kappa}_{k,d}$ is the κ-stream split inherited from W_v and the sign of X (not the standard scalar positive part $[v_{k,d}]^{+} = \max(0, v_{k,d})$); the two coincide only when $v_{k,d}$ has a single-signed predecessor decomposition.

Target rule. Let $R^O_{t,d}$ denote the relevance arriving at the output entry $O_{t,d}$ from the surrounding LRP recursion. We define the target rule as the $\alpha\beta$-LRP rule (1) on the composite map $X \mapsto O$, written in two stages that carry the κ-stream split $\tilde{v}^\kappa_{k,d}$ inherited from W_v via (103) through the Value entries. It is not sequential (layer-wise) $\alpha\beta$-LRP on $X \to v \to O$, which splits by the sign of the netted Value entry $v_{k,d}$ and applies (α, β) at both stages; the two rules differ whenever a Value entry has mixed-sign predecessors. The sequential variant is the alternative that this paper does not use: the implementation (Appendix D.4) and every ViT result reported here apply the composite rule stated below, and the numerical identity between the implementation and (107) is checked by the tests of Appendix I.8.

AV mixing $O \to v$. Applied to $O_{t,d} = \sum_k A_{t,k}\, v_{k,d}$, the rule gives the relevance at the Value entry $v_{k,d}$ as the weighted difference of two per-stream contributions:

$$R^{\mathrm{val}}_{k,d} \;=\; \alpha\, R^{\mathrm{val},+}_{k,d} \;-\; \beta\, R^{\mathrm{val},-}_{k,d}, \qquad R^{\mathrm{val},\kappa}_{k,d} \;\coloneqq\; \sum_t \frac{A_{t,k}\, \tilde{v}^\kappa_{k,d}}{Z^\kappa_{t,d}}\, R^O_{t,d}, \quad \kappa \in \{+,-\}. \tag{105}$$

Value projection $v \to X$. Applied to $v_{k,d} = \sum_e W_{v,e,d}\, X_{k,e} + b_{v,d}$, the rule routes each κ-stream contribution $R^{\mathrm{val},\kappa}_{k,d}$ to the input dim e via the matching κ-stream ratio $[X_{k,e}W_{v,e,d}]^\kappa / \tilde{v}^\kappa_{k,d}$ (the bias share is absorbed):

$$R^X_{k,e} \;=\; \sum_d \left(\alpha\, \frac{[X_{k,e}W_{v,e,d}]^+}{\tilde{v}^+_{k,d}}\, R^{\mathrm{val},+}_{k,d} \;-\; \beta\, \frac{[X_{k,e}W_{v,e,d}]^-}{\tilde{v}^-_{k,d}}\, R^{\mathrm{val},-}_{k,d} \right). \tag{106}$$

Aggregate. Substituting (105) into (106) and canceling $\tilde{v}^\kappa_{k,d}$ in numerator and denominator expresses $R^X_{k,e}$ directly in terms of $R^O_{t,d}$:

$$R^X_{k,e} \;=\; \sum_{t,d} \left(\alpha\, \frac{A_{t,k}\, [X_{k,e}W_{v,e,d}]^+}{Z^+_{t,d}} \;-\; \beta\, \frac{A_{t,k}\, [X_{k,e}W_{v,e,d}]^-}{Z^-_{t,d}} \right) R^O_{t,d}, \tag{107}$$

which coincides with what the standard $\alpha\beta$-LRP rule produces directly on the conditioned effective map $X \mapsto O$ with effective weight $W^{\mathrm{eff}}_{(k,e),(t,d)} = A_{t,k}\, W_{v,e,d}$ and bias $b_{v,d}$.

D.1.2 State space

We embed the attention block into the routing-graph state space of §C using the tuple-superscript notation $s_\bullet^{(\mathrm{att},\,\mathrm{Side},\,\mathrm{type})}$ throughout, where $\mathrm{Side} \in \{\mathrm{O}, \mathrm{val}, \mathrm{X}\}$ identifies the activation slot inside the attention block and $\mathrm{type} \in \{\mathrm{act}, \mathrm{lin}\}$ matches state types of §C. The state indices encode the tensor structure of the AV mixing and the Value projection, with $t, k \in \{1, \dots, S\}$ ranging over output/key tokens (sequence axis), $d \in \{1, \dots, d_h\}$ over the within-head feature axis of the focused head (i.e. the head index h is fixed and d runs over that head's d_h feature dimensions), and $e \in \{1, \dots, D\}$ over the per-token input embedding dimensions.

States for the composite $Av{+}W_v$ block. The explanatory walk through the attention block visits three state types beyond the surrounding ReLU-network's; all are parametrized by the player label in turn.

- **Output activation states** $s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{act})}_{t,d,p}$ at the head's output entry $O_{t,d}$, indexed by output token $t \in \{1, \dots, S\}$, within-head feature dim $d \in \{1, \dots, d_h\}$, and player $p \in \{+, -\}$. These are the entry points of the explanatory walk: the surrounding linear-layer recursion of §C.2 delivers occupation mass to $\{s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{act})}_{t,d,p}\}_{t,d,p}$ from the layer behind the head.

- **Output sign-branch (linear) states** $s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{lin})}_{(t,d),p,\kappa}$, indexed by (t, d, p, κ). These mirror the linear routing states $s^{(l,\mathrm{lin})}_{j,q,\kappa}$ of Definition 7 for the AV mixing at output entry (t, d): *this is where the QK logits enter the routing* via the reference $\bar{\pi}_t = A_{t,\cdot}$. The active player picks a key token k by a mixed action $\pi^\star_{t,d,\kappa} \in \Delta(\{1, \dots, S\})$ that combines the AV reference $\bar{\pi}_t$ with the κ-stream Value mass $\tilde{v}^\kappa_{k,d}$.

- **Value-projection linear states** $s^{(\mathrm{att},\,\mathrm{val},\,\mathrm{lin})}_{(k,d),p,\kappa}$, indexed by key token k, within-head feature dim d, player p, and sign κ. These are the linear-routing states of the per-token Value projection: the player picks an input dimension e (or the bias node) and the game value satisfies $V_p\big(s^{(\mathrm{att},\,\mathrm{val},\,\mathrm{lin})}_{(k,d),p,\kappa}\big) = \ell(\tilde{v}^\kappa_{k,d})$ with $\tilde{v}^\kappa_{k,d}$ from (104).

- **Input activation states** $s^{(\mathrm{att},\,\mathrm{X},\,\mathrm{act})}_{(k,e),p}$ at the input embedding entry $X_{k,e}$, indexed by token–embedding-dim pair (k,e) and player p. X lives upstream of the Value projection and carries the input-embedding axis e; the within-head feature axis d enters first at the Value projection's output downstream. They are the exit points of the attention block: the surrounding network's recursion resumes from the per-player occupation on $\{s^{(\mathrm{att},\,\mathrm{X},\,\mathrm{act})}_{(k,e),p}\}_{(k,e),p}$.

Transitions. The trajectory through the attention block has three steps; all transitions inside the block carry trajectory discount 1 except the sign-oracle split, which uses the same $\{2\alpha, 2\beta\}$ as a ReLU activation state of Definition 7. There is only *one* sign-oracle split per output entry (t,d), since $A_{t,k} \geq 0$ absorbs the AV mixing into the W_v sign split.

1. *Sign-oracle split (output activation → output sign-branch, fixed (t,d)).* At $s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{act})}_{t,d,p}$ an unobserved Oracle transitions uniformly to $s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{lin})}_{(t,d),p,+}$ (player p retains the turn, trajectory discount 2α) or to $s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{lin})}_{(t,d),p',-}$ (turn switches to opponent p', discount 2β), each with probability $\frac{1}{2}$. The feature index d is preserved.

2. *Value-routing policy (output sign-branch → Value-projection linear).* At $s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{lin})}_{(t,d),p,\kappa}$ the active player picks a key token k by the mixed action

$$\pi^{\star}_{t,d,\kappa}(k) \;=\; \frac{A_{t,k}\,\tilde{v}^{\kappa}_{k,d}}{Z^{\kappa}_{t,d}}, \tag{108}$$

   derived in §D.1.4 as the equilibrium of a KL-regularized log-payoff problem against the reference $\bar{\pi}_t = A_{t,\cdot}$. The trajectory transitions to $s^{(\mathrm{att},\,\mathrm{val},\,\mathrm{lin})}_{(k,d),p,\kappa}$ *with the player label, the sign κ, and the within-head feature dim d preserved*; the only thing changing is the sequence-axis index from output token t to key token k. The trajectory-discount on this transition is 1.

3. *Value-projection routing (Value-projection linear → input activation).* At $s^{(\mathrm{att},\,\mathrm{val},\,\mathrm{lin})}_{(k,d),p,\kappa}$ the active player picks an input dim e by the standard linear-state Gibbs policy of Definition 7 on the κ-stream masses $[X_{k,e}W_{v,e,d}]^{\kappa}$ (the bias node receives $[b_{v,d}]^{\kappa}/\tilde{v}^{\kappa}_{k,d}$ and is absorbed):

$$\pi^{\star}_{\mathrm{val},k,d,\kappa}(e) \;=\; \frac{[X_{k,e}W_{v,e,d}]^{\kappa}}{\tilde{v}^{\kappa}_{k,d}}. \tag{109}$$

   The trajectory transitions to $s^{(\mathrm{att},\,\mathrm{X},\,\mathrm{act})}_{(k,e),p}$ with the player label preserved and trajectory-discount factor 1.

The discounted occupation measure $\Gamma_x(\,\cdot\,)$ of §C.2 is evaluated at all four state types along the explanatory walk. The recovery theorem of §D.1.5 then compares the per-input-neuron player-difference of Γ_x at the input activation states with the target rule (107).

D.1.3 Legendre–Fenchel transform of the KL divergence

The Value-routing equilibrium below is a direct corollary of the convex-conjugate identity for the KL divergence; we record it as a stand-alone Lemma to make the inheritance explicit.

Lemma 2 (Legendre–Fenchel transform of KL divergence)**.** *Let $\mathcal{X}$ be a finite set, $\bar{\pi} \in \Delta(\mathcal{X})$ a reference distribution with full support on $\mathcal{X}$, and $r : \mathcal{X} \to \mathbb{R}$ a real-valued payoff. The KL-regularized linear maximization*

$$F(r;\bar{\pi}) \;\coloneqq\; \max_{\pi\in\Delta(\mathcal{X})} \Big\{ \mathbb{E}_{\pi}[r] \;-\; \mathrm{KL}(\pi \,\|\, \bar{\pi}) \Big\} \tag{110}$$

has the following optimizer and optimum:

$$\pi^{\star}_{r}(x) \;=\; \frac{\bar{\pi}(x)\,\exp\big(r(x)\big)}{Z(r;\bar{\pi})}, \qquad F(r;\bar{\pi}) \;=\; \log Z(r;\bar{\pi}) \;=\; \log \sum_{x\in\mathcal{X}} \bar{\pi}(x)\,\exp\big(r(x)\big). \tag{111}$$

Two specializations are central below. When $r(x) = \ell(u(x))$ for some non-negative $u : \mathcal{X} \to \mathbb{R}_{\geq 0}$ (so $\exp(r(x)) = u(x)$ with the convention $\exp(-\infty) := 0$), the value collapses to the $\bar{\pi}$-weighted linear sum and the optimizer becomes the $\bar{\pi}$-tilted normalization of u:

$$F\big(\ell(u);\, \bar{\pi}\big) \;=\; \ell\Big(\sum_x \bar{\pi}(x)\, u(x)\Big), \qquad \pi^\star_{\ell(u)}(x) \;=\; \frac{\bar{\pi}(x)\, u(x)}{\sum_{x'} \bar{\pi}(x')\, u(x')}. \tag{112}$$

When $\bar{\pi}$ is uniform on $\mathcal{X}$, $\mathrm{KL}(\pi \,\|\, \mathrm{Unif}_{\mathcal{X}}) = \log|\mathcal{X}| - \mathcal{H}(\pi)$, so subtracting the KL penalty in (110) flips into a $+\mathcal{H}$ entropy bonus:

$$\mathbb{E}_\pi[r] \;-\; \mathrm{KL}\big(\pi \,\|\, \mathrm{Unif}_{\mathcal{X}}\big) \;=\; \mathbb{E}_\pi[r] \;+\; \mathcal{H}(\pi) \;-\; \log|\mathcal{X}|. \tag{113}$$

Up to the additive constant $-\log|\mathcal{X}|$, (110) therefore reduces to the entropy-regularized maximization of (62), with value $\log\sum_x \exp(r(x))$ and optimizer $\pi^\star(x) \propto \exp(r(x))$.

Proof. This is the Gibbs variational principle (Boyd and Vandenberghe, 2004): for every π, $\mathbb{E}_\pi[r] - \mathrm{KL}(\pi \,\|\, \bar{\pi}) = \log Z(r;\bar{\pi}) - \mathrm{KL}(\pi \,\|\, \pi^\star_r)$, which is maximal, with value $\log Z(r;\bar{\pi})$, exactly at $\pi = \pi^\star_r$. □

D.1.4 Value-routing equilibrium

Fix an output entry (t, d), a sign $\kappa \in \{+, -\}$, and the reference $\bar{\pi}_t = A_{t,\cdot}$ from §D.1.2 (the *same* $\bar{\pi}_t$ for every d, since the AV mixing reuses one row across all feature dims). At the output sign-branch state $s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{lin})}_{(t,d),p,\kappa}$ the active player p solves the KL-regularized log-payoff maximization

$$V_p\big(s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{lin})}_{(t,d),p,\kappa}\big) \;\coloneqq\; \max_{\pi \in \Delta(\{1,\ldots,S\})} \left\{ \sum_{k=1}^{S} \pi(k)\, \ell\big(\tilde{v}^\kappa_{k,d}\big) \;-\; \mathrm{KL}\big(\pi \,\|\, \bar{\pi}_t\big) \right\}, \tag{114}$$

where $\tilde{v}^\kappa_{k,d} \geq 0$ is the κ-stream split of the Value entry $v_{k,d}$ from (104) and ℓ is the extended logarithm of §C. The payoff $\ell(\tilde{v}^\kappa_{k,d})$ on action k is the active-player value of the predecessor Value-projection linear state, $V_p(s^{(\mathrm{att},\,\mathrm{val},\,\mathrm{lin})}_{(k,d),p,\kappa}) = \ell(\tilde{v}^\kappa_{k,d})$, so combining with the AV log-weight $\ell(A_{t,k})$ gives composite payoff $\ell(A_{t,k}\, \tilde{v}^\kappa_{k,d})$ at the (t, d)-output. Define the per-output-entry $\bar{\pi}_t$-weighted Value mass

$$Z^\kappa_{t,d} \;\coloneqq\; \sum_{m=1}^{S} \bar{\pi}_t(m)\, \tilde{v}^\kappa_{m,d}. \tag{115}$$

The dependence on d enters only through $\tilde{v}^\kappa_{m,d}$; the AV reference $\bar{\pi}_t$ is shared across all feature dims.

Proposition 3 (Value-side optimal policy)**.** *If $Z^\kappa_{t,d} > 0$, the unique optimizer of* (114) *is*

$$\pi^\star_{t,d,\kappa}(k) \;=\; \frac{\bar{\pi}_t(k)\, \tilde{v}^\kappa_{k,d}}{Z^\kappa_{t,d}}, \tag{116}$$

with active-player value $V_p(s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{lin})}_{(t,d),p,\kappa}) = \ell(Z^\kappa_{t,d})$ and inactive-player value $V_{p'}(s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{lin})}_{(t,d),p,\kappa}) = -\infty$ as in Definition 7. If $Z^\kappa_{t,d} = 0$, no key contributes mass on stream κ and $V_p(s^{(\mathrm{att},\,\mathrm{O},\,\mathrm{lin})}_{(t,d),p,\kappa}) = -\infty$.

Proof. Apply the logit specialization (112) of Lemma 2 with $\mathcal{X} = \{1, \ldots, S\}$, $\bar{\pi} = \bar{\pi}_t$, and $u(k) = \tilde{v}^\kappa_{k,d}$, so that the payoff $r(k) = \ell(\tilde{v}^\kappa_{k,d})$ has $\exp r(k) = \tilde{v}^\kappa_{k,d}$. Equation (112) immediately gives the optimizer $\pi^\star_{t,d,\kappa}$ of (116) and value $\ell(Z^\kappa_{t,d})$ when $Z^\kappa_{t,d} > 0$. The inactive-player value $-\infty$ is the Routing-Game convention of Definition 7, inherited from the $r = q'$ branch of (65). If $Z^\kappa_{t,d} = 0$ then $\bar{\pi}_t(k)\, \tilde{v}^\kappa_{k,d} = 0$ for every k and the supremum of (114) over the simplex is $-\infty$, consistent with $\ell(0) = -\infty$. □

Specialization to the softmax reference. Setting $\bar{\pi}_t = A_{t,\cdot}$ in (116) yields the closed-form per-(t, d, κ) policies used in the RG's backward walk through the attention block,

$$\pi^\star_{t,d,+}(k) \;=\; \frac{A_{t,k}\, \tilde{v}^+_{k,d}}{Z^+_{t,d}}, \qquad \pi^\star_{t,d,-}(k) \;=\; \frac{A_{t,k}\, \tilde{v}^-_{k,d}}{Z^-_{t,d}}, \tag{117}$$

with $Z^\kappa_{t,d} = \sum_m A_{t,m}\, \tilde{v}^\kappa_{m,d}$ from (115). The numerical specification $\bar{\pi}_t = A_{t,\cdot}$ does not require the oracle interpretation of §D.2; it is enough that the cached softmax row is treated as a fixed input to (114).

D.1.5 $\alpha\beta$-LRP recovery for the attention block

We now show that, under the trajectory through the four state types of §D.1.2 and the trajectory discount of (88), the player-difference of Γ_x at the *input activation states* (the predecessors of the Value-projection linear states, where the W_v weights actually fire) recovers exactly the target rule (107) of §D.1.1. We keep the game side and the attribution side strictly separate:

- the *game-side* object is the discounted occupation measure $\Gamma_x(\,\cdot\,)$ of §C.2 evaluated at the four attention-block state types of §D.1.2. Following the per-neuron difference convention (87) of the surrounding Routing-Game recursion, we form the per-output-entry difference $\Gamma^O_{t,d}$ at the entry point (defined in (120) below) and the per-input-neuron difference at the exit

$$\Gamma^X_{k,e} \;\coloneqq\; \Gamma_x\big(s^{(\mathrm{att,\,X,\,act})}_{(k,e),+}\big) - \Gamma_x\big(s^{(\mathrm{att,\,X,\,act})}_{(k,e),-}\big). \tag{118}$$

- the *attribution-side* object is the standard $\alpha\beta$-LRP relevance vector $R^X_{k,e}$ given by the input-level target rule (107), seeded by $R^O_{t,d}$ from the surrounding linear-layer recursion above the head.

Theorem 9 ($\alpha\beta$-LRP recovery for attention)**.** *For signed inputs X, under the SPNE of §D.1.2 with reference $\bar\pi_t = A_{t,\cdot}$ and the trajectory discount of* (88) *restricted to the (sole) sign-oracle split at $s^{(\mathrm{att,\,O,\,act})}$, the per-input-neuron player-difference $\Gamma^X_{k,e}$ from* (118) *satisfies*

$$\Gamma^X_{k,e} \;=\; \sum_{t,d}\left(\alpha\,\frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^+}{Z^+_{t,d}} \;-\; \beta\,\frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^-}{Z^-_{t,d}}\right)\Gamma^O_{t,d}, \tag{119}$$

where the per-output-entry input difference is

$$\Gamma^O_{t,d} \;\coloneqq\; \Gamma_x\big(s^{(\mathrm{att,\,O,\,act})}_{t,d,+}\big) \;-\; \Gamma_x\big(s^{(\mathrm{att,\,O,\,act})}_{t,d,-}\big), \tag{120}$$

with $\Gamma^O_{t,d} = R^O_{t,d}$ by inductive hypothesis from the surrounding recursion, for the seed $R^{(L)}_u = 1$ (for the seed $f(x)$ every relevance equals $f(x)$ times the corresponding Γ, as in Theorem 8). The right-hand side is exactly the target rule (107)*, hence $\Gamma^X_{k,e} = R^X_{k,e}$. Aggregating over e and the bias node at fixed (k,d) collects the relevance routed through the Value entry $v_{k,d}$:*

$$\Gamma^{\mathrm{val}}_{k,d} \;=\; \sum_{t}\left(\alpha\,\frac{A_{t,k}\,\tilde v^+_{k,d}}{Z^+_{t,d}} \;-\; \beta\,\frac{A_{t,k}\,\tilde v^-_{k,d}}{Z^-_{t,d}}\right)\Gamma^O_{t,d} \;=\; R^{\mathrm{val}}_{k,d}, \tag{121}$$

recovering the per-Value-entry formula (105)*.*

Proof. The statement is block-local: by hypothesis the surrounding recursion delivers the output-side per-entry differences $\Gamma^O_{t,d} = R^O_{t,d}$ at every output position (t,d).

Per-state recursion at the input activation states. Fix an input embedding (k,e) and a player label $r \in \{+,-\}$. A trajectory reaches $s^{(\mathrm{att,\,X,\,act})}_{(k,e),r}$ from an output activation $s^{(\mathrm{att,\,O,\,act})}_{t,d,p}$ in three stages, with all transitions inside the attention block carrying their discounts and player-label switches as in the kernel of §D.1.2. The path goes through one specific within-head feature dim $d \in \{1,\dots,d_h\}$ (an index along the focused head's d_h-wide feature axis): the sign-oracle is per (t,d), the Value-routing policy $\pi^\star_{t,d,\kappa}$ depends on d via $\tilde v^\kappa_{k,d}$, and the Value-projection routing $\pi^\star_{\mathrm{val},k,d,\kappa}$ also depends on d. The d index is dropped at the final transition into X, where the input embedding $X_{k,e}$ has indices (k,e) and the within-head feature axis d enters downstream at the Value projection's output.

- $\kappa = +$ *branch (per (t,d)).* Sign-oracle takes $s^{(\mathrm{O,act})}_{t,d,p} \to s^{(\mathrm{O,lin})}_{(t,d),p,+}$ with probability $\frac12$ and trajectory discount 2α (player retained). The Value-routing policy $\pi^\star_{t,d,+}(k)$ takes the trajectory to $s^{(\mathrm{val,lin})}_{(k,d),p,+}$ (player, sign, d preserved, discount 1). The Value-projection routing $\pi^\star_{\mathrm{val},k,d,+}(e)$ takes the trajectory to $s^{(\mathrm{X,act})}_{(k,e),p}$ (player

preserved, d dropped, discount 1). The composite per-(t,d) contribution to $\Gamma_x(s^{(\mathrm{X,act})}_{(k,e),r})$ is

$$\tfrac{1}{2}(2\alpha)\,\pi^\star_{t,d,+}(k)\,\pi^\star_{\mathrm{val},k,d,+}(e)\,\Gamma_x\big(s^{(\mathrm{O,act})}_{t,d,r}\big) \;=\; \alpha\,\frac{A_{t,k}\,\tilde v^+_{k,d}}{Z^+_{t,d}}\,\frac{[X_{k,e}W_{v,e,d}]^+}{\tilde v^+_{k,d}}\,\Gamma_x\big(s^{(\mathrm{O,act})}_{t,d,r}\big) \tag{122}$$

$$\;=\; \alpha\,\frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^+}{Z^+_{t,d}}\,\Gamma_x\big(s^{(\mathrm{O,act})}_{t,d,r}\big), \tag{123}$$

where the $\tilde v^+_{k,d}$ in numerator and denominator cancel.

- $\kappa = -$ *branch (per (t,d)).* Sign-oracle takes $s^{(\mathrm{O,act})}_{t,d,p} \to s^{(\mathrm{O,lin})}_{(t,d),p',-}$ with probability $\frac{1}{2}$, discount 2β, turn switched to opponent p'; the rest of the path preserves p', ending at $s^{(\mathrm{X,act})}_{(k,e),p'}$. The contribution to $\Gamma_x(s^{(\mathrm{X,act})}_{(k,e),r})$ comes from sources with player $p = r'$ (so the switch yields r):

$$\tfrac{1}{2}(2\beta)\,\pi^\star_{t,d,-}(k)\,\pi^\star_{\mathrm{val},k,d,-}(e)\,\Gamma_x\big(s^{(\mathrm{O,act})}_{t,d,r'}\big) \;=\; \beta\,\frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^-}{Z^-_{t,d}}\,\Gamma_x\big(s^{(\mathrm{O,act})}_{t,d,r'}\big). \tag{124}$$

Summing over (t,d) and over the two branches,

$$\Gamma_x\big(s^{(\mathrm{att,\,X,\,act})}_{(k,e),r}\big) \;=\; \sum_{t,d}\Big[\alpha\,\frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^+}{Z^+_{t,d}}\,\Gamma_x\big(s^{(\mathrm{O,act})}_{t,d,r}\big) \;+\; \beta\,\frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^-}{Z^-_{t,d}}\,\Gamma_x\big(s^{(\mathrm{O,act})}_{t,d,r'}\big)\Big], \tag{125}$$

where the $\frac{1}{2}$ from the sign-oracle splits cancels against the 2 in $2\alpha, 2\beta$.

Player difference. Differencing (125) for $r{=}+$ minus $r{=}-$ uses $\Gamma_x(s^{(\mathrm{O,act})}_{t,d,+}) - \Gamma_x(s^{(\mathrm{O,act})}_{t,d,-}) = \Gamma^O_{t,d}$ in the $+$ branch (player kept) and the opposite $-\Gamma^O_{t,d}$ in the $-$ branch (player switched):

$$\Gamma^X_{k,e} \;=\; \alpha\sum_{t,d}\frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^+}{Z^+_{t,d}}\,\Gamma^O_{t,d} \;-\; \beta\sum_{t,d}\frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^-}{Z^-_{t,d}}\,\Gamma^O_{t,d}, \tag{126}$$

which is (119).

Target-rule identification. The right-hand side matches (107) verbatim. Since $\Gamma^O_{t,d} = R^O_{t,d}$, $\Gamma^X_{k,e} = R^X_{k,e}$. Aggregating over e and the bias node at fixed (k,d) uses $\sum_e [X_{k,e}W_{v,e,d}]^\kappa + [b_{v,d}]^\kappa = \tilde v^\kappa_{k,d}$, giving (121), which matches (105). □

D.2 The reference as the policy of a QK oracle player

§D.1 treated the softmax row $A_{t,\cdot}$ as a fixed reference policy. It is itself the entropy-regularized best response of an auxiliary *oracle* player whose payoff is the detached query–key logit $e_{tk} = Q_t K_k^\top/\sqrt{d_h}$: by (62) at temperature 1,

$$A_{t,\cdot} \;=\; \mathrm{argmax}_{\pi\in\Delta(\{1,\dots,S\})}\Big\{\textstyle\sum_k \pi(k)\,e_{tk} + \mathcal{H}(\pi)\Big\}, \qquad \max_\pi\Big\{\textstyle\sum_k \pi(k)\,e_{tk} + \mathcal{H}(\pi)\Big\} = \log\textstyle\sum_k \exp(e_{tk}). \tag{127}$$

The oracle acts in the forward pass only and hands $\bar\pi_t = A_{t,\cdot}$ to the Value-routing player; its states carry no occupation mass, so no relevance flows through Q or K (§D.4).

D.3 Risk-Averse Attention via ADF

When ADF variance propagation is active, let s_{tk} denote the uncertainty score attached to attending from query token t to key token k (in the current implementation, $s_{tk} = \hat\sigma_k$, the square root of the feature-averaged ADF variance of token k's query projection; it depends only on the key token and is a proxy for, not the standard deviation of, e_{tk}). Risk-averse attention shifts the QK-logits in the direction of reduced uncertainty,

$$\tilde e_{tk} = e_{tk} + \lambda_{\mathrm{sm}}\,s_{tk}, \qquad \lambda_{\mathrm{sm}} \le 0, \tag{128}$$

yielding the risk-modified policy

$$\tilde{\pi}_t(k) = \frac{\exp(\tilde{e}_{tk})}{\sum_{k'} \exp(\tilde{e}_{tk'})}. \tag{129}$$

The sign convention $\lambda_{\mathrm{sm}} \leq 0$ is used throughout the paper and in all tables: more-negative values penalize high-uncertainty keys more strongly, so the shift always redirects relevance toward tokens whose QK-interactions are stable under input noise. The shift is computed in the AV-backward hook from cached pre-softmax logits, so the forward pass and model predictions remain unchanged.

The naming *softmax-shift risk aversion* is a deliberate analog of the activation-side risk aversion λ but acts in a different direction: λ is risk-averse on the value side (it routes through stable activations by lowering mean$-|\lambda|\sigma$), whereas $\lambda_{\mathrm{sm}} < 0$ exponentially tilts the routing policy, $\tilde{\pi}_t(k) \propto A_{t,k} \exp(\lambda_{\mathrm{sm}} s_{tk})$, toward *low*-uncertainty keys: $\sum_k \tilde{\pi}_t(k)\, s_{tk}$ is non-decreasing in λ_{sm}, while the entropy of $\tilde{\pi}_t$ may rise or fall. We retain the "risk-averse" label because what is being avoided is uncertainty in the routed pathway, but the reader should note that the shift acts on the routing policy, not on the player's utility curvature.

D.4 Implementation details

The forward and backward passes of the RG on ViT-B/16 share four operator-level conventions that together preserve LRP conservation through the non-linearities of the ViT and align the explanatory walk with the state-space construction of §D.1.2. We refer to this collectively as the DAVE-style *conditioned forward pass* (Wróbel et al., 2026), augmented with the QK gradient detachment that the QK oracle of §D.2 requires.

LayerNorm detachment. Standard backpropagation through $\mathrm{LN}(x) = w_{\mathrm{LN}} \odot (x - m(x))/s(x) + b_{\mathrm{LN}}$ routes gradients through the data-dependent statistics $m(x), s(x)$, which couple the feature coordinates of each token (LayerNorm acts per token; it does not mix tokens) and violate LRP conservation (Ali et al., 2022). Following the detached-operator construction of Wróbel et al. (2026), we compute m, s from a detached copy $c = x.\mathrm{detach}()$, so the backward pass sees only the pointwise map $x \mapsto w_{\mathrm{LN}} \odot (x - m(c))/s(c) + b_{\mathrm{LN}}$ and conservation is preserved.

GELU gate detachment. GELU gates are conditioned analogously: the gate is implemented as $x \cdot \Phi(c)$ with $c = x.\mathrm{detach}()$ and Φ the standard normal CDF, so the backward pass treats the gate as a pointwise scaling by the (constant) activation $\Phi(c)$.

Softmax detachment. Softmax outputs are fully detached. This matches the QK oracle interpretation of §D.2: the cached softmax row $A_{t,\cdot}$ is the equilibrium policy of an auxiliary entropy-regularized problem, supplied as a fixed input to the Value-routing player; no trajectory of the explanatory walk enters Q or K.

QK gradient detachment. The softmax and QK-matmul backward hooks return zero gradients, ensuring that the QK pathway is used *only* for computing the detached kernel $A_{t,\cdot}$ and that no relevance flows through Q or K. The QK oracle therefore supplies the KL reference, and all explanatory mass through the attention block flows through the Value tensor. The implemented Value rule is the composite rule of Theorem 9: each Value entry keeps the stream masses $\tilde{v}^{\kappa}_{k,d}$ of its Value-projection products, the attention step routes stream κ in proportion to $A_{t,k}\tilde{v}^{\kappa}_{k,d}$, and the Value projection routes stream κ only to the contributions $[X_{k,e}W_{v,e,d}]^{\kappa}$ of the same sign, so the discounts (α, β) act once per block, as in (107). This composite rule is the only attention–Value rule in the released code (option `av_rule='merged'`); the sequential variant of §D.1.1 was not used for any reported number. The identity tests of Appendix I.8 check the implementation against an independent reference of (107).

Together the four conventions realize the detached-operator semantics of Wróbel et al. (2026) for the RG backward pass, so that its explanatory mass flows only through the linear effective transformation. They implement the explanation question of §D.1.1 (Value-only attribution conditioned on the realized attention matrix, as in CP-LRP (Ali et al., 2022)) and are not the rules of AttnLRP (Achtibat et al., 2024), which propagates relevance through the softmax and the QK product as well.

D.5 Equivariant Spatial Averaging

The 16×16 patch tokenization breaks translation equivariance. Following DAVE's transform-averaging strategy (Wróbel et al., 2026), RG+Eq computes explanations on transformed inputs, maps each explanation back to

the original image coordinates, and averages the aligned maps (a Monte-Carlo Reynolds average),

$$\mathcal{M}_N(x) \;=\; \frac{1}{N}\sum_{n=1}^{N} T_n^{-1}\big[\mathcal{M}(T_n x)\big], \tag{130}$$

where $\mathcal{M}$ is the RG attribution map and the T_n are i.i.d. random rotations (uniform in $\pm 10°$), translations (up to $\pm 5\%$ of the image extent per axis) and horizontal flips (probability 0.5), with $N = 50$ and no input noise. The conditioned forward pass of §D.4 separates operator variation from the effective transformation, in the same spirit as DAVE's conditioned-operator argument (Wróbel et al., 2026). The transforms do not form a group, so the average is an optional smoothing mode without an exactness claim.

E ANALYTICAL MODES

This appendix collects the formal derivations for the analytical modes that the game framework exposes on top of the $\alpha\beta$-LRP anchor: the local temperature deformation τ on the linear-state Gibbs readout (Appendix C.3), the activation-side risk-aversion λ via ADF variance propagation, the noisy-observation gate $\tilde{\Phi}$ on the player's stop/continue comparison, and the fan-in entropy bias λ_{ent}. Each mode maps a concept in game space (focus, risk aversion) to an explicit modification of the backward propagation rule; none carries an exactness claim, and each recovers the anchor at its default ($\tau = 1$, $\lambda = 0$, $\tilde{\Phi}$ off, $\lambda_{\text{ent}} = 0$). The mode-by-mode empirical sweeps live in Appendix J.3.

E.1 Defaults, costs and order of use

Table 4 lists every mode with its default, its effect and its measured cost (median seconds per image, one A100 GPU, batch size 1; Appendix I.7). At the defaults, SG is the gradient and RG is $\alpha\beta$-LRP. A practical order of use: (i) start at the anchor; (ii) vary τ first, since it is free and moves routing between focused and diffuse maps (Table 17); (iii) add risk aversion (λ on Convolutional Neural Networks (CNNs), λ_{sm} on ViTs, with the input variance σ^2) when the maps should discount routes through high-variance units, at the price of one ADF pass; (iv) use equivariant averaging on ViTs only when token-grid artifacts matter, since it multiplies the cost by N. The validation selections of Appendix J.1 show which combination the protocol picks per model and criterion.

Table 4: Modes: default (anchor) value, effect and measured cost per image.

Mode	Default	Effect	Cost (s/image)
(α, β), ε	$(2, 1)$, 0.5	LRP stream weights and stabilizer (exact)	none
τ	1	sharper ($\tau < 1$) or more diffuse ($\tau > 1$) routing	none (VGG-16 0.035 $\rightarrow$ 0.037, ResNet-50 0.089 $\rightarrow$ 0.094)
λ, σ^2	λ=0	attenuates routes through high-variance units	one ADF pass (ResNet-50 0.089 $\rightarrow$ 0.140)
λ_{sm}, σ^2	0	downweights high-variance keys in attention routing	one ADF pass (ViT-B/16 0.097 $\rightarrow$ 0.172)
$\tilde{\Phi}$	off	smooth stop/continue gate	needs the ADF σ (one ADF pass)
N	off	averages over input transformations (token-grid artifacts)	$\times N$ (ViT-B/16, N=50: 4.93; with λ_{sm}: 8.66)

E.2 Activation-Side Risk Aversion λ via ADF Variance Propagation

The activation-side risk-aversion mode replaces the deterministic activation $a_i^{(l)} = \text{ReLU}(z_i^{(l)})$ that the player observes by a mean–variance confidence bound $V_\lambda = \mu + \lambda\sigma$ ($\lambda \leq 0$) computed from an ADF variance propagation pass (Gast and Roth, 2018). We propagate the exact mean $\mathbb{E}[\text{ReLU}(X)]$ and the variance approximation $\sigma^2\Phi(\mu/\sigma)$ of $\text{Var}[\text{ReLU}(X)]$ for $X \sim \mathcal{N}(\mu, \sigma^2)$ (Gast and Roth, 2018).

The propagated variance is initialized at the input by the scale σ^2 at every input neuron. Negative λ shifts the player's observed game value at stopping layers (not linear layers!) by a lower-confidence amount, attenuating routes through high-variance neurons; this is the activation-side analog of confidence-bound exploration in bandit / RL settings (Auer et al., 2002; Fei et al., 2020). The softmax-shift variant λ_{sm} on the attention-logit oracle (ViT) is treated separately in Appendix D.3.

E.3 Noisy-Observation Gate $\tilde{\Phi}$

The noisy-observation mode replaces the player's pure stop/continue policy at every activation state by a *mixed* policy. At $s_{i,q}^{(l,\text{act})}$, the player continues with probability

$$\tilde{\Phi}(z_i^{(l)}) \;\coloneqq\; \Phi\big(\mu_i^{(l)}/\sigma_i^{(l)} + \lambda\big) \tag{131}$$

and stops with the complementary probability $1 - \tilde{\Phi}(z_i^{(l)})$, where Φ is the standard normal CDF and $\mu_i^{(l)}$, $\sigma_i^{(l)}$ are the ADF mean and standard deviation of $z_i^{(l)}$ (Appendix E.2). The analysis below takes $\mu_i^{(l)} = z_i^{(l)}$, $\sigma_i^{(l)} = 1$ and

$\lambda = 0$; the general case follows with $z \mapsto \mu + \lambda\sigma$ and $\mathcal{R}_\Phi \mapsto \sigma\mathcal{R}_\Phi$. The mixed policy (131) arises as the unique solution of a regularized maximization that the player can solve directly. Define the probit regularizer

$$\mathcal{R}_\Phi(\pi) \;\coloneqq\; \phi\big(\Phi^{-1}(\pi)\big), \qquad \pi \in [0,1], \tag{132}$$

where $\phi(y) = (2\pi)^{-1/2}\, e^{-y^2/2}$ is the standard normal probability density and Φ^{-1} is the quantile function (inverse of the standard normal CDF).

Lemma 3 (Probit dual of the soft gate)**.** *For every $z \in \mathbb{R}$, the regularized maximization*

$$\max_{\pi\in[0,1]} \big\{\, \pi z \;+\; \mathcal{R}_\Phi(\pi) \,\big\} \tag{133}$$

admits the unique maximizer $\pi^\star = \Phi(z)$ and optimum value

$$z\,\Phi(z) + \phi(z) \;=\; \mathbb{E}_{\eta\sim\mathcal{N}(0,1)}\big[\max(z+\eta,\,0)\big]. \tag{134}$$

Proof. Using $\phi'(y) = -y\,\phi(y)$ and the inverse-function derivative $(\Phi^{-1})'(\pi) = 1/\phi(\Phi^{-1}(\pi))$,

$$\frac{\mathrm{d}\mathcal{R}_\Phi}{\mathrm{d}\pi} \;=\; \phi'\big(\Phi^{-1}(\pi)\big)\cdot\frac{1}{\phi(\Phi^{-1}(\pi))} \;=\; -\Phi^{-1}(\pi). \tag{135}$$

The first-order condition $z - \Phi^{-1}(\pi) = 0$ yields $\pi^\star = \Phi(z)$; substituting back and using $\phi(\Phi^{-1}(\Phi(z))) = \phi(z)$ gives the left-hand side of (134). The right-hand identity is the standard Gaussian smoothing $\mathbb{E}[\max(z+\eta,0)] = z\Phi(z) + \phi(z)$. □

Remark 5 (Optimal policy under perceived payoff noise)**.** *The optimum $\pi^\star = \Phi(z_i^{(l)})$ admits an equivalent interpretation as the marginal of an optimal pure rule: the player perceives the continuation payoff at $s_{i,q}^{(l)}$ as the noisy quantity $z_i^{(l)} + \eta$ with $\eta \sim \mathcal{N}(0,1)$ independent across states, and applies the noiseless-optimal pure rule* "continue iff the perceived payoff is strictly positive" *on this perception. Marginalizing the rule over η at fixed $z_i^{(l)}$ recovers the continue probability,*

$$\mathbb{P}_{\eta\sim\mathcal{N}(0,1)}\big(z_i^{(l)} + \eta > 0\big) \;=\; \Phi(z_i^{(l)}). \tag{136}$$

The realized continuation payoff is the deterministic $z_i^{(l)}$ from the actual network forward pass: the noise η lives in the player's perception of the payoff while the environment pays out $z_i^{(l)}$ on continuation, so the expected backward Bellman comparison at $s_{i,q}^{(l)}$ is the soft mass $\Phi(z_i^{(l)}) \cdot z_i^{(l)}$ in place of the deterministic $\mathbf{1}[z_i^{(l)} > 0] \cdot z_i^{(l)}$.

The regularizer $\mathcal{R}_\Phi$ is symmetric around $\pi = \frac{1}{2}$ and peaks there at $\phi(0) = 1/\sqrt{2\pi}$ (where $\Phi^{-1}(\frac{1}{2}) = 0$), tapering to 0 at the boundary $\pi \in \{0,1\}$. It rewards *indecisiveness* — exactly the role Shannon entropy $\mathcal{H}(\pi)$ plays in the Softplus variant of §F, where the entropy bonus is the active player's surplus from being allowed to mix. The optimum $\pi^\star = \Phi(z) = \mathbb{E}_{\eta\sim\mathcal{N}(0,1)}[\,\mathbf{1}(z+\eta>0)\,]$ is the hard ReLU gate averaged over a Gaussian shift of its threshold, aligning with the original stochastic-gate motivation for GeLU (Hendrycks and Gimpel, 2016). The optimal value (134) is the Gaussian-smoothed ReLU $z\Phi(z) + \phi(z)$, whose derivative is Φ; it is not $\mathrm{GELU}(z) = z\Phi(z)$, which is not convex and hence not the conjugate of any regularizer.

On ReLU networks the forward pass still computes $\mathrm{ReLU}(z_i^{(l)})$ exactly, so the downstream game values inherited from Theorem 5 are unchanged: the player evaluates the soft gate $\tilde{\Phi}$ on the *current* layer and reuses the original network activations for everything below. There, $\tilde{\Phi}$ is an optional smoothing mode (used in the VGG-16 SG row) without an exactness claim.

GeLU networks. On a GeLU network, $\tilde{\Phi}$ multiplies the occupation-measure recursion by $\Phi(z_i^{(l)})$ at every layer, which is the chain rule of the gate-conditioned map $a_i^{(l)} = \Phi(\bar{z}_i^{(l)})\, z_i^{(l)}$ with the gate argument $\bar{z}_i^{(l)}$ held at its forward value, the conditioning used for ViTs (Appendix D.4). It is not the GeLU gradient, which needs $\mathrm{GELU}'(z) = \Phi(z) + z\phi(z)$; the substitution $\mathbf{1}[z > 0] \to \Phi(z)$ is exact only for the Gaussian-smoothed-ReLU network, as the entropy-regularized gate is for Softplus (Appendix F).

E.4 Fan-in entropy bias λ_{ent}

The fan-in entropy mode lowers the bias, i.e. the continuation payoff (SG) or bias-node payoff (RG), of neurons with diffuse fan-in: $b_j^{(l)} \mapsto b_j^{(l)} - \lambda_{\text{ent}} \tilde{\mathcal{H}}_j^{(l)}$ with $\tilde{\mathcal{H}}_j^{(l)} = -\sum_i p_{ij} \log p_{ij} / \log n_j$, $p_{ij} = |W_{ij}^{(l)}| / \gamma_j^{(l)}$ and n_j the fan-in of neuron j. It depends on the weights only (negligible cost) and changes the forward pass of the explained network.

F SOFTPLUS VARIANT OF THE STOPPING GAME

We introduce the entropy regularization that Gaubert and Vlassopoulos (2025) apply at activation states to model networks with Softplus instead of ReLU activations and, similarly to the ReLU case, adapt it to the Stopping Decomposition of Appendix B.1: the player-specific game values agree with the activations of a Softplus Decomposition, and the gradient of the Softplus network is recovered by a linear combination of occupation measures.

Modified rule: entropy-regularized activation-state policy. The technical tool throughout this section, and re-used by the Routing Game in Appendix C and the attention block of Appendix D, is the standard Legendre–Fenchel identity for Shannon entropy: for every $\theta > 0$, $n \in \mathbb{N}$, and payoff vector $r \in \mathbb{R}^n$,

$$\max_{\pi \in \Delta_n} \left\{ \sum_{i=1}^{n} \pi(i)\, r_i + \theta\, \mathcal{H}(\pi) \right\} = \theta\, \log \sum_{i=1}^{n} \exp\Big(\frac{r_i}{\theta}\Big), \qquad \pi^\star(i) \propto \exp\Big(\frac{r_i}{\theta}\Big), \tag{137}$$

with $\mathcal{H}(\pi) \coloneqq -\sum_i \pi(i) \log \pi(i)$; the unique maximizer is the Gibbs measure at inverse temperature $1/\theta$. Identity (137) is the Fenchel conjugate of the (scaled) negative entropy (Boyd and Vandenberghe, 2004).

Only the active player's stop/continue choice changes. At every activation state $s = s_{j,q}^{(l,\mathrm{act})}$, replace the binary maximization (31) by an entropy-regularized maximization over mixed policies $\pi_q(s) \in \Delta(\{\mathrm{stop}, \mathrm{cont}\})$ at temperature θ. With $\tilde{Q}_q^\pi(s, \mathrm{stop}) = \tilde{Q}_{q'}^\pi(s, \mathrm{stop}) = 0$ (cf. (27)) and $\tilde{Q}_p^\pi(s, \mathrm{cont})$ as in (28), the soft Bellman recursion for the active-player advantage is

$$V_q^\pi(s) \coloneqq \max_{\nu \in \Delta(\{\mathrm{stop},\mathrm{cont}\})} \Big\{ \textstyle\sum_a \nu(a) \big(\tilde{Q}_q^\pi(s,a) - \tilde{Q}_{q'}^\pi(s,a) \big) + \theta\, \mathcal{H}(\nu) \Big\}. \tag{138}$$

Applying (137) to the binary action set $\{\mathrm{stop}, \mathrm{cont}\}$ with payoffs 0 and $\tilde{Q}_q^\pi(s, \mathrm{cont}) - \tilde{Q}_{q'}^\pi(s, \mathrm{cont})$ immediately yields the unique optimizer as the Gibbs policy at inverse temperature $1/\theta$ and the closed-form soft advantage,

$$\pi_q^\star\big(\mathrm{cont} \,\big|\, s\big) = \mathrm{sp}'_\theta\big(\tilde{Q}_q^\pi(s, \mathrm{cont}) - \tilde{Q}_{q'}^\pi(s, \mathrm{cont})\big), \qquad V_q^\pi(s) = \mathrm{sp}_\theta\big(\tilde{Q}_q^\pi(s, \mathrm{cont}) - \tilde{Q}_{q'}^\pi(s, \mathrm{cont})\big), \tag{139}$$

with $\mathrm{sp}_\theta(z) = \theta \log(1 + e^{z/\theta})$ and $\mathrm{sp}'_\theta(z) = 1/(1 + e^{-z/\theta})$. The transition kernel (23)–(24), the payoffs r_p, the Bellman equations (26)–(28) for $\tilde{Q}_p^\pi$ and the value recursion (31) for $\tilde{V}_p^\pi$ are unchanged, except that π is no longer pure and the entropy bonus $\theta \mathcal{H}(\pi_q(s))$ is credited to the active player q.

Theorem 10 (Softplus forward equivalence)**.** *The entropy-regularized Stopping Game of* (138) *admits an SPNE whose activation-state policy is the Gibbs policy* (139)*; it is unique on networks without max-pooling nodes, since* (138) *is strictly concave (ties at max-pooling states are resolved as in Appendix A). Under this equilibrium, at every activation state* $s_{j,q}^{(l,\mathrm{act})}$*,*

$$\tilde{V}_p^{\pi^\star}\big(s_{j,q}^{(l,\mathrm{act})}\big) \;=\; a_{\mathrm{soft},j}^{(l,\,p\oplus q,\,\theta)}, \qquad V_q^{\pi^\star}\big(s_{j,q}^{(l,\mathrm{act})}\big) \;=\; \mathrm{sp}_\theta\big(z_j^{(l)}\big), \tag{140}$$

where $a_{\mathrm{soft},j}^{(l,\pm,\theta)}$ *are the Softplus-Decomposition activations of the Softplus network with activation* sp_θ*, propagated by* (8)–(9) *from* $a_{\mathrm{soft},k}^{(0,\pm,\theta)} = x_k^\pm$ *with the decomposition function*

$$f_{\mathrm{soft}}^{\theta}(z^+, z^-) \;\coloneqq\; \big(\mathrm{sp}'_\theta(z^+ - z^-)\, z^+ + \theta\, \mathcal{H}_\theta(z^+ - z^-),\;\; \mathrm{sp}'_\theta(z^+ - z^-)\, z^- \big), \tag{141}$$

and $\mathcal{H}_\theta(z) \coloneqq \mathcal{H}(\mathrm{sp}'_\theta(z))$ *the binary entropy of the Gibbs weight.*

Proof. By backward induction in l, mirroring the proof of Theorem 5, with the same base case $\tilde{V}_p^{\pi^\star}(s_{k,q}^{(0)}) = x_k^{p\oplus q} = a_{\mathrm{soft},k}^{(0,p\oplus q,\theta)}$. For $l \geq 1$, assume (140) holds at layer $l-1$, so the decomposed activations obey $a_{\mathrm{soft},i}^{(l-1,+,\theta)} - a_{\mathrm{soft},i}^{(l-1,-,\theta)} = a_i^{(l-1)}$. Substituting into (28), the Q-difference of (139) computes to $b_j^{(l)} + \sum_{i,r} W_{ij}^{(l,q\oplus r)}\, \xi_{q,r}\, a_i^{(l-1)} = z_j^{(l)}$, using $\sum_r W_{ij}^{(l,q\oplus r)} \xi_{q,r} = W_{ij}^{(l)}$ exactly as in the proof of Theorem 5. Hence, the Gibbs policy is $\pi_q^\star(\mathrm{cont} \mid \cdot) = \mathrm{sp}'_\theta(z_j^{(l)})$, and (139) gives $V_q^{\pi^\star}(s_{j,q}^{(l,\mathrm{act})}) = \mathrm{sp}_\theta(z_j^{(l)})$, the second identity of (140).

For the first identity, for the active player ($p = q$, so $p \oplus q = +$), the entropy-regularized optimization (138) attributes both the routed expectation and the entropy bonus to player q: $\tilde{V}_q^{\pi^\star}(s_{j,q}^{(l,\mathrm{act})}) = \mathrm{sp}'_\theta(z_j^{(l)})\, z_j^{(l,+)} +$

$\theta\,\mathcal{H}_\theta(z_j^{(l)}) = a_{\mathrm{soft},j}^{(l,+,\theta)}$, where $\tilde{Q}_q^{\pi^\star}(\cdot,\mathrm{cont}) = z_j^{(l,+)}$ follows from the inductive hypothesis via (8). For the opponent ($p = q'$, so $p \oplus q = -$), the entropy bonus is absent (it is the active player's surplus from being allowed to mix) and only the routed expectation contributes: $\tilde{V}_{q'}^{\pi^\star}(s_{j,q}^{(l,\mathrm{act})}) = \mathrm{sp}'_\theta(z_j^{(l)})\, z_j^{(l,-)} = a_{\mathrm{soft},j}^{(l,-,\theta)}$ by (9). The two identities together close the induction. □

Gradient recovery on Softplus networks. The discounted occupation measure of Definition 6 is unchanged in form, but the trajectory law Π_x^θ is now generated by the Gibbs policies (139) in place of the deterministic stop/continue rule.

Theorem 11 (Softplus gradient recovery)**.** *Under the SPNE of Theorem 10 on the Softplus network with activation* sp_θ*, the discounted occupation difference recovers the Softplus-network gradient exactly as in Theorem 6:*

$$\frac{\partial a_j^{(m)}}{\partial a_i^{(l)}} \;=\; \xi_{q,+}\Big(\Gamma_x\big(s_{i,+}^{(l,\mathrm{act})} \mid s\big) - \Gamma_x\big(s_{i,-}^{(l,\mathrm{act})} \mid s\big)\Big), \qquad s = s_{j,q}^{(m,\mathrm{act})}, \tag{142}$$

with Γ_x *now driven by* Π_x^θ*.*

Proof. Write $D(s_{h,r}^{(m',\mathrm{act})}) \coloneqq \Gamma_x(s_{i,r}^{(l,\mathrm{act})} \mid s_{h,r}^{(m',\mathrm{act})}) - \Gamma_x(s_{i,r'}^{(l,\mathrm{act})} \mid s_{h,r}^{(m',\mathrm{act})})$, the right-hand side above. The one-step occupation recursion in the proof of Theorem 6 holds with the gate $\mathbf{1}[z_j^{(m)} > 0]$ replaced by the Gibbs continuation probability $\mathrm{sp}'_\theta(z_j^{(m)})$; taking its difference over the two targets gives $D(s_{j,q}^{(m,\mathrm{act})}) = \mathrm{sp}'_\theta(z_j^{(m)}) \sum_h \big(W_{hj}^{(m,+)} D(s_{h,q}^{(m-1,\mathrm{act})}) - W_{hj}^{(m,-)} D(s_{h,q'}^{(m-1,\mathrm{act})})\big)$. By induction on $m-l$, $D(s_{h,r}^{(m-1,\mathrm{act})}) = \partial a_h^{(m-1)}/\partial a_i^{(l)}$ for both r, so the right-hand side is $\mathrm{sp}'_\theta(z_j^{(m)}) \sum_h W_{hj}^{(m)}\, \partial a_h^{(m-1)}/\partial a_i^{(l)}$, the chain rule at a Softplus gate; the base case $D(s_{i,r}^{(l,\mathrm{act})}) = 1$ and the addition and max-pooling steps are unchanged. Unlike Part 1 of Theorem 6, the individual occupation measures are not derivatives of the decomposed activations (141), because the soft gate depends on them; only their difference is. □

Probit analog and GELU. Replacing $\theta\mathcal{H}$ in (138) by the probit regularizer $\mathcal{R}_\Phi(\pi) = \phi(\Phi^{-1}(\pi))$ gives, by Lemma 3, the continuation probability $\Phi(z)$ and the value $z\Phi(z) + \phi(z) = \mathbb{E}_{\eta\sim\mathcal{N}(0,1)}[\mathrm{ReLU}(z+\eta)]$, so the proofs above carry over verbatim with sp_θ replaced by this Gaussian-smoothed ReLU, sp'_θ by its derivative Φ and $\theta\mathcal{H}_\theta(z)$ by $\phi(z)$: the probit-regularized game recovers Gaussian-smoothed-ReLU networks and their exact gradient. GeLU, $z\Phi(z)$, is *not* covered: it is not convex, so it is not the value $\max_\pi\{\pi z + \mathcal{R}(\pi)\}$ (a Fenchel conjugate) of any regularized stop/continue choice. Substituting the gate $\Phi(z)$ into Theorem 6 on a GeLU network yields the gradient of the gate-conditioned map $a = \Phi(z^{\mathrm{fwd}})\, z$, not the GeLU gradient $\Phi(z) + z\phi(z)$ (Appendix E.3).

Relation to the curvature-based defense of Dombrowski et al. Dombrowski et al. (2022) train Softplus networks from scratch as a defense against gradient-explanation manipulation. Their argument bounds the Frobenius norm of the network Hessian: replacing ReLU, whose distributional second derivative $\delta(z)$ leaves $\|\nabla^2 f\|_F$ unbounded, by sp_θ with $|\mathrm{sp}''_\theta(z)| \le 1/(4\theta)$ yields a uniform bound on $\|\nabla^2 f\|_F$, and the gradient theorem then bounds the maximal change of the gradient explanation under input perturbations. Theorems 10–11 place this Softplus replacement inside our framework as a single change in the game: entropy-regularizing the activation-state stop/continue policy at temperature θ. Whether the Hellinger response of the resulting trajectory law (Appendix G) yields bounds sharper than, or differently targeted from, the Lipschitz bounds of Dombrowski et al. (2022) is left for future work.

G HELLINGER DISTANCE AND DISAGREEMENT MAP

We compare two layered Markov processes on a common state graph and seek both a scalar divergence of the trajectory distributions and an exact per-pixel decomposition of this divergence at the input layer. We use the squared Hellinger distance because the geometric mean $\sqrt{PQ}$ preserves the multiplicative structure of Markov trajectory distributions and yields an exact linear-time dynamic program. Unlike Jeffreys divergence, it remains finite under support mismatch; unlike Jensen–Shannon, it admits an exact recursion on the layered graph. Later, we decompose the squared Hellinger distance as a sum of non-negative per-pixel contributions over all input-layer states, giving a disagreement map at the input layer (Theorem 12).

G.1 Layered Markov Process Foundations

We collect here the layered-Markov-process abstraction that underlies every Hellinger statement in this appendix: a finite layered graph augmented with a cemetery †, and the act-state Markov Processes (MPs) that the two games of the main paper instantiate on it. The natural RG carries an extra linear-routing state per neuron between consecutive activation states; we marginalize it out and check in Remark 6 that this preserves trajectory Bhattacharyya and Hellinger exactly.

G.1.1 Layered Markov Process with Cemetery

State graph. Fix an integer $L \geq 1$. For each $l \in \{0, 1, \ldots, L\}$ let $\mathcal{S}^{(l)}$ be a finite set of *ordinary states* at layer l. Adjoin a single absorbing *cemetery state* † shared across all layers and write $\mathcal{S}_+^{(l)} \coloneqq \mathcal{S}^{(l)} \cup \{\dagger\}$. The full state graph is the disjoint union $\bigsqcup_l \mathcal{S}_+^{(l)}$, and transitions are only permitted from layer l to layer $l-1$ (or into the cemetery, which then absorbs the trajectory forever). The states at layer 0, the input layer, are referred to as *pixels* below; the per-pixel decomposition will sum exactly over $\mathcal{S}_+^{(0)}$.

Transition kernel. For each layer $l \geq 1$, a *backward transition kernel* $T^{(l)} : \mathcal{S}_+^{(l)} \to \Delta\big(\mathcal{S}_+^{(l-1)}\big)$ assigns to each state $s \in \mathcal{S}_+^{(l)}$ a probability distribution $T^{(l)}(s \to \cdot)$ over $\mathcal{S}_+^{(l-1)}$, with the constraint that the cemetery is absorbing:

$$T^{(l)}(\dagger \to \dagger) = 1, \qquad T^{(l)}(\dagger \to s') = 0 \ \text{ for } \ s' \in \mathcal{S}^{(l-1)}. \tag{143}$$

Ordinary states may transition either to the next layer's ordinary states or into the cemetery (early termination). We write $T^{(l)}(s \to \dagger)$ for the probability that an ordinary state at layer l jumps to the cemetery rather than to a layer-$(l-1)$ ordinary state.

Initial distribution and trajectories. Fix a deterministic starting state $s_0 \in \mathcal{S}^{(L)}$. A *trajectory* (indexed by layer, $\zeta_l \in \mathcal{S}_+^{(l)}$) is a sequence

$$\zeta = (\zeta_L, \zeta_{L-1}, \ldots, \zeta_0) \ \in \ \mathcal{S}_+^{(L)} \times \mathcal{S}_+^{(L-1)} \times \cdots \times \mathcal{S}_+^{(0)}, \tag{144}$$

and the kernel $T^{(\cdot)}$ together with δ_{s_0} at layer L induces a probability distribution on trajectories via

$$\Pi(\zeta_L, \zeta_{L-1}, \ldots, \zeta_0) \ = \ \mathbf{1}\{\zeta_L = s_0\} \prod_{k=1}^{L} T^{(k)}(\zeta_k \to \zeta_{k-1}). \tag{145}$$

For any $l \in \{0, \ldots, L\}$ the prefix distribution $\Pi^{(l)}$ over $(\zeta_L, \ldots, \zeta_l)$ is the marginal of Π over the trailing coordinates.

Two parallel processes. We compare two such processes that share the state graph and the start state s_0, but use different kernels: $M = A$ with $T_A^{(l)}, \Pi_A^{(l)}$, and $M = B$ with $T_B^{(l)}, \Pi_B^{(l)}$. The kernels may differ either because the underlying network weights differ (e.g. randomization), the activations differ (e.g. different inputs), or both.

G.1.2 How the Two Games Fit This Abstraction

We instantiate the layered MP of Section G.1.1 for both games in this subsection. State labels $s_{i,p,\kappa}^{(l,\text{type})}$ follow Appendices B and C. We evaluate the SPNE policies of Theorems 5 (SG) and 7 (RG) at $\tau = 1$ (so activation-state decisions are deterministic, continue iff $z_j^{(l)} > 0$); for the RG we further fix $\varepsilon = 0$.

SG MP. Ordinary states at layers $l \in \{1, \dots, L\}$ are activation states $s_{i,p}^{(l,\mathrm{act})}$, and layer-0 ordinary states are pixel-player terminal pairs $s_{j,q}^{(0)}$ (each pixel splits into two terminals, mirroring the player tag the trajectory carries to it). The deterministic start state is $s_0 = s_{u,+}^{(L,\mathrm{act})}$. The SPNE backward kernel keeps the player on positive weights and flips it on negative weights:

$$T_M^{(l)}\big(s_{j,p}^{(l,\mathrm{act})} \to s_{i,p}^{(l-1,\mathrm{act})}\big) = \frac{W_M^{(l,+)}[i,j]}{\gamma_{M,j}^{(l)}}\,\mathbf{1}\{z_j^{(l)} > 0\},$$
$$T_M^{(l)}\big(s_{j,p}^{(l,\mathrm{act})} \to s_{i,p'}^{(l-1,\mathrm{act})}\big) = \frac{W_M^{(l,-)}[i,j]}{\gamma_{M,j}^{(l)}}\,\mathbf{1}\{z_j^{(l)} > 0\}, \qquad 2 \le l \le L, \tag{146}$$

with $\gamma_{M,j}^{(l)} = \sum_i |W_M^{(l)}[i,j]|$, $p' = -p$, cemetery row $T_M^{(l)}(s_{j,p}^{(l,\mathrm{act})} \to \dagger) = \mathbf{1}\{z_j^{(l)} \le 0\}$, and the analogous form at $l = 1$ landing on the pixel-player terminals $s_{i,p}^{(0)}, s_{i,p'}^{(0)}$. This is the natural SG MP of Appendix B verbatim.

RG MP. Ordinary states at layers $l \in \{1, \dots, L\}$ are activation states $s_{i,p}^{(l,\mathrm{act})}$, and layer-0 ordinary states are pixel-stream terminal pairs $s_{j,\kappa}^{(0)}$ (each pixel splits into two terminals tagged by the layer-1 routing-edge stream). The deterministic start state is $s_0 = s_{u,+}^{(L,\mathrm{act})}$. The kernel composes the activation-state stop/continue decision and the subsequent κ-branched Gibbs routing into one act-to-act transition:

$$T_M^{(l)}\big(s_{j,p}^{(l,\mathrm{act})} \to s_{i,p}^{(l-1,\mathrm{act})}\big) = \tfrac{1}{2}\,\pi_{M,+}^{(l)}(i \mid j)\,\mathbf{1}\{z_j^{(l)} > 0\},$$
$$T_M^{(l)}\big(s_{j,p}^{(l,\mathrm{act})} \to s_{i,p'}^{(l-1,\mathrm{act})}\big) = \tfrac{1}{2}\,\pi_{M,-}^{(l)}(i \mid j)\,\mathbf{1}\{z_j^{(l)} > 0\}, \qquad 2 \le l \le L, \tag{147}$$

and at $l = 1$ the destinations are pixel-stream terminals carrying the routing-edge κ,

$$T_M^{(1)}\big(s_{j,p}^{(1,\mathrm{act})} \to s_{i,\kappa}^{(0)}\big) \;=\; \tfrac{1}{2}\,\pi_{M,\kappa}^{(1)}(i \mid j)\,\mathbf{1}\{z_j^{(1)} > 0\}, \qquad \kappa \in \{+,-\}, \tag{148}$$

with the same cemetery row $\mathbf{1}\{z_j^{(l)} \le 0\}$. The Gibbs policy is the temperature-τ distribution

$$\pi_{M,\kappa}^{(l)}(i \mid j) \;\propto\; \big[\, a_{M,i}^{(l-1)}\, W_{M,ij}^{(l,\kappa)} \,\big]^{1/\tau}, \qquad i \in \mathrm{Pred}_j^{(l)}, \tag{149}$$

(at $l = 1$, $a_{M,i}^{(0)}\, W_{M,ij}^{(1,\kappa)}$ is replaced by $[x_i\, W_{M,ij}^{(1)}]^{\kappa}$; with $\varepsilon > 0$, mass $\varepsilon^{1/\tau}/Z_{j,\kappa}^{(l)}$ is added to the cemetery row; as in the implementation, the bias node of Appendix C is omitted). Equations (147)–(148) are the act-to-act marginal of the natural RG MP of Appendix C, which carries an extra linear-routing state per neuron between consecutive act states; Remark 6 gives a one-paragraph derivation and shows that this marginalization preserves Bhattacharyya and Hellinger distances on trajectories exactly.

G.1.3 Marginal Occupation Measures

For each model $M \in \{A, B\}$, the marginal probability that the chain visits state s at layer l is

$$m_M^{(l)}(s) \;=\; \sum_{\zeta : \zeta_l = s} \Pi_M^{(l)}(\zeta), \qquad s \in \mathcal{S}_+^{(l)}. \tag{150}$$

This satisfies the deterministic boundary condition

$$m_M^{(L)}(s_0) = 1, \qquad m_M^{(L)}(s) = 0 \;\text{ for }\; s \ne s_0, \tag{151}$$

and the one-step recursion

$$m_M^{(l-1)}(s') \;=\; \sum_{s \in \mathcal{S}_+^{(l)}} m_M^{(l)}(s)\, T_M^{(l)}(s \to s'), \qquad s' \in \mathcal{S}_+^{(l-1)}. \tag{152}$$

We refer to the single backward propagation (152) as the *Marginal Pass*. At the input layer, $m_M^{(0)}(s)$ is the probability that a backward trajectory from s_0 terminates at pixel s under model M, and $m_M^{(0)}(\dagger)$ is the total mass absorbed by the cemetery along the way.

G.2 Hellinger Distance from Bhattacharyya Coefficients

Properties. $\mathrm{BC}^{(l)}$ and $\mathrm{H}^{(l)}$ lie in $[0,1]$ for every l, even under support mismatch. If a state is alive in one model and dead in the other, its geometric-mean row vanishes and its mass contributes directly to the Hellinger increment. Since H is a metric and satisfies $\mathrm{H}^2 \le d_{\mathrm{TV}} \le \sqrt{2}\,\mathrm{H}$ and $\mathrm{H}^2 \le \frac{1}{2}\mathrm{KL}$, small Hellinger distance controls TV and small KL controls H as well. *Normalization conventions.* Here $d_{\mathrm{TV}}(\Pi_A, \Pi_B) = \frac{1}{2}\sum_\zeta |\Pi_A(\zeta) - \Pi_B(\zeta)| \in [0,1]$ is the probability-metric (half-L^1) total-variation distance, and KL is measured in nats; the displayed inequalities follow this convention. Readers using the unnormalized L^1 total variation $\|\Pi_A - \Pi_B\|_1 = 2d_{\mathrm{TV}}$ should absorb the factor of 2 accordingly.

Alternative divergences. KL and Jeffreys divergence have an exact Markov-chain chain rule but are unbounded and become infinite under support mismatch, e.g. when randomization opens or closes routes. Jensen–Shannon is bounded and finite, but $\frac{1}{2}(\Pi_A + \Pi_B)$ is not Markov on the layered graph, so no exact linear-time dynamic program is available. Wasserstein distances need a ground cost between neuron-labeled trajectories (across layers, signs and permuted neurons), for which there is no canonical choice, and the resulting transport problem has no exact local recursion. Hellinger is bounded, finite, and exactly tractable: the overlap factorizes along the layered law (Lemma 4), which gives a dynamic program linear in the number of edges (Algorithm 1) and the non-negative per-state decomposition of Theorem 12. Note that H compares the *undiscounted* law Π: the discount d_t does not enter the kernels, so H is blind to the path weights (α, β) and to the SG weight magnitudes $\gamma_j^{(l)}$, which enter only through d_t; it compares *where* trajectories go, not how much weight they carry.

Computation. Compute the marginal occupations $m_M^{(l)}$, run the Hellinger pass (Algorithm 1), and read off $\mathrm{H}^2(s)$ from (167); the total cost is linear in the number of edges.

G.2.1 Bhattacharyya Coefficient and the Geometric-Mean Kernel

Definitions. For two probability distributions P, Q on a common finite set, the *Bhattacharyya coefficient* is the geometric overlap

$$\mathrm{BC}(P,Q) \;\coloneqq\; \sum_y \sqrt{P(y)\,Q(y)} \;\in\; [0,1], \tag{153}$$

and the *squared Hellinger distance* is

$$\mathrm{H}^2(P,Q) \;\coloneqq\; \tfrac{1}{2}\sum_y \big(\sqrt{P(y)} - \sqrt{Q(y)}\big)^2 \;=\; 1 - \mathrm{BC}(P,Q) \;\in\; [0,1]. \tag{154}$$

The Hellinger distance $\mathrm{H}(P,Q) = \sqrt{1 - \mathrm{BC}(P,Q)}$ is a true metric: $0 \le \mathrm{H} \le 1$, with $\mathrm{H} = 0$ iff $P = Q$ and $\mathrm{H} = 1$ iff P and Q have disjoint supports. Crucially, $\sqrt{0 \cdot Q(y)} = 0$, so mismatched supports cause no infinity—only a smaller overlap.

Geometric-mean kernel. For each layer $l \ge 1$ and pair of states $(s, s') \in \mathcal{S}_+^{(l)} \times \mathcal{S}_+^{(l-1)}$, define the *geometric-mean kernel*

$$\widetilde{T}^{(l)}(s,s') \;\coloneqq\; \sqrt{T_A^{(l)}(s \to s')\; T_B^{(l)}(s \to s')}. \tag{155}$$

This kernel is non-negative but in general sub-stochastic. Its row sum

$$\widetilde{R}^{(l)}(s) \;\coloneqq\; \sum_{s' \in \mathcal{S}_+^{(l-1)}} \widetilde{T}^{(l)}(s,s') \tag{156}$$

satisfies $0 \le \widetilde{R}^{(l)}(s) \le 1$ by Cauchy–Schwarz, with equality iff $T_A^{(l)}(s \to \cdot) = T_B^{(l)}(s \to \cdot)$. Cemetery handling is automatic: $T_A^{(l)}(\dagger \to \dagger) = T_B^{(l)}(\dagger \to \dagger) = 1$, so $\widetilde{T}^{(l)}(\dagger,\dagger) = 1$ and $\widetilde{R}^{(l)}(\dagger) = 1$.

G.2.2 Multiplicative Chain Rule

Lemma 4 (Chain rule)**.** *For every $l \in \{0, 1, \dots, L\}$, the Bhattacharyya coefficient of the prefix distributions factorizes along trajectories:*

$$\mathrm{BC}\big(\Pi_A^{(l)}, \Pi_B^{(l)}\big) \;=\; \sum_{(\zeta_L,\dots,\zeta_l)} \mathbf{1}\{\zeta_L = s_0\} \prod_{k=l+1}^{L} \widetilde{T}^{(k)}(\zeta_k, \zeta_{k-1}). \tag{157}$$

Proof. Substitute the Markov factorization $\Pi_M^{(l)}(\zeta) = \mathbf{1}\{\zeta_L = s_0\} \prod_{k=l+1}^{L} T_M^{(k)}(\zeta_k, \zeta_{k-1})$ for $M \in \{A, B\}$ into the definition of BC:

$$\mathrm{BC}\big(\Pi_A^{(l)}, \Pi_B^{(l)}\big) \;=\; \sum_{(\zeta_L,\dots,\zeta_l)} \mathbf{1}\{\zeta_L = s_0\} \sqrt{\prod_{k=l+1}^{L} T_A^{(k)}(\zeta_k, \zeta_{k-1})\, T_B^{(k)}(\zeta_k, \zeta_{k-1})} \tag{158}$$

$$= \sum_{(\zeta_L,\dots,\zeta_l)} \mathbf{1}\{\zeta_L = s_0\} \prod_{k=l+1}^{L} \widetilde{T}^{(k)}(\zeta_k, \zeta_{k-1}), \tag{159}$$

using $\sqrt{\prod_k a_k b_k} = \prod_k \sqrt{a_k b_k}$ for non-negative factors. □

Lemma 5 (Monotonicity)**.** *Let* $\mathrm{BC}^{(l)} \coloneqq \mathrm{BC}(\Pi_A^{(l)}, \Pi_B^{(l)})$*. Then* $\mathrm{BC}^{(l)}$ *is monotonically non-increasing as l decreases from L to 0:*

$$1 \;=\; \mathrm{BC}^{(L)} \;\geq\; \mathrm{BC}^{(L-1)} \;\geq\; \cdots \;\geq\; \mathrm{BC}^{(0)} \;\geq\; 0. \tag{160}$$

Equivalently, the squared Hellinger distance $\mathrm{H}^2_{(l)} = 1 - \mathrm{BC}^{(l)}$ *is monotonically non-decreasing.*

Proof. Define

$$g^{(l)}(s) \;\coloneqq\; \sum_{(\zeta_L,\dots,\zeta_l)} \mathbf{1}\{\zeta_L = s_0\} \prod_{k=l+1}^{L} \widetilde{T}^{(k)}(\zeta_k, \zeta_{k-1})\, \mathbf{1}\{\zeta_l = s\}. \tag{161}$$

From Lemma 4, $\mathrm{BC}^{(l)} = \sum_{s \in \mathcal{S}_+^{(l)}} g^{(l)}(s)$. Grouping the layer-$(l-1)$ sum by $\zeta_l = s$,

$$\mathrm{BC}^{(l-1)} \;=\; \sum_{s \in \mathcal{S}_+^{(l)}} g^{(l)}(s)\, \widetilde{R}^{(l)}(s). \tag{162}$$

Since $\widetilde{R}^{(l)}(s) \in [0, 1]$ and $g^{(l)}(s) \geq 0$, we get $\mathrm{BC}^{(l-1)} \leq \mathrm{BC}^{(l)}$. The base case $\mathrm{BC}^{(L)} = 1$ holds because $\Pi_A^{(L)} = \Pi_B^{(L)} = \delta_{s_0}$ and $\sqrt{1 \cdot 1} = 1$. □

G.2.3 Hellinger backward pass

The chain rule (Lemma 4) yields a backward DP that propagates the per-state cumulative $\widetilde{T}$-mass $g^{(l)}(s)$ from the start state s_0 down through the layers. The prefix Bhattacharyya coefficient at each layer is the row sum $\mathrm{BC}^{(l)} = \sum_s g^{(l)}(s)$, the prefix Hellinger distance is $\mathrm{H}^{(l)} = \sqrt{1 - \mathrm{BC}^{(l)}}$, and the per-terminal-state disagreement map is read off at $l = 0$ from (167). The map is defined on the full terminal set $\mathcal{S}_+^{(0)} = \mathcal{S}^{(0)} \cup \{\dagger\}$; the visualized pixel attribution map is its restriction to ordinary pixels $\mathcal{S}^{(0)}$, and its sum is therefore $\mathrm{H}^2_{\mathrm{ord}} = \mathrm{H}^2 - \mathrm{H}^2(\dagger)$, not the full H^2 (Theorem 12(iii)).

Algorithm 1 Hellinger backward pass (with per-terminal-state disagreement map)

Require: Backward kernels $T_A^{(l)}, T_B^{(l)}$ for all l; start state s_0; marginal occupations $m_A^{(0)}, m_B^{(0)}$ from the Marginal Pass.

Ensure: Bhattacharyya coefficients $\mathrm{BC}^{(l)}$, Hellinger distances $\mathrm{H}^{(l)}$, and per-terminal-state disagreement map $\mathrm{H}^2(\cdot)$ on $\mathcal{S}_+^{(0)}$.

$g^{(L)}(s_0) \leftarrow 1$; $\quad g^{(L)}(s) \leftarrow 0$ for every $s \neq s_0$
$\mathrm{BC}^{(L)} \leftarrow 1$
for $l = L, L-1, \dots, 1$ **do**
  for each state $s' \in \mathcal{S}_+^{(l-1)}$ **do**
    $g^{(l-1)}(s') \leftarrow \sum_{s \in \mathcal{S}_+^{(l)}} g^{(l)}(s)\, \widetilde{T}^{(l)}(s, s')$ ▷ $\widetilde{T}^{(l)}(s,s') = \sqrt{T_A^{(l)}(s \to s')\, T_B^{(l)}(s \to s')}$
  end for
  $\mathrm{BC}^{(l-1)} \leftarrow \sum_{s' \in \mathcal{S}_+^{(l-1)}} g^{(l-1)}(s')$
end for
for $l = L, L-1, \dots, 0$ **do**
  $\mathrm{H}^{(l)} \leftarrow \sqrt{1 - \mathrm{BC}^{(l)}}$
end for
// **Per-terminal-state disagreement map (Theorem 12).**
// **The visualized pixel attribution map is the restriction to $\mathcal{S}^{(0)} \subset \mathcal{S}_+^{(0)}$.**
for each terminal state $s \in \mathcal{S}_+^{(0)}$ **do**
  $\mathrm{H}^2(s) \leftarrow \frac{1}{2}\big(m_A^{(0)}(s) + m_B^{(0)}(s)\big) - g^{(0)}(s)$
end for

Lemma 6 (Backward pass correctness)**.** *Algorithm 1 computes $g^{(l)}$, hence $\mathrm{BC}^{(l)}$, $\mathrm{H}^{(l)}$, and $\mathrm{H}^2(s)$ correctly for every l and s. In particular,*

$$\sum_{s \in \mathcal{S}_+^{(0)}} \mathrm{H}^2(s) \;=\; \mathrm{H}^2_{(0)}. \tag{163}$$

Proof. Correctness of $g^{(l)}$ by downward induction on l. At $l = L$, the initialization $g^{(L)}(s_0) = 1$ and $g^{(L)}(s) = 0$ for $s \neq s_0$ matches the definition $g^{(L)}(s) = \mathbf{1}\{s = s_0\}$ in Lemma 5 (the empty product equals 1).

Assume $g^{(l)}$ stored by the algorithm equals the $g^{(l)}$ defined in Lemma 5. In iteration l, the algorithm computes

$$g^{(l-1)}(s') \;=\; \sum_{s \in \mathcal{S}_+^{(l)}} g^{(l)}(s)\, \widetilde{T}^{(l)}(s, s'), \tag{164}$$

which by direct expansion equals

$$\begin{aligned} &\sum_{s} \Big(\sum_{(\zeta_L, \dots, \zeta_l)} \mathbf{1}\{\zeta_L = s_0\} \prod_{k=l+1}^{L} \widetilde{T}^{(k)}(\zeta_k, \zeta_{k-1})\, \mathbf{1}\{\zeta_l = s\} \Big) \widetilde{T}^{(l)}(s, s') \\ &\qquad = \sum_{(\zeta_L, \dots, \zeta_{l-1})} \mathbf{1}\{\zeta_L = s_0\} \prod_{k=l}^{L} \widetilde{T}^{(k)}(\zeta_k, \zeta_{k-1})\, \mathbf{1}\{\zeta_{l-1} = s'\}. \end{aligned} \tag{165}$$

This equals $g^{(l-1)}(s')$ from Lemma 5, closing the induction.

Correctness of $\mathrm{BC}^{(l)}$. After the loop, the algorithm sets $\mathrm{BC}^{(l)} = \sum_{s'} g^{(l)}(s')$, which by the proof of Lemma 5 equals $\mathrm{BC}\big(\Pi_A^{(l)}, \Pi_B^{(l)}\big)$. Then $\mathrm{H}^{(l)} = \sqrt{1 - \mathrm{BC}^{(l)}}$ by definition (154).

Correctness of $\mathrm{H}^2(\cdot)$. At $l = 0$, the stored value satisfies

$$g^{(0)}(s) = \sum_{\zeta : \zeta_0 = s} \sqrt{\Pi_A^{(0)}(\zeta) \Pi_B^{(0)}(\zeta)}, \tag{166}$$

which is exactly the quantity used in the proof of Theorem 12. Substituting into (167) gives the per-pixel disagreement values from Theorem 12, and (168) guarantees they sum to $\mathrm{H}^2_{(0)}$. □

Remark 6 (RG act-only MP matches the natural game)**.** *The natural RG MP of Appendix C alternates activation and linear-routing states: each $s^{(l,\mathrm{act})}_{j,p}$ either stops to † (if $z^{(l)}_j \le 0$) or branches uniformly to $s^{(l,\mathrm{lin})}_{j,p,+}$ or $s^{(l,\mathrm{lin})}_{j,p',-}$ (intra-layer act-to-lin), and each $s^{(l,\mathrm{lin})}_{j,q,\kappa}$ routes to a layer-$(l-1)$ activation (or, at $l=1$, to the layer-0 pixel-stream terminal $s^{(0)}_{i,\kappa}$) via the Gibbs policy $\pi^{(l)}_{M,\kappa}(\cdot \mid j)$ retaining the player (inter-layer lin-to-act). Multiplying these two kernels along the unique intermediate lin state yields the act-to-act rows (147)–(148) term by term. Deleting every lin state from a natural trajectory is a bijection between natural and act-only trajectories starting at $s_0 = s^{(L,\mathrm{act})}_{u,+}$ (each lin coordinate is recovered: j, q from the act states it sits between, and κ from the player update—kept ⇒ +, switched ⇒ −—or, at $l=1$, directly from the pixel-stream terminal). A bijection on trajectory supports preserves Bhattacharyya and Hellinger by the trivial reindexing $\sum_\zeta \sqrt{\Pi_A(\zeta)\Pi_B(\zeta)} = \sum_{\widehat{\zeta}} \sqrt{\widehat{\Pi}_A(\widehat{\zeta})\widehat{\Pi}_B(\widehat{\zeta})}$, so all $\mathrm{BC}^{(l)}$ and $\mathrm{H}^{(l)}$ computed from the act-only kernel coincide with the natural-game ones.*

G.3 Per-Pixel Disagreement Map

The squared Hellinger distance H^2 admits an exact non-negative decomposition over input-layer states: each input pixel carries a calibrated share of the total trajectory disagreement (Theorem 12), giving a disagreement *map* that is itself an attribution of the trajectory mismatch back to the input. The conditioned-on-survival variant removes the cemetery contribution that otherwise bounds plain H from below on randomized models.

G.3.1 Per-Pixel Decomposition: The Disagreement Map

The squared Hellinger distance at layer 0 decomposes exactly over terminal pixels.

Theorem 12 (Per-pixel decomposition)**.** *Let $\mathrm{H}^2 \coloneqq \mathrm{H}^2\big(\Pi^{(0)}_A, \Pi^{(0)}_B\big)$. For $s \in \mathcal{S}^{(0)}_+$, define*

$$\mathrm{H}^2(s) \;\coloneqq\; \tfrac{1}{2}\big(m^{(0)}_A(s) + m^{(0)}_B(s)\big) \;-\; g^{(0)}(s), \tag{167}$$

where $g^{(0)}(s)$ is the cumulative $\widetilde{T}$-mass at s. Then:

(i) $\mathrm{H}^2(s) \ge 0$ *for every* $s \in \mathcal{S}^{(0)}_+$.

(ii)

$$\mathrm{H}^2 \;=\; \sum_{s \in \mathcal{S}^{(0)}_+} \mathrm{H}^2(s). \tag{168}$$

(iii)

$$\mathrm{H}^2_{\mathrm{ord}} \;\coloneqq\; \sum_{s \in \mathcal{S}^{(0)}} \mathrm{H}^2(s) \;=\; \mathrm{H}^2 - \mathrm{H}^2(\dagger) \tag{169}$$

Proof. Apply the algebraic identity $(\sqrt{a} - \sqrt{b})^2 = a + b - 2\sqrt{ab}$ to each trajectory:

$$\mathrm{H}^2 \;=\; \tfrac{1}{2}\sum_\zeta \big(\sqrt{\Pi^{(0)}_A(\zeta)} - \sqrt{\Pi^{(0)}_B(\zeta)}\big)^2 \;=\; \tfrac{1}{2}\sum_\zeta \big[\Pi^{(0)}_A(\zeta) + \Pi^{(0)}_B(\zeta) - 2\sqrt{\Pi^{(0)}_A(\zeta)\Pi^{(0)}_B(\zeta)}\big]. \tag{170}$$

Group the trajectory sum by the terminal pixel $\zeta_0 = s$:

$$\mathrm{H}^2 \;=\; \tfrac{1}{2}\sum_{s \in \mathcal{S}^{(0)}_+} \Big[\sum_{\zeta:\zeta_0 = s} \Pi^{(0)}_A(\zeta) + \sum_{\zeta:\zeta_0 = s} \Pi^{(0)}_B(\zeta) - 2\sum_{\zeta:\zeta_0 = s} \sqrt{\Pi^{(0)}_A(\zeta)\Pi^{(0)}_B(\zeta)}\Big]. \tag{171}$$

By definition, $\sum_{\zeta:\zeta_0=s}\Pi_M^{(0)}(\zeta)=m_M^{(0)}(s)$ for $M\in\{A,B\}$. By Lemma 4 restricted to trajectories terminating at s (i.e. tracing through Lemma 5's definition of $g^{(l)}$), $\sum_{\zeta:\zeta_0=s}\sqrt{\Pi_A^{(0)}(\zeta)\Pi_B^{(0)}(\zeta)}=g^{(0)}(s)$. Substituting,

$$\mathrm{H}^2 \;=\; \tfrac{1}{2}\sum_s\big[m_A^{(0)}(s)+m_B^{(0)}(s)-2g^{(0)}(s)\big] \;=\; \sum_{s\in\mathcal{S}_+^{(0)}}\mathrm{H}^2(s), \tag{172}$$

proving (ii).

For (i), apply Cauchy–Schwarz to the sum defining $g^{(0)}(s)$:

$$g^{(0)}(s)=\sum_{\zeta:\zeta_0=s}\sqrt{\Pi_A^{(0)}(\zeta)\Pi_B^{(0)}(\zeta)}\le\sqrt{\Big(\sum_{\zeta:\zeta_0=s}\Pi_A^{(0)}(\zeta)\Big)\Big(\sum_{\zeta:\zeta_0=s}\Pi_B^{(0)}(\zeta)\Big)}=\sqrt{m_A^{(0)}(s)\,m_B^{(0)}(s)}. \tag{173}$$

Combined with the AM–GM inequality $\sqrt{m_A^{(0)}(s)m_B^{(0)}(s)}\le\frac{1}{2}(m_A^{(0)}(s)+m_B^{(0)}(s))$, this gives $g^{(0)}(s)\le\frac{1}{2}(m_A^{(0)}(s)+m_B^{(0)}(s))$, so $\mathrm{H}^2(s)\ge 0$. Equality holds iff $\Pi_A^{(0)}(\cdot\mid\zeta_0=s)=\Pi_B^{(0)}(\cdot\mid\zeta_0=s)$ *and* $m_A^{(0)}(s)=m_B^{(0)}(s)$.

Item (iii) is immediate from (ii) by separating the cemetery from the ordinary pixels. □

The disagreement map. Theorem 12 defines $\mathrm{H}^2(\cdot)$ on the full terminal set $\mathcal{S}_+^{(0)}=\mathcal{S}^{(0)}\cup\{\dagger\}$: this is the *full terminal-state decomposition*, with $\sum_{s\in\mathcal{S}_+^{(0)}}\mathrm{H}^2(s)=\mathrm{H}^2_{(0)}$, and the cemetery contribution $\mathrm{H}^2(\dagger)$ is part of it. The *Hellinger disagreement map* is the ordinary-pixel restriction $s\mapsto\mathrm{H}^2(s)$ on $\mathcal{S}^{(0)}$, which is the quantity we visualize. It is non-negative, sums to $\mathrm{H}^2_{\mathrm{ord}}$ over ordinary pixels, and is computed from $g^{(0)},m_A^{(0)},m_B^{(0)}$. The quantity $\mathrm{H}^2_{\mathrm{ord}}$ is an unnormalized ordinary-state subsum; the conditioned-on-survival Hellinger distance that normalizes it is introduced below. Recall from Section G.1.2 that layer-0 ordinary states of the two games are *tagged* pixel pairs $s_{j,\pm}^{(0)}$, tagged by the player q for the SG and by the stream sign κ for the RG; the per-pixel attribution displayed in figures aggregates the two tags,

$$\mathrm{H}^2_{\mathrm{pixel}}(j) \;=\; \mathrm{H}^2\big(s_{j,+}^{(0)}\big)+\mathrm{H}^2\big(s_{j,-}^{(0)}\big), \tag{174}$$

which remains non-negative and sums to $\mathrm{H}^2_{\mathrm{ord}}$ over pixels. By the Cauchy–Schwarz step in the proof of Theorem 12(i), $g^{(0)}(s)\le\sqrt{m_A^{(0)}(s)\,m_B^{(0)}(s)}$, so $\mathrm{H}^2(s)$ is bounded below by the marginal surrogate $\frac{1}{2}\big(m_A^{(0)}(s)+m_B^{(0)}(s)\big)-\sqrt{m_A^{(0)}(s)\,m_B^{(0)}(s)}$, the per-pixel squared Hellinger distance between the layer-0 marginal occupations; the trajectory map is strictly larger whenever the conditional distributions of trajectories reaching pixel s differ between models.

G.3.2 Conditioning on Survival to the Ordinary Input Layer

When the cemetery absorbs a large fraction of trajectory mass, the total Hellinger distance mixes two effects: disagreement about *whether* a trajectory survives and disagreement about *where surviving trajectories go*. To isolate the second effect, we condition each model's trajectory law on the event that the trajectory reaches an ordinary input-layer state. The resulting scalar compares the two *surviving* trajectory distributions directly and removes the common cemetery floor. Because survival to the input layer factorizes along the layered graph, this conditioning remains Markov and is implemented by a per-state Doob h-transform (with $h=\psi_M$).

Theorem 13 (Conditioned-survival trajectory law and conditioned Hellinger distance)**.** *Let* $\mathsf{Surv}:=\{(\zeta_L,\dots,\zeta_0):\zeta_0\in\mathcal{S}^{(0)}\}$ *be the event, in the trajectory space* $\prod_{l=0}^{L}\mathcal{S}_+^{(l)}$*, that the backward trajectory ends in an ordinary input-layer state — equivalently, the event that the trajectory never enters the cemetery* †*. For each model* $M\in\{A,B\}$ *define the survival function*

$$\psi_M^{(l)}(s) \;:=\; \mathbb{P}_M(\mathsf{Surv}\mid\zeta_l=s),\qquad s\in\mathcal{S}_+^{(l)},\;\; l\in\{0,\dots,L\}. \tag{175}$$

Then:

(i) The boundary and recursion are

$$\psi_M^{(0)}(s)=1\;\text{ for } s\in\mathcal{S}^{(0)},\qquad \psi_M^{(0)}(\dagger)=0, \tag{176}$$

and for every $l \geq 1$,

$$\psi_M^{(l)}(s) \;=\; \sum_{s' \in \mathcal{S}_+^{(l-1)}} T_M^{(l)}(s \to s')\, \psi_M^{(l-1)}(s') \;=\; \sum_{s' \in \mathcal{S}^{(l-1)}} T_M^{(l)}(s \to s')\, \psi_M^{(l-1)}(s'). \tag{177}$$

In particular,

$$Z_M \;\coloneqq\; \mathbb{P}_M(\mathsf{Surv}) \;=\; \psi_M^{(L)}(s_0) \;=\; \sum_{s \in \mathcal{S}^{(0)}} m_M^{(0)}(s) \;=\; 1 - m_M^{(0)}(\dagger). \tag{178}$$

(ii) Assume $Z_M > 0$. For every ordinary state $s \in \mathcal{S}^{(l)}$ with $\psi_M^{(l)}(s) > 0$, define the conditioned kernel on ordinary states by

$$T_{M,\mathrm{surv}}^{(l)}(s \to s') \;\coloneqq\; \frac{T_M^{(l)}(s \to s')\, \psi_M^{(l-1)}(s')}{\psi_M^{(l)}(s)}, \qquad s' \in \mathcal{S}^{(l-1)}. \tag{179}$$

Then $T_{M,\mathrm{surv}}^{(l)}(s \to \cdot)$ is row-stochastic on $\mathcal{S}^{(l-1)}$, and the conditional trajectory distribution $\Pi_{M,\mathrm{surv}} \coloneqq \Pi_M(\,\cdot \mid \mathsf{Surv}\,)$ factorizes as the layered Markov law

$$\Pi_{M,\mathrm{surv}}(\zeta_L, \ldots, \zeta_0) = \mathbf{1}\{\zeta_L = s_0\} \prod_{k=1}^{L} T_{M,\mathrm{surv}}^{(k)}(\zeta_k \to \zeta_{k-1}), \qquad (\zeta_L, \ldots, \zeta_0) \in \mathsf{Surv}. \tag{180}$$

Its layer-l marginals satisfy

$$m_{M,\mathrm{surv}}^{(l)}(s) \;=\; \frac{m_M^{(l)}(s)\, \psi_M^{(l)}(s)}{Z_M}, \qquad s \in \mathcal{S}^{(l)}. \tag{181}$$

(iii) Define the conditioned geometric-mean kernel

$$\widetilde{T}_{\mathrm{surv}}^{(l)}(s, s') \coloneqq \sqrt{T_{A,\mathrm{surv}}^{(l)}(s \to s')\, T_{B,\mathrm{surv}}^{(l)}(s \to s')} = \frac{\widetilde{T}^{(l)}(s, s') \sqrt{\psi_A^{(l-1)}(s')\, \psi_B^{(l-1)}(s')}}{\sqrt{\psi_A^{(l)}(s)\, \psi_B^{(l)}(s)}}. \tag{182}$$

The Bhattacharyya coefficient of the conditioned trajectory laws is

$$\mathrm{BC}\big(\Pi_{A,\mathrm{surv}}, \Pi_{B,\mathrm{surv}}\big) \;=\; \sum_{(\zeta_L, \ldots, \zeta_0) \in \mathsf{Surv}} \mathbf{1}\{\zeta_L = s_0\} \prod_{k=1}^{L} \widetilde{T}_{\mathrm{surv}}^{(k)}(\zeta_k, \zeta_{k-1}), \tag{183}$$

so it is computed by the same Hellinger pass (Algorithm 1), but on the ordinary-state graph with the conditioned kernel.

(iv) Let

$$\mathrm{BC}_{\mathrm{ord}}^{(0)} \;\coloneqq\; \sum_{s \in \mathcal{S}^{(0)}} g^{(0)}(s) \;=\; \sum_{(\zeta_L, \ldots, \zeta_0) \in \mathsf{Surv}} \sqrt{\Pi_A(\zeta_L, \ldots, \zeta_0)\, \Pi_B(\zeta_L, \ldots, \zeta_0)}. \tag{184}$$

Then

$$\mathrm{BC}\big(\Pi_{A,\mathrm{surv}}, \Pi_{B,\mathrm{surv}}\big) \;=\; \frac{\mathrm{BC}_{\mathrm{ord}}^{(0)}}{\sqrt{Z_A Z_B}}, \tag{185}$$

and the conditioned-survival Hellinger distance is

$$\mathrm{H}_{\mathrm{surv}}^2 \;\coloneqq\; \mathrm{H}^2\big(\Pi_{A,\mathrm{surv}}, \Pi_{B,\mathrm{surv}}\big) \;=\; 1 - \frac{\mathrm{BC}_{\mathrm{ord}}^{(0)}}{\sqrt{Z_A Z_B}}. \tag{186}$$

Proof. For (i), (176) is immediate from the definition of Surv: at layer 0 every ordinary state survives and the cemetery does not. For $l \geq 1$, conditioning on the next step gives

$$\psi_M^{(l)}(s) = \sum_{s' \in \mathcal{S}_+^{(l-1)}} T_M^{(l)}(s \to s')\, \mathbb{P}_M(\mathsf{Surv} \mid \zeta_{l-1} = s') = \sum_{s' \in \mathcal{S}_+^{(l-1)}} T_M^{(l)}(s \to s')\, \psi_M^{(l-1)}(s'). \tag{187}$$

Since $\psi_M^{(l-1)}(\dagger) = 0$, the cemetery term drops out and (177) follows. Evaluating at s_0 gives $Z_M = \psi_M^{(L)}(s_0)$. The identities with $m_M^{(0)}$ are just the layer-0 marginal of the event Surv.

For (ii), row-stochasticity follows from (177):

$$\sum_{s' \in \mathcal{S}^{(l-1)}} T_{M,\mathrm{surv}}^{(l)}(s \to s') = \frac{1}{\psi_M^{(l)}(s)} \sum_{s' \in \mathcal{S}^{(l-1)}} T_M^{(l)}(s \to s')\, \psi_M^{(l-1)}(s') = 1. \tag{188}$$

Now fix $(\zeta_L, \dots, \zeta_0) \in \mathsf{Surv}$. By definition of conditional probability,

$$\Pi_{M,\mathrm{surv}}(\zeta_L, \dots, \zeta_0) = \frac{\Pi_M(\zeta_L, \dots, \zeta_0)}{Z_M} = \frac{\mathbf{1}\{\zeta_L = s_0\}}{Z_M} \prod_{k=1}^{L} T_M^{(k)}(\zeta_k \to \zeta_{k-1}). \tag{189}$$

Multiplying the conditioned kernels and telescoping the survival factors gives

$$\prod_{k=1}^{L} T_{M,\mathrm{surv}}^{(k)}(\zeta_k \to \zeta_{k-1}) = \prod_{k=1}^{L} T_M^{(k)}(\zeta_k \to \zeta_{k-1}) \frac{\psi_M^{(k-1)}(\zeta_{k-1})}{\psi_M^{(k)}(\zeta_k)} \tag{190}$$

$$= \frac{\psi_M^{(0)}(\zeta_0)}{\psi_M^{(L)}(s_0)} \prod_{k=1}^{L} T_M^{(k)}(\zeta_k \to \zeta_{k-1}) \tag{191}$$

$$= \frac{1}{Z_M} \prod_{k=1}^{L} T_M^{(k)}(\zeta_k \to \zeta_{k-1}), \tag{192}$$

because $\zeta_0 \in \mathcal{S}^{(0)}$ implies $\psi_M^{(0)}(\zeta_0) = 1$. This is (180). For the marginals,

$$m_{M,\mathrm{surv}}^{(l)}(s) = \mathbb{P}_M(\zeta_l = s \mid \mathsf{Surv}) = \frac{\mathbb{P}_M(\zeta_l = s, \mathsf{Surv})}{Z_M} = \frac{m_M^{(l)}(s)\, \psi_M^{(l)}(s)}{Z_M}, \tag{193}$$

which is (181).

For (iii), substitute (179) into the square root to obtain (182). Then apply Lemma 4 to the conditioned trajectory laws (180) on the ordinary-state graph. This gives (183) directly.

For (iv), expand the Bhattacharyya coefficient of the conditioned laws:

$$\mathrm{BC}\big(\Pi_{A,\mathrm{surv}}, \Pi_{B,\mathrm{surv}}\big) = \sum_{(\zeta_L, \dots, \zeta_0) \in \mathsf{Surv}} \sqrt{\frac{\Pi_A(\zeta_L, \dots, \zeta_0)}{Z_A} \frac{\Pi_B(\zeta_L, \dots, \zeta_0)}{Z_B}} \tag{194}$$

$$= \frac{1}{\sqrt{Z_A Z_B}} \sum_{(\zeta_L, \dots, \zeta_0) \in \mathsf{Surv}} \sqrt{\Pi_A(\zeta_L, \dots, \zeta_0)\, \Pi_B(\zeta_L, \dots, \zeta_0)}. \tag{195}$$

Grouping the last sum by the ordinary terminal state $\zeta_0 = s$ and using the definition of $g^{(0)}(s)$ exactly as in the proof of Theorem 12 yields (184), hence (185). Equation (186) is then the identity $\mathrm{H}^2 = 1 - \mathrm{BC}$ applied to the conditioned trajectory laws. □

Implementation consequence. Theorem 13 gives two exactly equivalent computations of the conditioned scalar:

(a) *Kernel-side conditioning:* replace each row by the reweighted kernel (179), form the geometric mean (182), and run the ordinary Hellinger backward pass on $\mathcal{S}^{(L)} \times \cdots \times \mathcal{S}^{(0)}$.

(b) *Post-hoc normalization:* run the original pass once, extract the ordinary overlap $\mathrm{BC}^{(0)}_{\mathrm{ord}} = \sum_{s\in\mathcal{S}^{(0)}} g^{(0)}(s)$ and the survival masses Z_A, Z_B, and apply (186).

The first route makes the conditioned trajectory law explicit; the second is cheaper when only the scalar is needed, but for deep networks it becomes numerically infeasible when the survival probability Z_M of either model $M \in \{A, B\}$ gets too small.

Conditioned disagreement map. Applying Theorem 12 to the conditioned kernels $T^{(l)}_{M,\mathrm{surv}}$ yields the conditioned disagreement map

$$\mathrm{H}^2_{\mathrm{surv}}(s) \;:=\; \tfrac{1}{2}\big(m^{(0)}_{A,\mathrm{surv}}(s) + m^{(0)}_{B,\mathrm{surv}}(s)\big) - g^{(0)}_{\mathrm{surv}}(s), \qquad s \in \mathcal{S}^{(0)}, \tag{196}$$

which is non-negative and satisfies $\sum_{s\in\mathcal{S}^{(0)}} \mathrm{H}^2_{\mathrm{surv}}(s) = \mathrm{H}^2_{\mathrm{surv}}$.

G.4 Experiments

This subsection puts the trajectory-space machinery to two empirical tests, holding the model fixed: how the Hellinger trajectory distance reacts to input perturbations, and whether the resulting distance separates inputs from different ImageNet-S classes from one another and from random noise. Here SG is the Stopping-Game kernel (146) without risk aversion ($\lambda = 0$; named "ADF" in the code) and RG is the Routing-Game kernel (147) at $\tau = 1$.

G.4.1 Input-Noise Sensitivity

For each architecture and each noise standard deviation $\sigma \in \{0.005, 0.01, 0.02, 0.03, 0.05, 0.1, 0.2, 0.3, 0.5\}$ we draw one standard-Gaussian tensor η per image and compute $\mathrm{H}(\Pi_A(x), \Pi_A(x + \sigma\eta))$ under both game kernels with the argmax target of the clean input. Figure 5 reports the mean H and $\mathrm{H}_{\mathrm{surv}}$ curves over the swept σ; both increase with σ for both games on all three architectures. On VGG-16 the RG H and $\mathrm{H}_{\mathrm{surv}}$ are equal, as the kernel routes no trajectories to states belonging to dead neurons. On ResNet-50 and ViT-B/16, skip connections route trajectories to dead-neuron states, and on ViT-B/16 the GELU gate also sends some trajectories to the cemetery at every state.

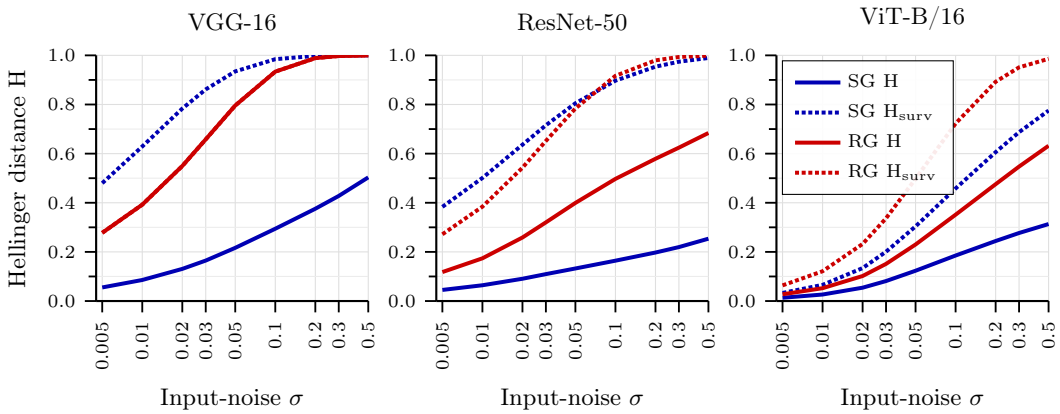

Figure 5: Input-noise Hellinger trajectory distance. Solid: plain H; dotted: $\mathrm{H}_{\mathrm{surv}}$, conditioned on survival to an ordinary input-layer state. x-axis is log-scale input-noise σ; same (A, A) model on both sides, only the input differs.

G.4.2 Class Separability and Reference Baselines

Can the Hellinger distance differentiate between inputs of different classes? Partly; the reference baselines in Table 5 make the answer architecture- and kernel-specific:

- On **VGG-16**, the SG H is the kernel that does the work: clean-intraclass H = 0.513 sits below clean-interclass H = 0.607. The RG kernel saturates at H = 1 on every row on VGG-16, so it cannot resolve class structure on this architecture.
- On **ResNet-50**, the RG kernel is the discriminating one: clean-intraclass H = 0.786 vs. clean-interclass H = 0.902. The SG kernel does not separate classes on ResNet-50 (intraclass 0.272 vs. interclass 0.268).
- On **ViT-B/16**, the RG kernel gives intraclass 0.789 vs. interclass 0.882; the SG gap (0.422 ± 0.023 vs. 0.435 ± 0.017) is smaller than the spread over pairs, so the SG does not separate classes there.

Random-vs-random is the lowest row in every column whose rows differ, and clean-vs-random is the highest row of every plain-H column except the SG on ResNet-50 (0.269 vs. intraclass 0.272) and the saturated RG on VGG-16, consistent with the model routing Gaussian inputs differently from in-distribution images. The conditioned variant H_{surv} is at least 0.99 on every row of VGG-16 and ResNet-50 and for the RG on ViT-B/16, so it carries no class-separability signal there. For the SG on ViT-B/16 it separates random-vs-random (0.852) from the clean rows (0.957–0.968), but neither intraclass from interclass pairs (0.963 vs. 0.968) nor clean-vs-random (0.957) from the clean pairs.

		Stopping Game		Routing Game	
Architecture	Baseline	H	H_{surv}	H	H_{surv}
VGG-16	clean-intraclass	0.513 ± 0.071	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
	clean-interclass	0.607 ± 0.032	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
	clean-vs-random	0.663 ± 0.020	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
	random-vs-random	0.320 ± 0.019	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
ResNet-50	clean-intraclass	0.272 ± 0.056	0.999 ± 0.000	0.786 ± 0.044	1.000 ± 0.000
	clean-interclass	0.268 ± 0.045	1.000 ± 0.000	0.902 ± 0.033	1.000 ± 0.000
	clean-vs-random	0.269 ± 0.035	1.000 ± 0.000	0.943 ± 0.010	1.000 ± 0.000
	random-vs-random	0.152 ± 0.008	0.992 ± 0.000	0.739 ± 0.011	0.999 ± 0.000
ViT-B/16	clean-intraclass	0.422 ± 0.023	0.969 ± 0.008	0.789 ± 0.067	1.000 ± 0.000
	clean-interclass	0.435 ± 0.017	0.974 ± 0.005	0.882 ± 0.010	1.000 ± 0.000
	clean-vs-random	0.439 ± 0.014	0.963 ± 0.006	0.891 ± 0.007	1.000 ± 0.000
	random-vs-random	0.377 ± 0.013	0.867 ± 0.013	0.476 ± 0.018	0.997 ± 0.000

Table 5: Reference baselines for the input-noise sweep, averaged over R pairs drawn from the 565-image ImageNet-S test split (clean rows: $R = 200$ sampled pairs; clean-vs-random: $R = 565$ images paired with a fresh Gaussian each; random-vs-random: $R = 200$ Gaussian pairs); entries are mean ± standard deviation over pairs. **clean-intraclass**: $H(\Pi_A(x_1), \Pi_A(x_2))$ with x_1, x_2 from the same ground-truth class; **clean-interclass**: same, with distinct classes; **clean-vs-random**: one side clean, other Gaussian; **random-vs-random**: both sides Gaussian.

H RANDOMIZATION SANITY CHECKS

This appendix collects the theoretical and empirical material behind the cascading-randomization experiments of Section 6. §H.1 carries the protocol of Adebayo et al. (2018) over to the trajectory-space Hellinger diagnostic of Appendix G; §H.2 is the pixel-level heatmap-similarity sanity check on the main-paper attribution methods; §H.3 gives the fully random reference; §H.4 compares trajectory and map diagnostics per image.

H.1 Cascading Randomization in Trajectory Space

We apply the Hellinger framework of Appendix G to the parameter randomization test of Adebayo et al. (2018). Replacing one weight matrix yields two kernels on the same layered graph, and the Hellinger pass quantifies the resulting trajectory-space change. Notation follows Appendix G: $\widetilde{T}^{(l)}$ is the geometric-mean kernel and $g^{(l)}(s)$ the overlap mass reaching state s, so $\mathrm{BC}^{(l)} = \sum_{s\in\mathcal{S}_+^{(l)}} g^{(l)}(s)$; $\mathrm{BC}^{(l)}_{\mathrm{ord}}$ denotes the same sum over ordinary states $s \in \mathcal{S}^{(l)}$ only.

Lemma 7 (Alive-set mismatch)**.** *For backward kernels $T_A^{(l)}, T_B^{(l)}$ governed by alive sets via $T_M^{(l)}(s_i \to \dagger) = \mathbf{1}\{i \notin \mathrm{alive}_M^{(l)}\}$, every state in the alive symmetric difference satisfies $\sum_{s'} \widetilde{T}^{(l)}(s_i, s') = 0$ and contributes nothing to $g^{(l-1)}$.*

Proof. WLOG $i \in \mathrm{alive}_A^{(l)} \setminus \mathrm{alive}_B^{(l)}$, so $T_A^{(l)}(s_i, \cdot)$ is supported on ordinary states and $T_B^{(l)}(s_i, \cdot) = \delta_\dagger$. Then $\widetilde{T}^{(l)}(s_i, s') = \sqrt{T_A^{(l)}(s_i, s') \cdot 0} = 0$ for ordinary s' and $\widetilde{T}^{(l)}(s_i, \dagger) = \sqrt{0 \cdot 1} = 0$. □

Fix a ReLU MLP with hidden widths $n_1, \ldots, n_L$, an input x, and start state s_0 at the target output. Let B replace $W^{(l^*)}$ by an i.i.d. matrix from a continuous distribution symmetric about zero, and assume that for each $l \geq l^*$ and every neuron i, $\mathbb{P}_B\big(i \in \mathrm{alive}_B^{(l)} \mid \text{kernels above layer } l\big) \leq 1 - \delta$ for some $\delta \in (0, \frac{1}{2}]$. Then $T_A^{(l)} = T_B^{(l)}$ for $l < l^*$; at $l = l^*$ the row weights differ and the alive sets may differ; for $l > l^*$ the perturbation propagates through the activations. The same reduction applies to convolutional and transformer blocks interpreted as single Layered Markov Process (LMP) layers. Write $\eta \coloneqq 1 - \delta \in \big[\frac{1}{2}, 1\big)$ and $N \coloneqq L - l^* + 1$.

Theorem 14 (Randomization in the Stopping Game)**.** $\mathbb{E}_B\big[\mathrm{BC}^{(0)}_{\mathrm{ord}}(\Pi_A, \Pi_B)\big] \leq \eta^N$. *Hence, for every $z > 0$, the conditioned-survival distance of Theorem 13 satisfies $\mathbb{E}_B\big[\mathrm{H}^2_{\mathrm{surv}}\, \mathbf{1}\{Z_A Z_B \geq z\}\big] \geq \mathbb{P}_B(Z_A Z_B \geq z) - \eta^N/\sqrt{z}$, while the plain distance is $\mathrm{H}^2 = 1 - \mathrm{BC}^{(0)}_{\mathrm{ord}} - g^{(0)}(\dagger)$.*

Proof. At a layer $l \geq l^*$, ordinary overlap mass at a state dead in both models moves into the common cemetery, mass on the alive symmetric difference vanishes (Lemma 7), and on states alive in both models the row mass is at most 1 by Cauchy–Schwarz. Hence $\mathrm{BC}^{(l-1)}_{\mathrm{ord}} \leq \sum_{s\in\mathcal{S}^{(l)}} g^{(l)}(s)\, \mathbf{1}\{s \in \mathrm{alive}_B^{(l)}\}$, and, since $g^{(l)}$ is fixed by the kernels above layer l, the conditional bound gives $\mathbb{E}_B[\mathrm{BC}^{(l-1)}_{\mathrm{ord}}] \leq \eta\, \mathbb{E}_B[\mathrm{BC}^{(l)}_{\mathrm{ord}}]$. Iterating from $\mathrm{BC}^{(L)}_{\mathrm{ord}} = 1$ over $l = L, \ldots, l^*$ and using that $\mathrm{BC}^{(l)}_{\mathrm{ord}}$ is non-increasing below l^* yields the bound. The $\mathrm{H}_{\mathrm{surv}}$ statement follows from $\mathrm{H}^2_{\mathrm{surv}} = 1 - \mathrm{BC}^{(0)}_{\mathrm{ord}}/\sqrt{Z_A Z_B} \geq 1 - \mathrm{BC}^{(0)}_{\mathrm{ord}}/\sqrt{z}$ on $\{Z_A Z_B \geq z\}$ (Theorem 13), the plain one from $\mathrm{BC}^{(0)} = \mathrm{BC}^{(0)}_{\mathrm{ord}} + g^{(0)}(\dagger)$. □

Remark 7 (Cemetery floor)**.** *Theorem 14 bounds only the ordinary-state overlap. Overlap mass that reaches a neuron dead in both models enters the cemetery in both, $g^{(l-1)}(\dagger) = g^{(l)}(\dagger) + \sum_{s\in\mathrm{dead}_A^{(l)}\cap\mathrm{dead}_B^{(l)}} g^{(l)}(s)$, and is never thinned again, so plain* H *saturates below one. In a mean-field approximation with uniform dead fraction q under A and alive probability p under B, the ordinary overlap shrinks by $(1-q)p$ per layer while a fraction $q(1-p)$ is absorbed, so $g^{(0)}(\dagger) \to q(1-p)/(1-(1-q)p)$ as $N \to \infty$; for illustrative values $q = 0.2$, $p = \frac{1}{2}$ this gives* $\mathrm{H} \approx 0.91$*, the same order as the observed* 0.94–0.97 *(Table 7). The conditioned-survival distance removes the cemetery floor exactly, and $\mathbb{E}_B[\mathrm{H}^2_{\mathrm{surv}} \mid Z_A Z_B \geq z] \to 1$ for every fixed $z > 0$ with $\mathbb{P}_B(Z_A Z_B \geq z)$ bounded away from zero. For the Routing Game the activation-weighted kernel concentrates $g^{(l)}$ on neurons with largest $|aW|$, which rarely coincide with $\mathrm{dead}_A \cap \mathrm{dead}_B$, so* H *is nearly identical to* $\mathrm{H}_{\mathrm{surv}}$ *in practice.*

Theorem 15 (Randomization in the Routing Game)**.** *Under the same setup without the alive-set assumption, take the RG kernel at temperature $\tau > 0$ and assume that for every $l \geq l^*$ each state alive in both models has row mass $\sum_{s'} \widetilde{T}^{(l)}(s, s') \leq \bar{b} < 1$. Then $\mathrm{BC}^{(0)}_{\mathrm{ord}}(\Pi_A, \Pi_B) \leq \bar{b}^N$.*

Proof. Since $a_{M,i}^{(l-1)} = 0$ for $i \notin \text{alive}_M^{(l-1)}$, the Gibbs policy (149) sends no mass to dead neurons, so for $1 \le l < L$ the overlap $g^{(l)}$ vanishes outside $\text{alive}_A^{(l)} \cap \text{alive}_B^{(l)}$, and the start state is alive in both, in the symmetric difference (Lemma 7) or dead in both. Hence $\text{BC}_{\text{ord}}^{(l-1)} \le \bar{b}\,\text{BC}_{\text{ord}}^{(l)}$ for every $l \ge l^*$, with no alive-set factor (the alive mismatch at layer $l-1$ is already part of the row masses at layer l); iterate as in Theorem 14. The assumption reflects the structure of the RG: each act-to-act row factors as the source-deadness indicator $\mathbf{1}\{z_j^{(l)} > 0\}$, the constant $\frac{1}{2}$ uniform κ-branch, and the Gibbs policy $\pi_{M,\kappa}^{(l)}(\cdot \mid j) \propto [a_M^{(l-1)}[\,\cdot\,]\, W_M^{(l,\kappa)}[\,\cdot, j]]^{1/\tau}$. At $l = l^*$ the two policies share $a^{(l^*-1)}$, but $W_B^{(l^*)}$ drawn from a continuous distribution makes the normalized Gibbs rows differ a.s.; for $l > l^*$ the activation factors differ as well. By the equality case of Cauchy–Schwarz every such row mass is then < 1 a.s., whereas SG rows on states alive in both models coincide for $l > l^*$. □

The input-layer marginal $m_M^{(0)}$ recovers gradient and $\alpha\beta$-LRP maps (Theorems 6 and 8). Theorems 14–15 show that Π_M changes drastically under randomization, while Table 6 shows that $m_M^{(0)}$ can remain similar. The two diagnostics therefore detect different aspects of parameter sensitivity: the map-level test of Adebayo et al. (2018) asks whether the explanation delivered to the user depends on the learned parameters, the trajectory-level test whether the calculation behind it does. A map can fail the first while its calculation passes the second; the trajectory-level result locates where the sensitivity is lost (between the calculation and its pixel projection), but it does not make such a map adequate for model debugging (Adebayo et al., 2018; Sixt et al., 2020).

H.2 Heatmap similarity at full randomization

We apply the cascading parameter randomization test of Adebayo et al. (2018) to the attribution methods of Table 6. Starting from a pretrained network, we cumulatively re-initialize the convolutional and linear layers from the output toward the input (weights Kaiming normal (He et al., 2015), biases $\mathcal{N}(0, 0.01^2)$); BatchNorm (ResNet-50) and LayerNorm, the packed query/key/value projection of the attention, the positional embedding and the class token (ViT-B/16) keep their pretrained values, here and in the Hellinger cascade of Section 6.1. At each randomization depth we compute the attribution for every test image under both the original and the (partially) randomized model, then measure the similarity between the two attribution maps. A method that genuinely reflects the learned parameters should produce increasingly dissimilar maps as more layers are randomized; a method that merely responds to low-level input structure (edges, textures) will show high similarity even at full randomization.

Metrics. We report two complementary similarity measures between attribution maps:

- **Spearman** $\rho(|a|)$: The Spearman rank correlation of the absolute attribution values, measuring whether the two maps rank pixels in the same order of importance. We use absolute values because sign conventions may differ between original and randomized attributions; the correlation itself is signed.
- **HOG** r: The Pearson correlation between Histograms of Oriented Gradients (HOG) descriptors (Dalal and Triggs, 2005) extracted from the attribution maps, capturing structural similarity at the level of local gradient orientations.

We intentionally do *not* report SSIM here: Binder et al. (2023) show that randomization-based similarity scores can remain deceptively high because input-dependent structure and activation scales survive top-down randomization. All values are averaged over the 565-image test split of ImageNet-S (50 classes); a map that is constant over the image (LIME and KernelSHAP on the fully randomized ResNet-50, whose output no longer depends on the superpixels) has similarity 0 by convention. Table 6 reports the similarity at *full randomization* (all convolutional and linear layers re-initialized).

Discussion. Several patterns emerge. First, Gradient and Grad-CAM stay closest to zero on the CNNs (and LayerCAM on VGG-16), and Gradient, SmoothGrad and GAE on ViT-B/16: they are sensitive to the parameters, though this does not imply that they are faithful. Second, LRP-ε keeps the most rank structure on VGG-16 (0.341); on ResNet-50 the LRP rules (LRP-γ 0.254, LRP-ε 0.250) are close to Integrated Gradients and LayerCAM, which keep the most there. Third, LIME and KernelSHAP keep a HOG correlation of about 0.86 on VGG-16 and ViT-B/16 at low Spearman correlation: their superpixels are computed from the input image, so the maps inherit

its edge structure; on ResNet-50 the fully randomized model's output no longer depends on the superpixels and their maps are constant (‡). Fourth, the RG rows lie between Gradient and LRP-ε on VGG-16, slightly below the LRP rules on ResNet-50, and keep the highest Spearman correlation of all methods on ViT-B/16 (0.454 for the anchor, 0.450 for the selected mode). Two maps become anti-correlated rather than unrelated (Grad-CAM++ on ResNet-50, TIBAV on ViT-B/16). The complementary trajectory-space analysis of §H.1 measures divergence of the underlying trajectory distributions rather than pixel-level map similarity.

Table 6: Parameter randomization sanity check (Adebayo et al., 2018). Each cell reports the average similarity between the attribution of the original pretrained model and the attribution of a fully randomized copy (every convolutional and linear layer re-initialized; normalization layers, the attention's query/key/value projection and the ViT embeddings stay pretrained). Values near zero indicate that the method detects the parameter change; the Spearman correlation is signed, so negative values occur. Bold marks the value closest to zero per metric within each architecture, among maps that are not constant; ‡: the map is constant at full randomization (similarity 0 by convention). Rows: the reported configuration of every method (Section 7).

Method	Spear. $\rho(\lvert a\rvert)$	HOG r
VGG-16		
Gradient	0.013	**0.009**
SmGrad	0.084	0.101
IntGrad	0.207	0.348
DeepLift	0.216	0.340
LRP-ε	0.341	0.457
LRP-γ	0.092	0.370
GCAM	**-0.007**	0.069
GCAM++	0.180	0.129
LCAM	-0.011	0.052
LIME	0.257	0.864
KernelSHAP	0.075	0.868
SG (λ=$-1, \sigma^2$=5, gate on)	0.289	0.456
RG anchor	0.299	0.535
RG (τ=0.3, ε=0.1)	0.284	0.287
ResNet-50		
Gradient	0.148	0.134
SmGrad	0.233	0.214
IntGrad	0.277	0.424
DeepLift	0.224	0.351
LRP-ε	0.250	0.393
LRP-γ	0.254	0.130
GCAM	**0.020**	**0.072**
GCAM++	-0.502	0.151
LCAM	0.266	0.145
LIME‡	0.000	0.000
KernelSHAP‡	0.000	0.000
RG anchor	0.217	0.344
RG (τ=0.9, ε=1)	0.180	0.316
ViT-B/16		
Gradient	**-0.003**	0.041
SmGrad	0.010	**0.038**
IntGrad	0.272	0.362
LRP-ε	0.165	0.300
LRP-γ	0.027	0.234
AttnLRP	0.124	0.176
DAVE	0.178	0.323
TIBAV	-0.175	0.150
GAE	-0.009	0.130
AR	0.120	0.153
LIME	0.248	0.861
KernelSHAP	0.131	0.867
RG anchor	0.454	0.469
RG (τ=0.4, λ_{sm}=$-10, \sigma^2$=5, ε=0.1)	0.450	0.310

Spear. $\rho(|a|)$ = Spearman rank correlation of absolute attribution values; HOG r = Pearson correlation of HOG descriptors. RG anchor = the $\alpha\beta$-LRP representation (τ=1, ε=0.5); the second RG row and the SG row are the localization-selected configurations and list the modes that differ (τ = temperature; λ, λ_{sm} = risk aversion; σ^2 = ADF input variance; ε = LRP stabilizer; $\tilde{\Phi}$ = noisy-observation gate).

H.3 Double-Random Empirical Baseline

To calibrate the asymptotic value that Theorems 14–15 approach under *full* randomization, we compare two *independently random-initialized* copies of each architecture. This endpoint matches the deepest step of the cascading test of Adebayo et al. (2018) while eliminating any residual correlation with the pretrained weights.

Protocol. For each architecture we run $R = 500$ independent rounds. Each round: *(i)* instantiate two copies A_r, B_r of the pretrained checkpoint and re-initialize every `Conv2d`/`Linear` layer of each copy independently as in the cascade, matching its endpoint; the remaining parameters (normalization layers, the attention's packed query/key/value projection, positional embedding and class token) keep their pretrained values in both copies (starting from the checkpoint rather than from `weights=None`, whose ViT classification head is identically zero and would make the two trajectory laws trivially indistinguishable; the same construction is used for all three architectures). *(ii)* Draw one image index with replacement from the 565-image ImageNet-S test split (the 500 draws hit 336 distinct images under the fixed seed we use). *(iii)* Compute the plain Hellinger $\mathrm{H}(\Pi_{A_r}, \Pi_{B_r})$ and conditioned-survival $\mathrm{H}_{\mathrm{surv}}$ for both the Stopping Game and Routing Game kernels using the ground-truth label as target class. Table 7 reports the round-to-round mean and standard deviation; the round-to-round variance mixes two sources (image choice and the random-pair realization).

	Stopping Game		Routing Game	
Architecture	H	$\mathrm{H}_{\mathrm{surv}}$	H	$\mathrm{H}_{\mathrm{surv}}$
VGG-16	0.939 ± 0.003	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
ResNet-50	0.938 ± 0.003	1.000 ± 0.000	0.973 ± 0.002	1.000 ± 0.000
ViT-B/16	0.958 ± 0.002	1.000 ± 0.000	0.956 ± 0.001	1.000 ± 0.000

Table 7: Hellinger trajectory distance between two independently random-initialized copies of the same architecture, averaged over $R = 500$ rounds (one fresh ImageNet-S test image per round). Plain H concentrates at an architecture-dependent cemetery floor of 0.94–0.97; conditioned-survival $\mathrm{H}_{\mathrm{surv}}$ saturates at unity. The VGG-16 Routing Game cell is the one case in which the plain distance also saturates at 1.000.

Interpretation. Plain H concentrates at a model-dependent floor of 0.94–0.97, matching the cemetery-floor prediction of Remark 7: once both random kernels produce uncorrelated dead masks, the shared dead-in-both coincidence rate bounds $\mathrm{BC}^{(0)}$ below by a strictly positive constant and plain H cannot reach unity. The conditioned-survival distance removes the cemetery floor by renormalizing on survival and correctly reports the two independent random trajectory laws as essentially support-disjoint on ordinary input states ($\mathrm{H}_{\mathrm{surv}} \approx 1$ throughout). The pretrained-vs-randomized Hellinger distance that the cascading sanity check reports should therefore be read against this architecture-specific asymptote rather than against unity: values close to the double-random baseline already indicate that the game-theoretic trajectory law has become fully decoupled from the pretrained one.

H.4 Per-Image Trajectory-vs-Map Independence

Theorems 14–15 and the double-random baseline of Table 7 bound the *aggregate* behavior of the Hellinger trajectory distance and the pixel-level attribution-map similarity. A sharper version of the argument of §H.1 is the claim that the two diagnostics disagree *per image*: at fixed randomization depth, whether a given image's trajectory distribution is strongly displaced by the randomization need not predict whether its attribution map becomes more dissimilar. Table 8 tests this directly.

Protocol. For each architecture we pick a cascading step at which the Hellinger run and the attribution-sanity run have randomized the same part of the network, so per-image values are directly comparable: the layers down to `features.26` on VGG-16 and down to `layer4.1.conv3` on ResNet-50 (identical sets of re-initialized layers), and the last encoder block on ViT-B/16, where the Hellinger cascade re-initializes the MLP and the output projection of the block and the attribution cascade additionally its Q, K and V projections. The mean H / $\mathrm{H}_{\mathrm{surv}}$ at these steps can be read off Figure 2. For each of the 565 ImageNet-S test images we then record four Hellinger scalars (H plain/surv for the Stopping and Routing Games) and four attribution-map scalars (Spearman $\rho(|a|)$ and HOG, each computed between the pretrained and partially-randomized model, for Gradient and for the

$\alpha\beta$-RG anchor at τ=1; these are the attribution methods whose marginal laws line up with the Hellinger game kernels). Table 8 reports Spearman rank correlations across the 565 images.

		Within Hellinger				Gradient		RG anchor	
Architecture	Hellinger variant	$H_{RG,pl}$	$H_{RG,surv}$	$H_{SG,pl}$	$H_{SG,surv}$	Spear. $\rho(\lvert a\rvert)$	HOG r	Spear. $\rho(\lvert a\rvert)$	HOG r
VGG-16	$H_{RG,plain}$	–	+1.00	−0.33	+0.85	+0.07	+0.10	−0.01	+0.04
	$H_{RG,surv}$	+1.00	–	−0.33	+0.85	+0.07	+0.10	−0.01	+0.04
	$H_{SG,plain}$	−0.33	−0.33	–	−0.30	+0.12	+0.11	+0.00	−0.02
	$H_{SG,surv}$	+0.85	+0.85	−0.30	–	+0.01	+0.05	−0.10	−0.10
ResNet-50	$H_{RG,plain}$	–	+0.58	−0.08	+0.70	+0.05	−0.01	−0.08	−0.19
	$H_{RG,surv}$	+0.58	–	+0.16	+0.89	+0.15	+0.08	+0.05	−0.05
	$H_{SG,plain}$	−0.08	+0.16	–	+0.16	−0.04	+0.07	−0.01	+0.02
	$H_{SG,surv}$	+0.70	+0.89	+0.16	–	+0.16	+0.07	−0.01	−0.10
ViT-B/16	$H_{RG,plain}$	–	+0.96	+0.29	+0.30	+0.02	+0.00	−0.04	−0.08
	$H_{RG,surv}$	+0.96	–	+0.25	+0.28	+0.02	+0.02	−0.07	−0.07
	$H_{SG,plain}$	+0.29	+0.25	–	+0.99	−0.12	−0.06	−0.03	−0.05
	$H_{SG,surv}$	+0.30	+0.28	+0.99	–	−0.12	−0.06	−0.04	−0.05

Table 8: Per-image Spearman rank correlations across the 565-image ImageNet-S test split at the aligned cascading steps of the Hellinger and attribution-sanity pipelines. Left block: within-Hellinger cross-correlations of the plain ($H_{\cdot,pl}$) and survival-conditioned ($H_{\cdot,surv}$) distances under the RG and SG kernels (self-entries shown as −; trivially +1). Right block: Hellinger-vs-map cross-correlations. The Hellinger-vs-map block is uniformly $|\rho| \leq 0.19$: per-image Hellinger and per-image attribution-map similarity rank images essentially independently.

Readings. The Hellinger-vs-map cross-block in Table 8 is uniformly near zero ($|\rho| \leq 0.19$ on all three architectures): per-image trajectory displacement and per-image attribution-map drift rank images essentially independently.

The within-Hellinger block shows that per-image H carries a stable ordering of the images. Where little trajectory mass reaches the cemetery, the plain and conditioned distances of the same game rank images almost identically: $\rho = +0.96$ (RG) and +0.99 (SG) on ViT-B/16, and identical values for the RG on VGG-16, whose kernel routes no trajectories to states of dead neurons. Under the SG on the ReLU backbones the two normalizations decouple ($\rho = -0.30$ on VGG-16, +0.16 on ResNet-50): the cemetery share of plain H varies with the image's dead masks, and the renormalization by $1/\sqrt{Z_A Z_B}$ removes exactly this share. The conditioned distances of the two games agree strongly on the ReLU backbones ($\rho = +0.85$ on VGG-16, +0.89 on ResNet-50) and only weakly on ViT-B/16 (+0.28), where the two kernels read partly different per-image structure.

A secondary observation from the within-method pattern: Spearman $\rho(|a|)$ and HOG within a single attribution method agree well ($\rho = 0.65$–0.82, computed from the same per-image joins), but Gradient and $\alpha\beta$-RG similarities barely agree across images (ρ between −0.10 and +0.30 depending on architecture and similarity metric). Together with the near-zero Hellinger-vs-map block, this tightens the conclusion of §H.1: trajectory-space and pixel-level diagnostics look at orthogonal per-image structure, and within pixel-level diagnostics the choice of attribution method changes *which* images a randomization sanity check flags.

I EVALUATION PROTOCOL

I.1 Data and splits

We evaluate on a *custom* 50-class subset of ImageNet-S (Gao et al., 2023), sampled once at the start of the project and held fixed across all experiments. The subset has two partitions: a **custom validation split of 50 images** (one image per class, seed 42) used only for configuration selection, and the **test split** of the remaining images, used for every reported test number. Images whose object mask lies entirely outside the 224×224 center crop are excluded after the split, because the mask-based metrics are undefined for them; this removes one test image and leaves **565 test images**. The Pascal VOC 2012 transfer split is described in Appendix J.4.

Backbone and dataset provenance. All main-paper experiments use the pretrained torchvision checkpoints shipped with the library release used for evaluation. Specifically, VGG-16 uses `VGG16_Weights.IMAGENET1K_V1` (`https://download.pytorch.org/models/vgg16-397923af.pth`; recipe `https://github.com/pytorch/vision/tree/main/references/classification#alexnet-and-vgg`), ResNet-50 uses `ResNet50_Weights.IMAGENET1K_V2` (`https://download.pytorch.org/models/resnet50-11ad3fa6.pth`; recipe `https://github.com/pytorch/vision/issues/3995#issuecomment-1013906621`), and ViT-B/16 uses `ViT_B_16_Weights.IMAGENET1K_V1` (`https://download.pytorch.org/models/vit_b_16-c867db91.pth`; recipe `https://github.com/pytorch/vision/tree/main/references/classification#vit_b_16`). These are ImageNet-1K classifiers trained on the ImageNet / ILSVRC benchmark (Deng et al., 2009; Russakovsky et al., 2015). Quantitative scoring is performed with Quantus (Hedström et al., 2023).

Licenses. ImageNet images are used under the ImageNet terms of access (non-commercial research and educational use) (Deng et al., 2009). The ImageNet-S masks (Gao et al., 2023) are released by their authors without an explicit license; their images are ImageNet images and fall under the same terms. The Pascal VOC 2012 images (Everingham et al., 2010) are subject to the Flickr terms of use, as stated by the dataset providers. The checkpoints are distributed with torchvision (BSD-3-Clause) and were trained on ImageNet-1K. Libraries: PyTorch and torchvision (BSD-3-Clause), Quantus (GPL-3.0), zennit (GPL-3.0/LGPL-3.0), lxt, captum and scikit-image (BSD-3-Clause), pytorch-grad-cam (MIT); the reference code of TIBAV and GAE (MIT) is included with its license notice.

I.2 Methods

Gradient-based block. **Gradient** is the raw input-gradient saliency map (Simonyan et al., 2014), the maximum over color channels of the absolute gradient. **SmGrad** is SmoothGrad, i.e. an average over noisy gradient maps (Smilkov et al., 2017). **IntGrad** is Integrated Gradients (Sundararajan et al., 2017) with a black-image baseline. **DeepLift** propagates activation differences relative to a reference input with the Rescale rule (Shrikumar et al., 2017) (CNNs only). **DAVE** is the distribution-aware ViT gradient decomposition of Wróbel et al. (2026). **SG** is our SG, included in this block because it recovers the gradient as a Stopping-Game occupation measure (Theorem 6).

LRP / relevance-propagation block. **LRP-ε** denotes the stabilized Layer-wise Relevance Propagation rule (Bach et al., 2015). **LRP-γ** is the positive-weight-amplified LRP variant used in best-practice comparisons (Kohlbrenner et al., 2020). **AttnLRP** is transformer LRP that propagates relevance through the softmax and the bilinear attention product as well as the Value path (Achtibat et al., 2024). **RG** is our RG. **RG+Eq** is the same method with equivariant Reynolds averaging, used mainly on ViT to suppress tokenization-grid artifacts. **TIBAV** (Chefer et al., 2021b), **GAE** (Chefer et al., 2021a) and Attention Rollout (**AR**) are the transformer-specific attention-map baselines.

CAM block. **GCAM** is Grad-CAM (Selvaraju et al., 2017), **GCAM++** is Grad-CAM++ (Chattopadhay et al., 2018), and **LCAM** is LayerCAM, which weights class-activation evidence per location (Jiang et al., 2021). All three are computed at the last convolutional block of the CNNs and upsampled bilinearly.

Perturbation block. **LIME** is Local Interpretable Model-agnostic Explanations (Ribeiro et al., 2016). **KShap** denotes KernelSHAP, the model-agnostic weighted-regression estimator of SHAP values (Lundberg and Lee,

2017). Both perturb SLIC superpixels (Achanta et al., 2012). These methods explain predictions by perturbing the input and fitting a local surrogate, so they are separated from the backward-propagation methods above.

I.3 Metric definitions

Let $x \in \mathbb{R}^{C\times H\times W}$ be an image, let y be its target class, and let $a \in \mathbb{R}^{H\times W}$ be the scalar attribution map produced by an explanation method after channel reduction. Write $d \coloneqq HW$ and let $\mathrm{vec}(a) \in \mathbb{R}^d$ denote the flattened map. For any binary segmentation mask $m \in \{0,1\}^{H\times W}$, write $\mathrm{vec}(m) \in \{0,1\}^d$. When a metric is configured with `abs=true` and `normalise=true`, the evaluator applies the max-abs normalization (not min–max)

$$\psi(a) = \frac{|\mathrm{vec}(a)|}{\max_i |\mathrm{vec}(a)_i|}, \tag{197}$$

with the convention that the all-zero map remains zero. We denote the preprocessed attribution vector by $\bar{a}$. All metrics below use $\bar{a}$; faithfulness metrics use raw logits (no softmax) to avoid saturation effects, and the perturbation baseline is the channel-wise mean.

Attribution localization (AL). The inside-mass ratio of Kohlbrenner et al. (2020), who compute it with ground-truth bounding boxes; we use the ImageNet-S pixel masks:

$$\mathrm{AL}(a,m) = \frac{\sum_{i=1}^{d} \bar{a}_i m_i}{\sum_{i=1}^{d} \bar{a}_i + 10^{-9}}. \tag{198}$$

Pointing game (PG). Following the Excitation Backprop protocol (Zhang et al., 2018), the pointing-game config uses `abs=true` and no normalization. Let $M(a) \coloneqq \{i \in \{1,\dots,d\} : \bar{a}_i = \max_j \bar{a}_j\}$ be the entire set of maximizers. Then

$$\mathrm{PG}(a,m) = \mathbf{1}\big\{M(a) \cap \{i : m_i = 1\} \neq \varnothing\big\}. \tag{199}$$

Ties are scored as a hit whenever at least one maximizer lies inside the object mask. PG is binary per image, so its test mean is a multiple of $1/565$.

Top-k intersection (TK). Following Theiner et al. (2022), with $k = 1000$ and the permutation π that sorts $\bar{a}$ in descending order:

$$\mathrm{TK}(a,m) = \frac{1}{1000} \sum_{r=1}^{1000} m_{\pi_r}. \tag{200}$$

Pixel-flipping AUC ($\mathbf{PF}_5$, $\mathbf{PF}_{20}$). Pixel flipping (Bach et al., 2015; Samek et al., 2017) removes the most relevant features first and tracks the class score. The evaluator uses `pixel_flipping_abs` with `features_in_step = 1568` and `perturb_baseline = mean`. For $T \coloneqq \lceil d/1568 \rceil$, define perturbed inputs $x^{(t)}$ by replacing features $\pi_{(t-1)\cdot 1568+1}, \dots, \pi_{\min\{t\cdot 1568, d\}}$ (and all earlier ones) with the per-sample mean baseline. The evaluator computes

$$f_t \coloneqq F(x^{(t)})_y, \qquad t = 1,\dots,T, \tag{201}$$

and PF_p is the trapezoidal area over the first $p\%$ of this flipping curve, i.e. over its first K_p points with the abscissa rescaled to $[0,1]$: $\mathrm{PF}_p = \mathrm{trapz}\big(\{(t-1)/(K_p-1), f_t\}_{t=1}^{K_p}\big)$, $K_p = \max(2, \lceil pT/100 \rceil)$; with $T = 96$ steps of 1568 features (the attribution is broadcast over the three color channels), $K_5 = 5$ and $K_{20} = 20$. Lower is better.

Selectivity (Sel). Following Montavon et al. (2018) and its Quantus implementation (Hedström et al., 2023), $\bar{a}$ is summed over square patches of 39×39 pixels (side $\lfloor\sqrt{1568}\rfloor$); non-overlapping patches are removed (replaced by the per-sample mean) in decreasing order of their summed attribution, and the target logit is recorded after each removal. Sel is the trapezoidal area under this curve, with the fraction of removed patches on the abscissa. Lower is better.

Max and average sensitivity (MaxS, AvgS). Following Yeh et al. (2019), the evaluator regenerates attributions on the same $N_s = 50$ inputs for both metrics, $x + \xi_k$ with uniform noise $\xi_k \sim \mathcal{U}[-r, r]^{C\times H\times W}$, $r = 0.2$, and reports the maximum (MaxS) or mean (AvgS) over k of $\|\psi(a(x+\xi_k)) - \psi(a(x))\|_F / \|\psi(a(x))\|_F$. Lower is better, as it indicates stability of the explanation to input noise. The noise is seeded per image, so every row sees the same perturbed inputs. MaxS and AvgS are computed on a fixed random subset of 200 test images (seed 0) that is identical for all rows and models; all other metrics use the full test split.

I.4 Declared validation grids

Table 9 lists the grid of every method. Functionally duplicate rows (e.g. different σ^2 when $\lambda = \lambda_{\text{sm}} = 0$) are evaluated once; in total 2,021 validation configurations are evaluated. On ViT-B/16 the activation-side λ is fixed to 0, which keeps GELU gates conditioned (Section 4.3). At attention–Value products, RG applies $\alpha\beta$-LRP-ε to the composite map of the attention product and the Value projection (Theorem 9; Appendix D.4). Baselines without a declared grid are run at the settings of their reference implementations.

Table 9: Per-method validation grids (all architectures unless stated). Symbols: Appendix J.

Method	Grid
Gradient	single config
SmGrad	N=50, noise level $\sigma/(x_{\max}-x_{\min}) \in \{0.1, 0.15, 0.2\}$, mean aggregation (3 configs)
IntGrad	black-image baseline, $N_{\text{steps}} \in \{20, 50, 100, 300\}$ (4 configs)
DeepLift	Rescale rule, single config (CNNs only)
DAVE	$N \in \{25, 50\} \times \theta_{\max} \in \{0, 10, 20\} \times s_{\max} \in \{0, 0.05, 0.1\} \times t_{\max} \in \{0, 0.25, 0.5\}$, horizontal flips with probability 0.5, averaged effective map multiplied by the input (54 configs; ViT-B/16 only)
SG	$\lambda \in \{-5, -2, -1, -0.3, 0\} \times \sigma^2 \in \{1, 2, 5, 10\}\times$ noisy gate $\tilde{\Phi} \in \{\text{off}, \text{on}\}$ (40 configs; VGG-16 only)
LRP-ε	$\varepsilon \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1, 2\}$ (7 configs)
LRP-γ	$(\varepsilon, \gamma) \in \{0.1, 0.25, 0.5\} \times \{0.1, 0.25, 0.5, 1, 2\}$; ViT-B/16: γ on all Linear layers (15 configs); CNNs: both zennit's EpsilonGammaBox composite ($z^{\mathcal{B}}$ input layer, γ on convolutions, ε on dense layers) and γ on all layers (30 configs)
AttnLRP	γ-composite of Achtibat et al. (2024): reference setting $(\gamma_{\text{conv}}, \gamma_{\text{lin}}, \varepsilon_{\text{attn}})$=$(0.25, 0.05, 10^{-6})$ + univariate sweeps $\gamma_{\text{lin}} \in \{0.01, 0.025, 0.1, 0.25\}$, $\gamma_{\text{conv}} \in \{0.1, 0.5, 1\}$, $\varepsilon_{\text{attn}} \in \{0.01, 0.05, 0.25\}$ (11 configs; ViT-B/16 only)
TIBAV	single config (ViT-B/16 only)
GAE	single config (ViT-B/16 only)
AR	residual weight $w_r \in \{0, 0.25, 0.5, 0.75\}$ (4 configs; ViT-B/16 only)
RG	$\tau \in \{0.3, 0.4, \ldots, 1.0\}$, (α, β)=$(2, 1)$, $\sigma^2 \in \{1, 2, 5, 10\}$, $\varepsilon \in \{0.1, 0.25, 0.5, 1\}$; ViT-B/16: λ=0, $\lambda_{\text{sm}} \in \{-50, -10, -5, -2, -1, 0\}$, composite Value rule of Theorem 9 (672 configs); CNNs: $\lambda \in \{-5, -2, -1, -0.3, 0\}$ (544 configs)
RG+Eq	$\tau \in \{0.3, 0.4, \ldots, 1.0\} \times \lambda_{\text{sm}} \in \{0, -10\}$, σ^2=2, ε=0.5, N=50, composite Value rule (16 configs; ViT-B/16 only)
GCAM, GCAM++, LCAM	last convolutional block (VGG-16: `features.29`, ResNet-50: `layer4`), single config (CNNs only)
LIME	80 SLIC superpixels, N=1000, ridge surrogate, exponential kernel of width 0.25 on the cosine distance of the binary superpixel vectors, removed superpixels filled with their mean color; fixed reference row
KShap	80 SLIC superpixels, N=$2M$+2048 for M superpixels, removed superpixels filled with the image mean; fixed reference row
Anchor	$\alpha\beta$-LRP anchor $(\alpha, \beta, \varepsilon, \tau)$=$(2, 1, 0.5, 1)$ of RG and RG+Eq, reported as the theorem-driven reference row (not selected)

I.5 Selection

For every tunable method, baselines and ours alike, two configurations are selected on the validation split: `loc` minimizes the rank-sum over $\{\text{AL}, \text{PG}, \text{TK}\}$ and `faith` the rank-sum over $\{\text{PF}_5, \text{PF}_{20}, \text{Sel}\}$. Ranks are computed within each method's grid, respecting metric directions; ties are broken in favor of the default (for RG, the anchor), then of fewer non-default parameters, then of grid order. A configuration is admissible only if its map is

non-zero and every metric is defined on every validation image; inadmissible configurations are excluded before ranking (5 of the 40 SG configurations on VGG-16, all with strong risk aversion and the deterministic gate, whose maps vanish on some images; no other configuration). MaxS and AvgS are not used for selection because each evaluation re-explains every image $2N_s$ times. The `loc` row is the one reported in the main tables, and the `faith` row of every method is reported alongside (Appendix J): localization is a natural target because the analytical modes are designed to control routing focus, because mask localization has an external ground truth whereas perturbation metrics evaluate off-distribution inputs, and because comparing LRP rule choices by object localization has precedent, there with bounding boxes (Kohlbrenner et al., 2020); localization is not faithfulness, so both rows are reported and the trade-off stays visible.

I.6 Statistics

Every test metric is reported with a 95% percentile bootstrap confidence interval over test images (Efron and Tibshirani, 1993) (10,000 resamples of the images; the same resampled image indices are used for all rows, so comparisons are paired). MaxS and AvgS are computed on the fixed 200-image subset of Appendix I.3, and their intervals resample the images of that subset. For each metric, every row is tested against the best row for that metric: continuous metrics with a two-sided paired bootstrap test of the mean per-image difference (the bootstrap distribution of the mean difference is centered at zero, and p is the fraction of centered resampled differences at least as large in absolute value as the observed one), and the binary PG with McNemar's test (McNemar, 1947) (χ^2 statistic with continuity correction). The p-values are Holm-corrected (Holm, 1979) within each metric: over the rows shown in the main table, over all rows of a column in the appendix tables, and, for the statement that a row is significantly ahead of every baseline row, over its comparisons with the baseline rows. We call a difference significant at level 0.05 after correction and describe all other differences as comparable.

I.7 Compute resources

The attribution evaluations are inference-only. The main cost is the repeated backward attribution pass over the validation and test images across the method grids, including the equivariant-Reynolds and noise-averaged configurations. Runtime measurements were taken on a single NVIDIA A100-SXM4-40GB with batch size 1; evaluation sweeps were distributed over several GPUs of the same type, with up to two evaluation processes per GPU. The full pipeline (validation grids, test evaluation, robustness, Pascal VOC, ablations and randomization diagnostics) required about 450 GPU-hours.

I.8 Code and correctness tests

The code (`https://github.com/JayPiZimmermann/Playing_The_Network_Backward_Official_Repository`) contains unit tests that pin the implemented backward rules to the statements of the appendix. Two of them concern the attention block. `tests/test_attention_ab_recovery.py` runs the RG hooks at $\tau = 1$ on a small attention block with signed inputs, a Value bias and $\varepsilon > 0$, in float64, and compares the result with an independent implementation of $\alpha\beta$-LRP-ε on the composite map $X \mapsto A(XW_v^\top + b_v)$, i.e. the target rule (107) of Theorem 9, for a single head and for a multi-head block in the layout of `nn.MultiheadAttention`; it also asserts that the result differs from the sequential variant of §D.1.1, which the released code does not offer (`av_rule` accepts only `'merged'`). `tests/test_hellinger_merged_attention.py` enumerates every backward trajectory of a small attention block under two models, at $\tau \in \{0.5, 1, 2\}$, and checks the per-input masses, the Bhattacharyya mass and the cemetery masses of the Hellinger pass (Appendix G) against this brute-force reference; a pre-norm residual block built from `nn.MultiheadAttention` must give the same distance as the same block with its attention matrix supplied as a constant.

J ADDITIONAL RESULTS

The validation, test and VOC tables of this appendix are generated by one pipeline from the declared grids and the selection rules of Appendix I.

Configuration symbols. The configuration columns of all tables use the following parameters. RG, RG+Eq and SG (Table 1): temperature τ, activation-side risk aversion λ, attention-logit risk aversion λ_{sm}, ADF input variance σ^2, LRP stabilizer ε, stream weights (α, β) (default $(2, 1)$), noisy-observation gate $\tilde{\Phi}$ ("gate on/off"), and the number N of transformed inputs averaged by RG+Eq. SmoothGrad: N noisy samples with noise standard deviation σ relative to the input range (σ/range). Integrated Gradients: N_{steps} integration steps from a black-image baseline. DAVE: N transformed inputs with rotations up to $\theta_{\max}$ degrees, shifts up to $s_{\max}$ of the image extent, and variance-preserving Gaussian noise of interpolation weight up to $t_{\max}$. LRP-ε and LRP-γ: stabilizer ε and positive-weight boost γ; on the CNNs, LRP-γ is zennit's composite with the $z^{\mathcal{B}}$ rule at the input layer. AttnLRP: γ-rule parameters γ_{conv} and γ_{lin} of the convolutional and linear layers and the stabilizer $\varepsilon_{\text{attn}}$ of the attention layers. AR (attention rollout): residual weight w_r. LIME and KernelSHAP: N perturbation samples ($2M$+2048 for M superpixels) on 80 SLIC superpixels; LIME fits a ridge surrogate. DeepLift: Rescale rule. Grad-CAM, Grad-CAM++ and LayerCAM: the target layer (torchvision module name). Row labels in parentheses give the selection criterion (`loc`, `faith`); single-configuration methods have no criterion.

J.1 Validation selections

Tables 10–12 list, for every method, the configurations selected on the 50-image validation split under the `loc` and `faith` criteria of Appendix I.5, together with both validation rank-sums.

Table 10: ViT-B/16: validation-selected configurations per method and criterion (`loc`: minimal rank-sum over AL, PG, TK; `faith`: over PF_5, PF_{20}, Sel; rank-sums over the configurations of one method, lower is better; the number of configurations is given after the method name). Symbols: Appendix J.

Method	Criterion	Selected configuration	Loc. rank-sum	Faith. rank-sum
Gradient (1)	fixed	–	–	–
SmGrad (3)	loc	N=50, σ/range=0.2	3.5	3
	faith	N=50, σ/range=0.2	3.5	3
IntGrad (4)	loc	N_{steps}=100, black baseline	3.5	4
	faith	N_{steps}=100, black baseline	3.5	4
LRP-ε (7)	loc	ε=0.25	5.5	7
	faith	ε=0.1	6.5	3
LRP-γ (15)	loc	ε=0.1, γ=2	6	25
	faith	ε=0.1, γ=0.1	32	12
AttnLRP (11)	loc	γ_{conv}=0.25, γ_{lin}=0.05, $\varepsilon_{\text{attn}}$=0.25	3	29
	faith	γ_{conv}=0.25, γ_{lin}=0.05, $\varepsilon_{\text{attn}}$=0.01	14	7
DAVE (54)	loc	N=50, $\theta_{\max}$=0, $s_{\max}$=0, $t_{\max}$=0.5	14	6
	faith	N=50, $\theta_{\max}$=0, $s_{\max}$=0, $t_{\max}$=0.5	14	6
TIBAV (1)	fixed	–	–	–
GAE (1)	fixed	–	–	–
AR (4)	loc	w_r=0.75	3.5	3
	faith	w_r=0.75	3.5	3
LIME (1)	fixed	N=1000, 80 superpixels, ridge	–	–
KernelSHAP (1)	fixed	N=2M+2048, 80 superpixels	–	–
RG (672)	loc	τ=0.4, λ_{sm}=−10, σ^2=5, ε=0.1	47	788
	faith	τ=0.7, λ_{sm}=−5, σ^2=5, ε=0.1	1156	22
	anchor	τ=1	1420	344
RG+Eq (16)	loc	τ=0.5, λ_{sm}=−10, σ^2=2, N=50	6.5	29
	faith	τ=0.7, λ_{sm}=−10, σ^2=2, N=50	14	11
	anchor	τ=1, N=50	47	31

Table 11: VGG-16: validation-selected configurations per method and criterion, as in Table 10.

Method	Criterion	Selected configuration	Loc. rank-sum	Faith. rank-sum
Gradient (1)	fixed	–	–	–
SmGrad (3)	loc	N=50, σ/range=0.1	3	3
	faith	N=50, σ/range=0.1	3	3
IntGrad (4)	loc	N_{steps}=20, black baseline	6	3
	faith	N_{steps}=20, black baseline	6	3
DeepLift (1)	fixed	Rescale	–	–
LRP-ε (7)	loc	ε=0.5	6	9
	faith	ε=1	6	4
LRP-γ (30)	loc	ε=0.5, γ=0.25, $z^{\mathcal{B}}/\gamma/\varepsilon$ composite	13	43
	faith	ε=0.25, γ=0.1	36	5
GCAM (1)	fixed	`features.29`	–	–
GCAM++ (1)	fixed	`features.29`	–	–
LCAM (1)	fixed	`features.29`	–	–
LIME (1)	fixed	N=1000, 80 superpixels, ridge	–	–
KernelSHAP (1)	fixed	N=2M+2048, 80 superpixels	–	–
SG (40)	loc	λ=-1, σ^2=5, gate on	37	10
	faith	λ=-0.3, σ^2=1, gate off	49.5	6
RG (544)	loc	τ=0.3, ε=0.1	78	194
	faith	τ=0.8, ε=0.1	173.5	5
	anchor	τ=1	339	60

Table 12: ResNet-50: validation-selected configurations per method and criterion, as in Table 10.

Method	Criterion	Selected configuration	Loc. rank-sum	Faith. rank-sum
Gradient (1)	fixed	–	–	–
SmGrad (3)	loc	N=50, σ/range=0.15	4	5
	faith	N=50, σ/range=0.1	8	4
IntGrad (4)	loc	N_{steps}=100, black baseline	3	6
	faith	N_{steps}=50, black baseline	7.5	6
DeepLift (1)	fixed	Rescale	–	–
LRP-ε (7)	loc	ε=0.5	5.5	9
	faith	ε=2	8	4
LRP-γ (30)	loc	ε=0.5, γ=2, $z^{\mathcal{B}}/\gamma/\varepsilon$ composite	12.5	33
	faith	ε=0.5, γ=0.5, $z^{\mathcal{B}}/\gamma/\varepsilon$ composite	36	22.5
GCAM (1)	fixed	`layer4`	–	–
GCAM++ (1)	fixed	`layer4`	–	–
LCAM (1)	fixed	`layer4`	–	–
LIME (1)	fixed	N=1000, 80 superpixels, ridge	–	–
KernelSHAP (1)	fixed	N=2M+2048, 80 superpixels	–	–
RG (544)	loc	τ=0.9, ε=1	19	224
	faith	τ=1, λ=-1, σ^2=1, ε=0.25	466	38
	anchor	τ=1	212.5	140

J.2 Test results with confidence intervals

Tables 13–15 list every row evaluated on the 565-image test split (the `loc`- and `faith`-selected configuration of every method, the anchors and the single-configuration methods) with 95% bootstrap confidence intervals and

the tests of Appendix I.6 (bold and dark shading: best row of a column; $^{\approx}$ and light shading: not significantly different from it). Per-image runtimes come from the single-GPU benchmark of Appendix I.7.

Table 13: ViT-B/16, ImageNet-S test split: all rows with 95% bootstrap confidence intervals. Metrics as in Table 2; configuration symbols and row labels: Appendix J.

Method	Config	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓	MaxS↓	AvgS↓
Gradient	–	0.411 [0.389,0.432]	0.361 [0.322,0.402]	0.386 [0.359,0.413]	8.45 [8.33,8.56]	8.08 [7.94,8.22]	6.033 [5.905,6.154]	1.290 [1.233,1.351]	0.922 [0.893,0.953]
SmGrad (loc/faith)	N=50, σ/range=0.2	0.517 [0.498,0.536]	0.706 [0.667,0.743]	0.675 [0.654,0.694]	8.10 [7.96,8.24]	7.02 [6.83,7.20]	5.029 [4.894,5.163]	0.597 [0.569,0.630]	0.468 [0.451,0.485]
IntGrad (loc/faith)	N_{steps}=100,black baseline	0.458 [0.437,0.479]	0.513 [0.473,0.554]	0.498 [0.472,0.524]	8.37 [8.24,8.48]	7.93 [7.78,8.08]	5.712 [5.577,5.842]	1.090 [1.041,1.140]	0.825 [0.796,0.855]
LRP-ε (loc)	ε=0.25	0.592 [0.575,0.610]	0.834 [0.804,0.864]	0.746 [0.728,0.764]	7.97 [7.81,8.11]	6.88 [6.69,7.07]	4.859 [4.733,4.983]	0.486 [0.468,0.505]	0.384 [0.372,0.395]
LRP-ε (faith)	ε=0.1	0.582 [0.564,0.599]	0.834 [0.804,0.864]	0.748 [0.730,0.765]	7.84 [7.68,7.99]	6.76 [6.57,6.96]	4.890 [4.763,5.015]	0.623 [0.597,0.650]	0.498 [0.482,0.515]
LRP-γ (loc)	ε=0.1, γ=2	0.644 [0.628,0.660]	0.873 [0.844,0.899]	0.816 [0.800,0.832]	7.93 [7.78,8.08]	6.64 [6.44,6.83]	4.451$^{\approx}$ [4.324,4.572]	0.300 [0.287,0.314]	0.218 [0.210,0.226]
LRP-γ (faith)	ε=0.1, γ=0.1	0.614 [0.597,0.630]	0.844 [0.814,0.873]	0.779 [0.763,0.795]	7.73 [7.56,7.89]	6.40 [6.20,6.60]	4.603 [4.478,4.724]	0.394 [0.378,0.410]	0.309 [0.299,0.319]
AttnLRP (loc)	γ_{conv}=0.25, γ_{lin}=0.05, $\varepsilon_{\text{attn}}$=0.25	0.644 [0.628,0.660]	0.855 [0.825,0.883]	0.809 [0.794,0.825]	7.77 [7.61,7.93]	6.43 [6.23,6.62]	4.582 [4.452,4.706]	0.456 [0.430,0.484]	0.342 [0.324,0.364]
AttnLRP (faith)	γ_{conv}=0.25, γ_{lin}=0.05, $\varepsilon_{\text{attn}}$=0.01	0.629 [0.613,0.645]	0.850 [0.819,0.878]	0.798 [0.783,0.814]	**7.64** [7.47,7.81]	**6.12** [5.92,6.32]	**4.441** [4.314,4.563]	0.416 [0.395,0.438]	0.317 [0.304,0.331]
DAVE (loc/faith)	N=50, $\theta_{\max}$=0, $s_{\max}$=0, $t_{\max}$=0.5	0.491 [0.472,0.510]	0.717 [0.678,0.752]	0.612 [0.591,0.633]	8.12 [7.98,8.25]	7.37 [7.19,7.54]	5.386 [5.256,5.514]	0.781 [0.755,0.810]	0.643 [0.626,0.661]
TIBAV	–	0.615 [0.597,0.634]	0.878 [0.851,0.904]	0.782 [0.759,0.805]	8.13 [8.01,8.24]	7.60 [7.46,7.74]	4.906 [4.774,5.033]	1.062 [1.001,1.126]	0.512 [0.486,0.540]
GAE	–	0.533 [0.514,0.551]	0.743 [0.708,0.779]	0.649 [0.623,0.677]	8.13 [8.01,8.25]	7.62 [7.47,7.76]	5.098 [4.961,5.231]	0.525 [0.494,0.558]	0.368 [0.346,0.391]
AR (loc/faith)	w_r=0.75	0.447 [0.428,0.467]	0.565 [0.522,0.605]	0.537 [0.510,0.564]	8.44 [8.34,8.53]	8.06 [7.94,8.17]	5.254 [5.128,5.374]	0.208 [0.195,0.222]	0.122 [0.116,0.128]
LIME	N=1000,80 superpixels,ridge	0.535 [0.515,0.553]	0.897 [0.871,0.922]	0.686 [0.656,0.716]	8.36 [8.25,8.46]	7.68 [7.54,7.82]	5.376 [5.242,5.505]	0.937 [0.905,0.972]	0.668 [0.647,0.689]
KernelSHAP	N=2M+2048,80 superpixels	0.565 [0.548,0.583]	**0.961** [0.945,0.977]	0.823$^{\approx}$ [0.799,0.848]	8.29 [8.18,8.40]	7.50 [7.35,7.65]	5.140 [5.012,5.261]	0.897 [0.868,0.928]	0.650 [0.632,0.667]
RG anchor	τ=1	0.509 [0.490,0.528]	0.724 [0.685,0.759]	0.699 [0.678,0.720]	8.19 [8.05,8.31]	6.97 [6.78,7.15]	4.854 [4.732,4.972]	**0.166** [0.161,0.172]	0.120 [0.117,0.124]
RG (loc)	τ=0.4, λ_{sm}=−10, σ^2=5, ε=0.1	**0.711** [0.692,0.730]	0.825 [0.793,0.855]	0.788 [0.768,0.808]	8.27 [8.14,8.39]	7.48 [7.31,7.64]	5.109 [4.987,5.226]	0.729 [0.685,0.777]	0.494 [0.471,0.518]
RG (faith)	τ=0.7, λ_{sm}=−5, σ^2=5, ε=0.1	0.549 [0.531,0.567]	0.770 [0.736,0.804]	0.741 [0.721,0.760]	8.11 [7.97,8.24]	6.78 [6.59,6.97]	4.808 [4.686,4.925]	0.257 [0.245,0.268]	0.174 [0.168,0.179]
RG+Eq anchor	τ=1, N=50	0.520 [0.501,0.539]	0.761 [0.726,0.796]	0.752 [0.729,0.775]	8.41 [8.29,8.52]	7.72 [7.56,7.87]	4.864 [4.742,4.981]	0.170$^{\approx}$ [0.165,0.175]	**0.115** [0.112,0.118]
RG+Eq (loc)	τ=0.5, λ_{sm}=−10, σ^2=2, N=50	0.698 [0.680,0.717]	0.855 [0.827,0.883]	**0.836** [0.815,0.857]	8.39 [8.28,8.50]	7.80 [7.65,7.94]	5.010 [4.889,5.127]	0.441 [0.421,0.463]	0.301 [0.290,0.313]
RG+Eq (faith)	τ=0.7, λ_{sm}=−10, σ^2=2, N=50	0.654 [0.636,0.672]	0.812 [0.781,0.844]	0.820 [0.799,0.841]	8.39 [8.28,8.51]	7.71 [7.56,7.86]	4.856 [4.735,4.974]	0.310 [0.297,0.325]	0.199 [0.194,0.206]

Table 14: VGG-16, ImageNet-S test split: all rows with 95% bootstrap confidence intervals, as in Table 13.

Method	Config	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓	MaxS↓	AvgS↓
Gradient	–	0.522 [0.501,0.542]	0.738 [0.701,0.773]	0.688 [0.665,0.711]	14.47 [13.95,14.99]	10.96 [10.55,11.37]	6.470 [6.212,6.731]	1.212 [1.162,1.269]	0.947 [0.917,0.980]
SmGrad (loc/faith)	N=50, σ/range=0.1	0.533 [0.515,0.552]	0.782 [0.747,0.816]	0.723 [0.705,0.741]	14.12 [13.60,14.65]	9.24 [8.85,9.63]	5.768 [5.526,6.007]	0.641 [0.625,0.659]	0.523 [0.515,0.531]
IntGrad (loc/faith)	N_{steps}=20,black baseline	0.550 [0.530,0.571]	0.770 [0.735,0.804]	0.706 [0.683,0.730]	15.39 [14.86,15.95]	11.89 [11.46,12.32]	6.438 [6.174,6.694]	1.132 [1.089,1.179]	0.917 [0.890,0.945]
DeepLift	Rescale	0.542 [0.523,0.562]	0.701 [0.664,0.738]	0.654 [0.631,0.677]	13.05 [12.59,13.52]	9.32 [8.96,9.69]	6.049 [5.806,6.291]	1.193 [1.145,1.249]	0.948 [0.919,0.985]
LRP-ε (loc)	ε=0.5	0.613 [0.595,0.632]	0.832 [0.800,0.862]	0.745 [0.725,0.764]	12.63 [12.18,13.10]	8.61 [8.26,8.96]	5.685 [5.458,5.908]	1.173 [1.127,1.223]	0.918 [0.893,0.947]
LRP-ε (faith)	ε=1	0.639 [0.621,0.657]	0.809 [0.777,0.841]	0.744 [0.725,0.762]	**12.43** [11.98,12.89]	8.17 [7.83,8.50]	5.489 [5.270,5.704]	1.100 [1.059,1.143]	0.840 [0.820,0.861]
LRP-γ (loc)	ε=0.5, γ=0.25, $z^{\mathcal{B}}/\gamma/\varepsilon$ composite	0.601 [0.584,0.617]	0.692 [0.655,0.729]	0.704 [0.687,0.721]	14.83 [14.28,15.39]	10.07 [9.68,10.46]	**5.010** [4.799,5.219]	0.515 [0.489,0.542]	0.368 [0.353,0.384]
LRP-γ (faith)	ε=0.25, γ=0.1	0.615 [0.598,0.633]	0.793 [0.759,0.827]	0.698 [0.681,0.716]	12.66 [12.18,13.13]	**7.89** [7.56,8.23]	5.078 [4.867,5.285]	0.702 [0.675,0.733]	0.561 [0.543,0.580]
GCAM	`features.29`	0.557 [0.538,0.576]	0.830 [0.798,0.860]	**0.821** [0.798,0.844]	16.36 [15.79,16.93]	12.47 [12.01,12.93]	5.640 [5.407,5.870]	0.430 [0.412,0.449]	0.334 [0.320,0.349]
GCAM++	`features.29`	0.551 [0.533,0.569]	0.823 [0.791,0.855]	0.814≈ [0.792,0.835]	16.68 [16.09,17.27]	12.87 [12.40,13.35]	5.700 [5.462,5.938]	0.314 [0.301,0.327]	0.243 [0.233,0.253]
LCAM	`features.29`	0.544 [0.526,0.562]	0.720 [0.683,0.756]	0.747 [0.725,0.768]	15.98 [15.42,16.54]	11.48 [11.05,11.91]	5.341 [5.113,5.565]	0.434 [0.416,0.452]	0.330 [0.319,0.341]
LIME	N=1000,80 superpixels,ridge	0.603 [0.585,0.622]	**0.972** [0.958,0.984]	0.814≈ [0.790,0.837]	16.66 [16.09,17.25]	12.62 [12.17,13.06]	5.785 [5.551,6.015]	0.848 [0.816,0.882]	0.621 [0.599,0.644]
KernelSHAP	N=2M+2048,80 superpixels	0.559 [0.541,0.577]	0.954≈ [0.936,0.970]	0.820≈ [0.795,0.844]	16.68 [16.10,17.27]	12.82 [12.37,13.26]	6.140 [5.912,6.366]	0.994 [0.955,1.035]	0.736 [0.712,0.761]
SG (loc)	λ=−1, σ^2=5,gate on	0.597 [0.577,0.616]	0.770 [0.735,0.804]	0.770 [0.747,0.793]	17.28 [16.69,17.87]	13.32 [12.85,13.79]	6.217 [5.969,6.459]	**0.278** [0.267,0.290]	**0.196** [0.188,0.204]
SG (faith)	λ=−0.3, σ^2=1,gate off	0.598 [0.577,0.618]	0.821 [0.789,0.851]	0.788≈ [0.764,0.812]	17.08 [16.50,17.67]	12.98 [12.53,13.44]	6.278 [6.025,6.524]	0.543 [0.516,0.571]	0.423 [0.401,0.446]
RG anchor	τ=1	0.619 [0.601,0.637]	0.759 [0.724,0.795]	0.691 [0.673,0.709]	14.27 [13.74,14.80]	9.49 [9.10,9.89]	5.179 [4.968,5.388]	0.551 [0.526,0.578]	0.398 [0.385,0.412]
RG (loc)	τ=0.3, ε=0.1	**0.685** [0.665,0.705]	0.736 [0.699,0.772]	0.673 [0.654,0.692]	14.48 [13.95,15.02]	9.64 [9.25,10.03]	5.632 [5.411,5.845]	1.349 [1.273,1.430]	0.879 [0.842,0.918]
RG (faith)	τ=0.8, ε=0.1	0.640 [0.623,0.658]	0.712 [0.673,0.749]	0.688 [0.670,0.706]	14.05 [13.53,14.58]	9.07 [8.70,9.46]	5.117 [4.906,5.325]	0.655 [0.621,0.690]	0.466 [0.448,0.485]

Table 15: ResNet-50, ImageNet-S test split: all rows with 95% bootstrap confidence intervals, as in Table 13.

Method	Config	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓	MaxS↓	AvgS↓
Gradient	–	0.481 [0.459,0.504]	0.556 [0.515,0.596]	0.568 [0.540,0.595]	6.33 [6.22,6.44]	5.91 [5.78,6.04]	5.027 [4.900,5.151]	1.448 [1.365,1.535]	1.072 [1.019,1.130]
SmGrad (loc)	N=50, σ/range=0.15	0.549 [0.530,0.569]	0.789 [0.756,0.821]	0.738 [0.719,0.755]	6.05 [5.93,6.17]	5.17 [5.02,5.31]	4.429 [4.303,4.554]	0.735 [0.718,0.753]	0.600 [0.593,0.609]
SmGrad (faith)	N=50, σ/range=0.1	0.544 [0.525,0.564]	0.773 [0.740,0.807]	0.735 [0.715,0.753]	6.06 [5.94,6.19]	5.19 [5.04,5.34]	4.489 [4.361,4.612]	0.726 [0.706,0.748]	0.586 [0.577,0.597]
IntGrad (loc)	N_{steps}=100,black baseline	0.510 [0.488,0.532]	0.635 [0.596,0.676]	0.600 [0.574,0.626]	6.27 [6.16,6.38]	5.83 [5.69,5.96]	4.949 [4.820,5.072]	1.203 [1.152,1.257]	0.947 [0.917,0.979]
IntGrad (faith)	N_{steps}=50,black baseline	0.507 [0.484,0.529]	0.604 [0.563,0.642]	0.595 [0.568,0.622]	6.28 [6.16,6.39]	5.82 [5.68,5.95]	4.952 [4.822,5.075]	1.207 [1.157,1.261]	0.951 [0.922,0.983]
DeepLift	Rescale	0.543 [0.523,0.564]	0.727 [0.690,0.765]	0.653 [0.629,0.677]	5.97 [5.85,6.10]	5.29 [5.15,5.43]	4.515 [4.390,4.637]	1.273 [1.203,1.351]	1.000 [0.955,1.052]

continued on next page

Method	Config	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓	MaxS↓	AvgS↓
LRP-ε (loc)	ε=0.5	0.603 [0.584,0.622]	0.775 [0.740,0.809]	0.714 [0.694,0.734]	5.94 [5.81,6.06]	5.14 [4.99,5.28]	4.318 [4.200,4.433]	1.246 [1.203,1.292]	0.982 [0.960,1.006]
LRP-ε (faith)	ε=2	0.637 [0.618,0.656]	0.699 [0.662,0.736]	0.683 [0.663,0.702]	5.94 [5.81,6.06]	4.91 [4.76,5.05]	4.280 [4.164,4.394]	1.262 [1.224,1.301]	0.967 [0.948,0.986]
LRP-γ (loc)	ε=0.5, γ=2, $z^{\mathcal{B}}/\gamma/\varepsilon$ composite	0.711 [0.695,0.726]	0.842 [0.812,0.873]	0.822 [0.807,0.836]	5.69 [5.55,5.83]	4.53 [4.38,4.68]	3.935≈ [3.814,4.053]	0.633 [0.601,0.666]	0.486 [0.464,0.507]
LRP-γ (faith)	ε=0.5, γ=0.5, $z^{\mathcal{B}}/\gamma/\varepsilon$ composite	0.700 [0.684,0.716]	0.834 [0.802,0.864]	0.796 [0.781,0.812]	**5.65** [5.51,5.79]	**4.43** [4.28,4.59]	**3.931** [3.809,4.050]	0.745 [0.714,0.777]	0.584 [0.562,0.605]
GCAM	`layer4`	0.664 [0.645,0.683]	0.873 [0.844,0.899]	0.850≈ [0.830,0.870]	6.42 [6.32,6.52]	6.15 [6.03,6.26]	4.344 [4.216,4.469]	0.292 [0.274,0.312]	0.207 [0.195,0.220]
GCAM++	`layer4`	0.627 [0.607,0.646]	0.834 [0.804,0.864]	0.836 [0.815,0.857]	6.43 [6.34,6.52]	6.18 [6.07,6.29]	4.483 [4.356,4.609]	0.294 [0.278,0.310]	0.207 [0.197,0.219]
LCAM	`layer4`	0.649 [0.630,0.668]	0.869 [0.841,0.897]	**0.851** [0.831,0.871]	6.44 [6.34,6.53]	6.18 [6.07,6.29]	4.437 [4.311,4.559]	**0.274** [0.260,0.288]	**0.194** [0.184,0.204]
LIME	N=1000,80 superpixels,ridge	0.553 [0.533,0.572]	0.926 [0.903,0.947]	0.716 [0.687,0.744]	6.37 [6.26,6.46]	5.99 [5.88,6.11]	4.759 [4.630,4.881]	0.916 [0.891,0.941]	0.672 [0.655,0.690]
KernelSHAP	N=2M+2048,80 superpixels	0.558 [0.540,0.575]	**0.965** [0.949,0.979]	0.818 [0.793,0.841]	6.26 [6.16,6.36]	5.81 [5.69,5.92]	4.727 [4.601,4.849]	0.924 [0.896,0.954]	0.688 [0.670,0.708]
RG anchor	τ=1	0.693 [0.676,0.710]	0.752 [0.717,0.788]	0.756 [0.738,0.774]	5.84 [5.70,5.97]	4.71 [4.56,4.86]	4.017 [3.898,4.133]	0.708 [0.651,0.775]	0.508 [0.481,0.541]
RG (loc)	τ=0.9, ε=1	**0.719** [0.701,0.737]	0.763 [0.727,0.796]	0.760 [0.741,0.778]	5.85 [5.72,5.98]	4.70 [4.55,4.85]	4.047 [3.929,4.163]	0.798 [0.749,0.855]	0.560 [0.535,0.586]
RG (faith)	τ=1, λ=−1, σ^2=1, ε=0.25	0.674 [0.656,0.691]	0.743 [0.708,0.779]	0.738 [0.719,0.756]	5.88 [5.74,6.01]	4.74 [4.59,4.89]	4.030 [3.911,4.144]	0.685 [0.629,0.755]	0.489 [0.460,0.525]

J.3 Trade-off sweeps

Table 17 includes one-at-a-time sweeps of τ and λ_{sm} around the `loc`-selected ViT-B/16 RG configuration. These sweep rows are evaluated on the test split only after selection and never feed back into it.

J.4 Cross-Dataset Evaluation: Pascal VOC 2012

This part details the protocol behind the ViT-B/16 cross-dataset row of the summary table in Section 7 (all rows in Table 16). The goal is twofold: (i) test whether the RG / RG+Eq configurations selected on ImageNet-S transfer to a different dataset *without* re-tuning, and (ii) put baselines on equal footing by re-running the same per-method validation selection on the new pool.

Dataset. We use Pascal VOC 2012 (Everingham et al., 2010) segmentation annotations: the trainval pool of 2,913 images with pixel-accurate `SegmentationClass` masks across 20 foreground classes plus background. Each image is expanded to one (image, foreground-class) sample per distinct VOC class label present in its mask, yielding 3,422 total samples after dropping the `person` class (no faithful ImageNet-1K target exists for it) and discarding sliver masks whose pixel area on the *original* (pre-resize) annotation is below 64 pixels. Each (image, class) sample carries the same input image and an explanation target equal to the most semantically central ImageNet-1K index for that VOC class (e.g. `horse` → n02389026 SORREL; `tvmonitor` → n03782006 MONITOR); the curated mapping is part of the released code.

Per-image disjoint val/test split. The trainval pool is partitioned randomly with seed 42 at the *image* level (not the (image, class) level) into a 1:5 val/test split: the trainval stems are shuffled deterministically and the first 1/6 is assigned to val. All (image, class) samples for a given image stay in the same partition, so leakage between val tuning and test evaluation is structurally impossible. As on ImageNet-S, samples whose object mask lies entirely outside the 224×224 center crop are excluded after the split (18 val and 105 test samples). The resulting partitions:

	Images	(image, class) samples
val (tuning)	460	556
test (eval)	2,293	2,743

Pixel masks for localization. AL, PG, and Top-k Intersection are computed against the per-(image, class) *binary VOC2012 segmentation pixel mask* for that class, not bounding boxes. The mask is resized via NEAREST interpolation and center-cropped to 224×224 to match the input pipeline (`Resize(256)` → `CenterCrop(224)`), then passed verbatim into Quantus (Hedström et al., 2023) alongside the attribution map. This avoids the well-known bbox over-credit issue (Selvaraju et al., 2017) where a method that puts mass anywhere inside the box wins the metric; pixel masks reward only mass on the actual foreground object.

Tuning protocol. The protocol mirrors the ImageNet-S protocol of Appendix I, with one deliberate asymmetry: RG and RG+Eq do *not* get a fresh grid search on VOC.

- **Baselines.** Every row of the per-method ViT-B/16 grid of Table 9 is evaluated on the 556-sample val pool, and the `loc` and `faith` configurations are selected per method with the criteria of Appendix I.5. The selected rows are re-evaluated on the 2,743-sample test pool.
- **RG / RG+Eq (transfer test).** The anchor and the `loc`- and `faith`-selected ImageNet-S configurations are evaluated on the VOC test pool unchanged, without any VOC-side re-tuning.

To save compute, VOC is evaluated on localization and faithfulness only (no MaxS/AvgS and no runtime column). Any tuning advantage flows to the baselines, not to us. Wins, ties and losses are counted per metric with the tests of Appendix I.6.

Table 16: ViT-B/16 on the Pascal VOC 2012 test pool (2,743 samples): all rows with 95% bootstrap confidence intervals. RG / RG+Eq use the ImageNet-S-selected configurations without re-tuning; baselines are selected on the VOC val pool. Metrics as in Table 2; configuration symbols and row labels: Appendix J.

Method	Config	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓	MaxS↓	AvgS↓
Gradient	–	0.295 [0.286,0.304]	0.360 [0.343,0.378]	0.355 [0.344,0.367]	3.45 [3.32,3.57]	3.17 [3.05,3.28]	2.316 [2.234,2.398]	–	–
SmGrad (loc)	N=50, σ/range=0.1	0.344 [0.335,0.353]	0.557 [0.539,0.576]	0.511 [0.500,0.521]	3.25 [3.13,3.37]	2.70 [2.60,2.81]	1.940 [1.870,2.010]	–	–
SmGrad (faith)	N=50, σ/range=0.15	0.347 [0.338,0.355]	0.555 [0.537,0.574]	0.511 [0.500,0.522]	3.24 [3.12,3.35]	2.67 [2.57,2.77]	1.907 [1.841,1.975]	–	–
IntGrad (loc)	N_{steps}=300,black baseline	0.292 [0.283,0.301]	0.388 [0.369,0.406]	0.358 [0.346,0.369]	3.43 [3.31,3.56]	3.21 [3.10,3.32]	2.278 [2.197,2.356]	–	–
IntGrad (faith)	N_{steps}=100,black baseline	0.292 [0.283,0.301]	0.389 [0.370,0.407]	0.358 [0.346,0.369]	3.44 [3.31,3.56]	3.21 [3.10,3.32]	2.279 [2.198,2.357]	–	–
LRP-ε (loc/faith)	ε=0.1	0.378 [0.368,0.387]	0.604 [0.586,0.622]	0.518 [0.507,0.529]	3.07 [2.96,3.19]	2.63 [2.54,2.73]	1.923 [1.857,1.990]	–	–
LRP-γ (loc/faith)	ε=0.1, γ=1	0.403 [0.394,0.412]	0.591 [0.572,0.609]	0.545 [0.533,0.556]	2.99 [2.88,3.10]	2.37 [2.27,2.46]	1.694≈ [1.634,1.756]	–	–
AttnLRP (loc/faith)	γ_{conv}=0.25, γ_{lin}=0.05, $\varepsilon_{\text{attn}}$=0.05	0.404 [0.394,0.413]	0.602 [0.583,0.620]	0.542 [0.530,0.553]	**2.91** [2.80,3.01]	**2.31** [2.22,2.40]	**1.690** [1.631,1.750]	–	–
DAVE (loc)	N=50, $\theta_{\max}$=0, $s_{\max}$=0, $t_{\max}$=0.25	0.334 [0.325,0.342]	0.547 [0.529,0.565]	0.456 [0.445,0.467]	3.37 [3.25,3.49]	3.01 [2.90,3.12]	2.172 [2.097,2.247]	–	–
DAVE (faith)	N=50, $\theta_{\max}$=0, $s_{\max}$=0, $t_{\max}$=0.5	0.332 [0.323,0.341]	0.535 [0.517,0.553]	0.454 [0.444,0.465]	3.37 [3.25,3.49]	2.98 [2.88,3.09]	2.153 [2.080,2.226]	–	–
TIBAV	–	0.413 [0.403,0.423]	0.729 [0.713,0.746]	**0.575** [0.561,0.589]	3.04 [2.92,3.15]	2.67 [2.56,2.78]	1.731 [1.666,1.798]	–	–
GAE	–	0.355 [0.346,0.364]	0.712 [0.695,0.729]	0.548 [0.535,0.561]	3.00 [2.88,3.11]	2.63 [2.52,2.74]	1.754 [1.685,1.823]	–	–
AR (loc/faith)	w_r=0.75	0.272 [0.264,0.281]	0.471 [0.452,0.489]	0.411 [0.398,0.424]	3.57 [3.45,3.68]	3.41 [3.30,3.51]	2.146 [2.075,2.216]	–	–
LIME	N=1000,80 superpixels,ridge	0.351 [0.342,0.360]	0.785≈ [0.769,0.799]	0.523 [0.508,0.537]	3.45 [3.33,3.56]	3.21 [3.11,3.32]	2.121 [2.051,2.191]	–	–
KernelSHAP	N=2M+2048,80 superpixels	0.341 [0.332,0.350]	**0.793** [0.778,0.808]	0.561≈ [0.546,0.576]	3.36 [3.25,3.48]	3.03 [2.93,3.13]	2.077 [2.009,2.147]	–	–

continued on next page

Method	Config	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓	MaxS↓	AvgS↓
RG anchor	τ=1	0.323 [0.314,0.332]	0.517 [0.498,0.536]	0.489 [0.477,0.501]	3.30 [3.18,3.41]	2.67 [2.57,2.77]	1.980 [1.914,2.046]	–	–
RG (loc)	τ=0.4, λ_{sm}=−10, σ^2=5, ε=0.1	**0.459** [0.448,0.471]	0.562 [0.544,0.580]	0.512 [0.500,0.526]	3.44 [3.33,3.56]	3.06 [2.96,3.17]	2.152 [2.082,2.222]	–	–
RG (faith)	τ=0.7, λ_{sm}=−5, σ^2=5, ε=0.1	0.351 [0.342,0.360]	0.538 [0.520,0.557]	0.508 [0.496,0.520]	3.27 [3.15,3.38]	2.63 [2.53,2.73]	1.965 [1.900,2.032]	–	–
RG+Eq anchor	τ=1, N=50	0.329 [0.320,0.338]	0.574 [0.556,0.592]	0.534 [0.521,0.548]	3.43 [3.31,3.55]	3.06 [2.95,3.17]	1.988 [1.922,2.055]	–	–
RG+Eq (loc)	τ=0.5, λ_{sm}=−10, σ^2=2, N=50	0.456 [0.445,0.467]	0.599 [0.581,0.617]	0.571≈ [0.556,0.585]	3.45 [3.33,3.57]	3.14 [3.03,3.25]	2.085 [2.017,2.154]	–	–
RG+Eq (faith)	τ=0.7, λ_{sm}=−10, σ^2=2, N=50	0.424 [0.414,0.434]	0.591 [0.573,0.609]	0.562≈ [0.548,0.576]	3.44 [3.32,3.55]	3.08 [2.98,3.19]	2.004 [1.937,2.071]	–	–

J.5 One-factor ablations

Tables 17–19 vary one parameter at a time around each validation-selected configuration of RG, RG+Eq and SG (bold: the selected value), over the values of the declared grid; all other parameters stay at their selected values. The rows are evaluated on the 565-image test split with the localization and faithfulness metrics and never feed back into the selection. On ViT-B/16, "+Eq" evaluates RG+Eq at the parameters of the selected RG row. Configurations that are inadmissible on the validation split (Appendix I.5) are not included.

Table 17: ViT-B/16, ImageNet-S test split: one-factor ablations around the selected configurations. Center: the selected configuration (criterion in parentheses) around which the parameter in the Factor column is varied over the Value column, all other parameters fixed; bold: the center's own value. Symbols: Appendix J.

Center	Factor	Value	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓
RG (faith)	τ	0.3	0.705	0.802	0.731	8.34	7.60	5.195
		0.4	0.688	0.827	0.773	8.26	7.41	5.095
		0.5	0.665	0.821	0.793	8.22	7.20	4.956
		0.6	0.613	0.814	0.774	8.13	6.88	4.820
		0.7	0.549	0.770	0.741	8.11	6.78	4.808
		0.8	0.507	0.742	0.712	8.14	6.85	4.881
		0.9	0.482	0.697	0.683	8.21	7.02	4.983
		1	0.467	0.648	0.648	8.29	7.25	5.115
RG (faith)	λ_{sm}	-50	0.467	0.556	0.541	8.51	8.13	5.928
		-10	0.556	0.777	0.763	8.16	6.97	4.876
		-5	0.549	0.770	0.741	8.11	6.78	4.808
		-2	0.547	0.743	0.720	8.11	6.81	4.793
		-1	0.546	0.733	0.713	8.11	6.83	4.794
		0	0.546	0.727	0.706	8.13	6.87	4.804
RG (faith)	σ^2	1	0.558	0.738	0.703	8.16	6.90	4.790
		2	0.555	0.743	0.719	8.13	6.81	4.783
		5	0.549	0.770	0.741	8.11	6.78	4.808
		10	0.545	0.765	0.749	8.12	6.82	4.833
RG (faith)	ε	**0.1**	0.549	0.770	0.741	8.11	6.78	4.808
		0.25	0.595	0.804	0.778	8.16	6.95	4.846
		0.5	0.621	0.809	0.792	8.24	7.21	4.944
		1	0.630	0.812	0.794	8.29	7.43	5.051
RG (faith)	+Eq (N=50)	–	0.557	0.800	0.791	8.38	7.69	4.816
RG (loc)	τ	0.3	0.723	0.816	0.746	8.33	7.64	5.201
		0.4	0.711	0.825	0.788	8.27	7.48	5.109
		0.5	0.691	0.830	0.812	8.23	7.31	4.969
		0.6	0.632	0.827	0.801	8.16	7.00	4.847
		0.7	0.556	0.777	0.763	8.16	6.97	4.876
		0.8	0.506	0.743	0.715	8.22	7.13	5.013

Center	Factor	Value	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓
		0.9	0.477	0.683	0.673	8.29	7.28	5.143
		1	0.461	0.628	0.629	8.36	7.42	5.247
RG (loc)	λ_{sm}	-50	0.520	0.543	0.528	8.52	7.99	5.745
		-10	0.711	0.825	0.788	8.27	7.48	5.109
		-5	0.688	0.827	0.773	8.26	7.41	5.095
		-2	0.676	0.827	0.764	8.26	7.39	5.086
		-1	0.673	0.818	0.761	8.26	7.38	5.080
		0	0.671	0.818	0.758	8.26	7.38	5.076
RG (loc)	σ^2	1	0.707	0.823	0.769	8.23	7.35	4.986
		2	0.712	0.823	0.780	8.24	7.39	5.032
		5	0.711	0.825	0.788	8.27	7.48	5.109
		10	0.705	0.837	0.787	8.31	7.58	5.168
RG (loc)	ε	**0.1**	0.711	0.825	0.788	8.27	7.48	5.109
		0.25	0.709	0.842	0.780	8.29	7.55	5.155
		0.5	0.708	0.828	0.776	8.30	7.57	5.171
		1	0.708	0.823	0.773	8.31	7.59	5.181
RG (loc)	+Eq (N=50)	–	0.711	0.846	0.842	8.38	7.82	5.077
RG+Eq (faith)	τ	0.3	0.721	0.805	0.786	8.37	7.75	5.129
		0.4	0.712	0.832	0.822	8.39	7.81	5.049
		0.5	0.698	0.855	0.836	8.39	7.80	5.010
		0.6	0.681	0.846	0.833	8.39	7.77	4.943
		0.7	0.654	0.812	0.820	8.39	7.71	4.856
		0.8	0.610	0.805	0.803	8.38	7.70	4.814
		0.9	0.567	0.796	0.787	8.41	7.71	4.825
		1	0.534	0.777	0.772	8.42	7.74	4.876
RG+Eq (faith)	λ_{sm}	**-10**	0.654	0.812	0.820	8.39	7.71	4.856
		0	0.613	0.814	0.815	8.37	7.73	4.914
RG+Eq (loc)	τ	0.3	0.721	0.805	0.786	8.37	7.75	5.129
		0.4	0.712	0.832	0.822	8.39	7.81	5.049
		0.5	0.698	0.855	0.836	8.39	7.80	5.010
		0.6	0.681	0.846	0.833	8.39	7.77	4.943
		0.7	0.654	0.812	0.820	8.39	7.71	4.856
		0.8	0.610	0.805	0.803	8.38	7.70	4.814
		0.9	0.567	0.796	0.787	8.41	7.71	4.825
		1	0.534	0.777	0.772	8.42	7.74	4.876
RG+Eq (loc)	λ_{sm}	**-10**	0.698	0.855	0.836	8.39	7.80	5.010
		0	0.652	0.848	0.828	8.40	7.85	5.073

Table 18: VGG-16, ImageNet-S test split: one-factor ablations around the selected configurations, as in Table 17.

Center	Factor	Value	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓
RG (faith)	τ	0.3	0.685	0.736	0.673	14.48	9.64	5.632
		0.4	0.645	0.657	0.641	14.40	9.36	5.574
		0.5	0.626	0.653	0.628	14.27	9.14	5.451
		0.6	0.633	0.651	0.646	14.07	9.00	5.258
		0.7	0.643	0.671	0.674	14.03	8.98	5.142
		0.8	0.640	0.712	0.688	14.05	9.07	5.117
		0.9	0.630	0.742	0.693	14.05	9.17	5.126
		1	0.618	0.749	0.695	14.06	9.25	5.137
RG (faith)	λ	-5	0.518	0.458	0.536	17.60	13.82	7.362
		-2	0.623	0.604	0.631	15.90	11.58	5.852
		-1	0.636	0.676	0.661	15.03	10.35	5.404
		-0.3	0.640	0.694	0.676	14.54	9.70	5.231
		0	0.640	0.712	0.688	14.05	9.07	5.117

Center	Factor	Value	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓
RG (faith)	ε	**0.1**	0.640	0.712	0.688	14.05	9.07	5.117
		0.25	0.639	0.692	0.684	14.18	9.23	5.146
		0.5	0.638	0.690	0.680	14.34	9.46	5.183
		1	0.639	0.690	0.676	14.63	9.79	5.249
RG (loc)	τ	**0.3**	0.685	0.736	0.673	14.48	9.64	5.632
		0.4	0.645	0.657	0.641	14.40	9.36	5.574
		0.5	0.626	0.653	0.628	14.27	9.14	5.451
		0.6	0.633	0.651	0.646	14.07	9.00	5.258
		0.7	0.643	0.671	0.674	14.03	8.98	5.142
		0.8	0.640	0.712	0.688	14.05	9.07	5.117
		0.9	0.630	0.742	0.693	14.05	9.17	5.126
		1	0.618	0.749	0.695	14.06	9.25	5.137
RG (loc)	λ	-5	0.561	0.543	0.555	17.05	13.21	6.898
		-2	0.657	0.696	0.634	15.31	10.80	5.885
		-1	0.674	0.735	0.656	14.85	10.13	5.731
		-0.3	0.680	0.727	0.666	14.66	9.87	5.675
		0	0.685	0.736	0.673	14.48	9.64	5.632
RG (loc)	ε	**0.1**	0.685	0.736	0.673	14.48	9.64	5.632
		0.25	0.691	0.747	0.673	14.66	9.91	5.659
		0.5	0.697	0.742	0.672	14.87	10.19	5.705
		1	0.702	0.742	0.671	15.13	10.55	5.767
SG (faith)	λ	-2	0.614	0.710	0.636	18.17	14.83	6.941
		-1	0.609	0.750	0.697	17.91	13.87	6.517
		-0.3	0.598	0.821	0.788	17.08	12.98	6.278
		0	0.547	0.781	0.764	17.63	14.35	6.789
SG (faith)	σ^2	**1**	0.598	0.821	0.788	17.08	12.98	6.278
		2	0.601	0.811	0.765	17.30	13.07	6.323
		5	0.604	0.798	0.731	17.66	13.50	6.417
		10	0.607	0.763	0.706	17.86	13.77	6.500
SG (faith)	gate $\tilde{\Phi}$	**off**	0.598	0.821	0.788	17.08	12.98	6.278
		on	0.568	0.823	0.805	17.30	13.74	6.428
SG (loc)	λ	-5	0.568	0.550	0.558	18.17	14.31	7.509
		-2	0.620	0.703	0.681	17.84	13.63	6.404
		-1	0.597	0.770	0.770	17.28	13.32	6.217
		-0.3	0.566	0.830	0.809	17.33	13.75	6.452
		0	0.554	0.827	0.802	17.55	14.16	6.618
SG (loc)	σ^2	1	0.596	0.793	0.766	17.32	13.35	6.246
		2	0.597	0.786	0.766	17.31	13.34	6.232
		5	0.597	0.770	0.770	17.28	13.32	6.217
		10	0.595	0.772	0.776	17.25	13.30	6.206
SG (loc)	gate $\tilde{\Phi}$	off	0.614	0.696	0.628	18.19	15.00	7.050
		on	0.597	0.770	0.770	17.28	13.32	6.217

Table 19: ResNet-50, ImageNet-S test split: one-factor ablations around the selected configurations, as in Table 17.

Center	Factor	Value	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓
RG (faith)	τ	0.3	0.688	0.727	0.684	6.10	5.10	4.357
		0.4	0.676	0.712	0.692	6.08	5.03	4.340
		0.5	0.677	0.720	0.706	6.06	5.00	4.308
		0.6	0.673	0.704	0.711	5.99	4.96	4.270
		0.7	0.667	0.680	0.707	5.95	4.89	4.219
		0.8	0.665	0.688	0.711	5.89	4.85	4.134
		0.9	0.674	0.720	0.728	5.88	4.79	4.075
		1	0.674	0.743	0.738	5.88	4.74	4.030

Center	Factor	Value	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓
RG (faith)	λ	-5	0.667	0.740	0.730	5.90	4.81	4.034
		-2	0.672	0.740	0.736	5.89	4.75	4.034
		-1	0.674	0.743	0.738	5.88	4.74	4.030
		-0.3	0.674	0.747	0.739	5.87	4.75	4.026
		0	0.675	0.747	0.740	5.86	4.74	4.023
RG (faith)	σ^2	**1**	0.674	0.743	0.738	5.88	4.74	4.030
		2	0.673	0.742	0.737	5.87	4.74	4.032
		5	0.671	0.736	0.736	5.88	4.75	4.031
		10	0.670	0.747	0.734	5.89	4.76	4.031
RG (faith)	ε	0.1	0.658	0.745	0.722	5.90	4.79	4.049
		0.25	0.674	0.743	0.738	5.88	4.74	4.030
		0.5	0.692	0.747	0.754	5.86	4.72	4.023
		1	0.712	0.766	0.761	5.85	4.72	4.038
RG (loc)	τ	0.3	0.733	0.761	0.719	6.03	5.04	4.272
		0.4	0.717	0.758	0.718	6.02	4.96	4.243
		0.5	0.709	0.740	0.720	5.98	4.92	4.216
		0.6	0.710	0.733	0.728	5.93	4.87	4.172
		0.7	0.716	0.745	0.741	5.91	4.80	4.125
		0.8	0.720	0.761	0.752	5.87	4.72	4.078
		0.9	0.719	0.763	0.760	5.85	4.70	4.047
		1	0.713	0.770	0.763	5.85	4.70	4.035
RG (loc)	λ	-5	0.708	0.765	0.745	5.89	4.80	4.069
		-2	0.716	0.773	0.755	5.87	4.74	4.055
		-1	0.718	0.777	0.757	5.86	4.72	4.054
		-0.3	0.719	0.768	0.758	5.86	4.72	4.050
		0	0.719	0.763	0.760	5.85	4.70	4.047
RG (loc)	ε	0.1	0.656	0.712	0.712	5.89	4.84	4.106
		0.25	0.674	0.717	0.730	5.87	4.77	4.067
		0.5	0.696	0.736	0.749	5.84	4.70	4.042
		1	0.719	0.763	0.760	5.85	4.70	4.047

J.6 Pointing game at matched granularity

The pointing game of Quantus counts a hit whenever any pixel attaining the map maximum lies in the mask. For piecewise-constant maps (superpixel methods; attention maps on the 14×14 patch grid) the maximum is a region of hundreds to thousands of pixels, for pixel-level maps usually a single pixel. Table 20 reports, for the reported rows, the median size of the maximum set, the expected hit rate of a single maximum pixel drawn uniformly from that set (PG_{1px}), and the pointing game after average-pooling every map to the same 16×16-pixel grid (PG_{16px}). Drawing a single maximum pixel removes most of the advantage of the superpixel methods LIME and KernelSHAP. At the common 16×16 grid the leading methods move closer together: on ViT-B/16 LRP-γ and AttnLRP lead, on VGG-16 the localization-selected SG row and the faithfulness-selected RG row are not significantly below the best baseline (Grad-CAM++), and on ResNet-50 Grad-CAM, LayerCAM and LRP-γ lead (McNemar tests against the best baseline row).

Table 20: Pointing game under the Quantus tie rule (PG, as in the main tables) and at matched granularity, ImageNet-S test split. max. set: median number of pixels attaining the map maximum; PG_{1px}: expected hit rate of one pixel drawn uniformly from that set; PG_{16px}: PG after average-pooling the map to 16×16-pixel cells. Rows: the `loc`-selected or fixed configuration of every method, plus the `faith` rows of our games. **Bold**: best per model and column.

Model	Method	PG	max. set (px)	PG_{1px}	PG_{16px}
ViT-B/16	Gradient	0.361	1	0.361	0.382
	SmGrad	0.706	1	0.706	0.835
	IntGrad	0.513	1	0.513	0.543
	LRP-ε	0.834	1	0.834	0.904
	LRP-γ	0.873	1	**0.873**	**0.968**
	AttnLRP	0.855	1	0.855	0.926
	DAVE	0.717	1	0.717	0.781
	TIBAV	0.878	256	0.784	0.878
	GAE	0.743	256	0.670	0.743
	AR	0.565	256	0.516	0.565
	LIME	0.897	1104	0.693	0.786
	KernelSHAP	**0.961**	1031	0.833	0.903
	RG (faith)	0.770	1	0.770	0.899
	RG	0.825	1	0.825	0.888
	RG+Eq (faith)	0.812	1	0.812	0.919
	RG+Eq	0.855	1	0.855	0.919
VGG-16	Gradient	0.738	1	0.738	0.812
	SmGrad	0.782	1	0.782	0.915
	IntGrad	0.770	1	0.770	0.860
	DeepLift	0.701	1	0.701	0.784
	LRP-ε	0.832	1	**0.832**	0.906
	LRP-γ	0.692	1	0.692	0.908
	GCAM	0.830	1	0.823	0.920
	GCAM++	0.823	1	0.815	**0.924**
	LCAM	0.720	1	0.715	0.834
	LIME	**0.972**	1103	0.826	0.899
	KernelSHAP	0.954	1011	0.826	0.894
	SG (faith)	0.821	1	0.821	0.883
	SG	0.770	1	0.770	0.908
	RG (faith)	0.712	1	0.712	0.915
	RG	0.736	1	0.736	0.873
ResNet-50	Gradient	0.556	1	0.556	0.630
	SmGrad	0.789	1	0.789	0.919
	IntGrad	0.635	1	0.635	0.743
	DeepLift	0.727	1	0.727	0.793
	LRP-ε	0.775	1	0.775	0.880
	LRP-γ	0.842	1	0.842	0.958
	GCAM	0.873	1	**0.862**	**0.959**
	GCAM++	0.834	1	0.827	0.945
	LCAM	0.869	1	0.860	0.958
	LIME	0.926	1058	0.729	0.844
	KernelSHAP	**0.965**	1014	0.825	0.912
	RG (faith)	0.743	1	0.743	0.903
	RG	0.763	1	0.763	0.912

J.7 Gradient-fallback attention

The RG backward operates by default through a routing rule that holds the attention policy fixed and routes relevance only through the Value path, with zero gradient on Q, K (Value-only routing conditioned on the realized attention matrix, as in CP-LRP (Ali et al., 2022); see Appendix D.4). The natural alternative is a gradient fallback: pure autograd through the attention block, recovering the zennit-style ε-LRP attention treatment. Table 21 swaps the attention rule at the $\alpha\beta$-LRP anchor and at the two validation-selected RG configurations and evaluates both rules on the 565-image test split. The softmax risk λ_{sm} acts on the Value-path routing and is therefore inactive under the gradient fallback. The gradient fallback lowers all three localization metrics at every

configuration, AL and TK significantly throughout, and it does not improve faithfulness at any configuration: PF_5, PF_{20} and Sel are worse in every cell, significantly so in four of the nine.

Table 21: ViT-B/16, RG: attention rule ablation on the ImageNet-S test split (565 images), at the anchor and at the loc- and faith-selected configurations. Each cell shows the gradient fallback on top and the composite Value rule of Theorem 9 (gray) below; configuration symbols: Appendix J; †: paired difference significant (paired bootstrap / McNemar for PG, Holm-corrected over the table, $p < 0.05$).

Center	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓
anchor (τ=1)	$0.504^\dagger$	0.722	$0.694^\dagger$	8.19	$6.98^\dagger$	$4.877^\dagger$
	0.509	0.724	0.699	8.19	6.97	4.854
loc (τ=0.4, λ_{sm}=−10, σ^2=5, ε=0.1)	$0.624^\dagger$	$0.743^\dagger$	$0.672^\dagger$	8.31	7.50	$5.364^\dagger$
	0.711	0.825	0.788	8.27	7.48	5.109
faith (τ=0.7, λ_{sm}=−5, σ^2=5, ε=0.1)	$0.543^\dagger$	$0.729^\dagger$	$0.706^\dagger$	8.13	$6.87^\dagger$	4.816
	0.549	0.770	0.741	8.11	6.78	4.808

J.8 Qualitative comparison

Figure 6 compares attribution maps for six ImageNet-S examples on ViT-B/16. Each column shows a method at its `loc`-selected configuration (Appendix I.5), or at its fixed configuration if the method has no tunable parameters (Gradient, TIBAV): RG with τ=0.4, λ_{sm}=−10, σ^2=5, ε=0.1 and RG+Eq with τ=0.5, λ_{sm}=−10, σ^2=2, N=50, both with the merged attention–Value rule, i.e. the composite rule of Theorem 9, which routes through the conditioned attention block as one map $X \mapsto O$; SmoothGrad with N=50, σ/range=0.2; Integrated Gradients with N_{steps}=100 and a black baseline; DAVE with N=50, $\theta_{\max}$=0, $s_{\max}$=0, $t_{\max}$=0.5; LRP-γ with ε=0.1, γ=2; and AttnLRP with γ_{conv}=0.25, γ_{lin}=0.05, $\varepsilon_{\text{attn}}$=0.25. As in the evaluation, every map explains the ImageNet-S label, which is the top-1 prediction for all images except the kuvasz (predicted: English setter). The first image is from the test split, the other five are from the 50-image validation split used for the selection. Heatmaps show the raw evaluator-side maps, before the absolute value and max-abs normalization of Appendix I.3 (the 14×14 TIBAV map is upsampled by nearest neighbor, as for the metrics): red marks positive and blue negative attribution; each sign is divided by its 99th percentile over the image and clipped, so the top 1% of pixels saturate and concentrated maps stay legible in print; the same display transform is used for every method.

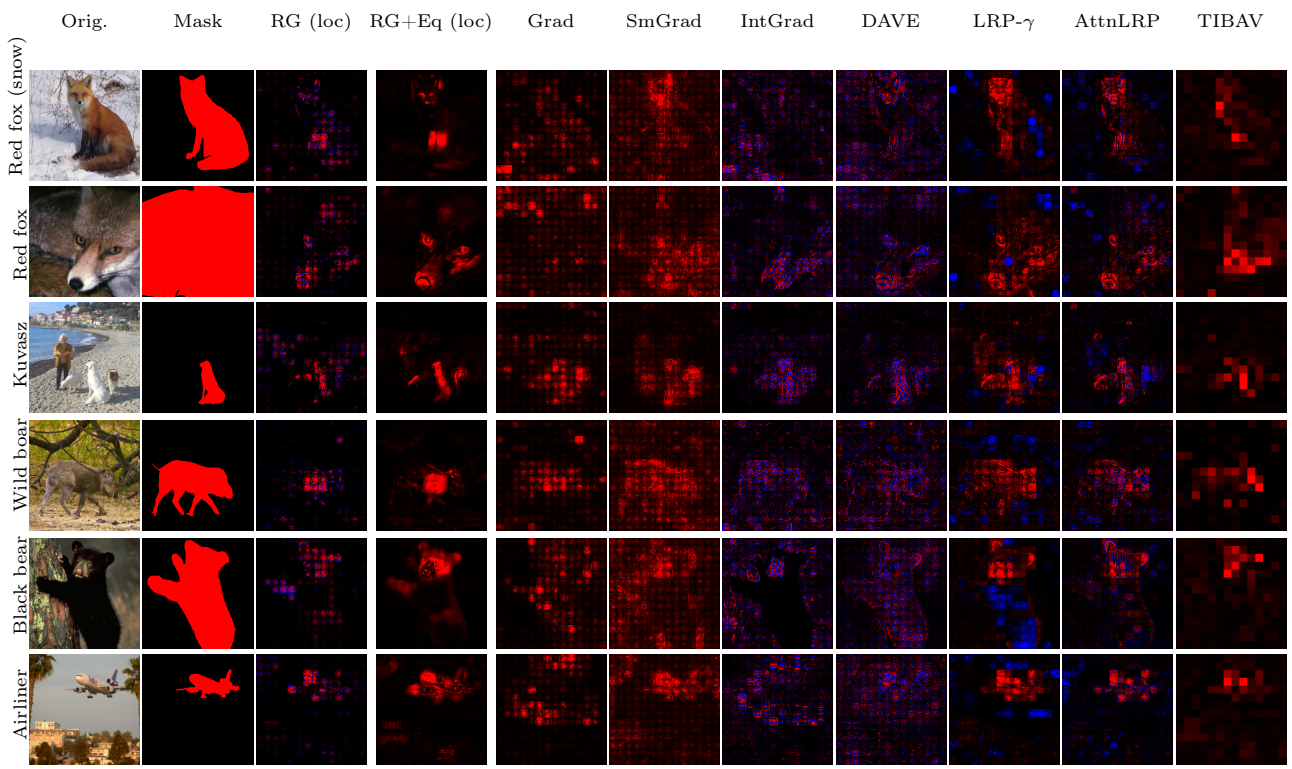

Figure 6: Dense qualitative comparison on six ImageNet-S examples (ViT-B/16). Columns: Original; ImageNet-S pixel Mask; RG and RG+Eq (merged attention–Value rule); Gradient, SmoothGrad, Integrated Gradients, DAVE, LRP-γ, AttnLRP and TIBAV. Every method is shown at its `loc`-selected configuration, Gradient and TIBAV at their fixed configurations (values in the text). Red: positive, blue: negative attribution, each sign divided by its 99th percentile over the image and clipped.

K BROADER IMPACT

The work is methodological: it studies attribution methods for existing image classifiers and does not introduce a deployed decision system, a new dataset of people, or a high-risk generative model. Its intended positive impact is to make explanation methods more diagnostically accountable. In particular, trajectory-level comparisons can reveal cases where visually similar attribution maps arise from different backward calculations, and can therefore support more careful auditing of explanation pipelines used in scientific or engineering workflows.

The same capability can also have negative or misleading uses. Sharper or more stable attribution maps may create unwarranted confidence in a model if users treat explanations as causal guarantees rather than as post-hoc diagnostics. In safety-critical or fairness-sensitive settings, improved-looking explanations could be used to justify a classifier whose predictions remain brittle, biased, or dataset-dependent. The Hellinger trajectory distance also compares internal calculation paths rather than social validity; small trajectory distance does not imply that the underlying model is fair, robust, or appropriate for deployment.

We therefore view the framework as an analysis tool rather than a deployment certificate. The experiments use public benchmark classifiers and ImageNet-derived evaluation data, and the results should be interpreted as evidence about attribution behavior under these benchmarks, not as evidence that the underlying models are suitable for consequential use.